STEPPERONLINE®

User Manual

DSY-C Series

CANopen DC Servo Drive

User manual (V1.0)



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Chapter 1 Selection of Servo System

1.1Drive specifications

Basic Specifications	Model		DSY-200C	DSY-400C	DSY-750C	DSY-1200C
	Continuous Output Current (Arms)		7.5	15	25	40
	Maximum Output Current (Arms)		24	48	80	120
	Power Supply Input		24VDC~70VDC			
	Operating Environment	Temperature	Operation Temperature: 0℃～55℃ (If Environment temperature over 55 ℃, plz lower it) Storage Temperature: -40℃～80℃			
		Humidity	0～90% RH（No dew environment）			
		Elevation	Elevation ≤1000m			
		Vibration/Strike	5.88m/s2 / 19.6m/s2			
		IP Grade /Contamination	IP20/ Contamination Rating 2			
	Control Method		MOSFET SVPWM Space vector pulse width modulation algorithm			
	Encoder		17-bit magnetic encoder; 23-bit optical encoder			
	Input/ Output Ports	IO Signal Input	3 pcs (General)			
		IO Signal Output	1 General DO Output, 1 Brake Output			
	Network	CAN	Canopen Communication			
		RS-232	Connect with PC to debug servo drive			
	Protection Function		Overcurrent, Overload, Overvoltage, Low Voltage, Overspeed, Overheat, Encoder error, Communication			

				error, Position deviation
	Control Mode			Homing Mode :(HM) Profile Position Mode :(PP) Profile Velocity Mode :(PV) Profile Torque Mode :(PT)
Functional specifications	Position Control Mode (PP/HM)	Location Command Format		CANopen Bus Digital Quantity
		Smoothing Filter		Smoothing process for position command to ensure motor run more stably
		Electronic Gear		One sets of electronic gear ratios available
		Feedforward Compensation		0 ～ 100% (default resolution ratio: 1%)
	Speed-Torque Control Mode (PV/PT)	Command Form		CANopen Bus Digital Quantity
		Speed Change Rate	Current control Precision	±5% (Rated torque)
			Overload Capacity	300% motor rated torque
			Temperature Fluctuation	25±25℃ (Rated speed)
	Common Spec	Self-tuning Function		Identifying inertia
		Over-distance (OT) Prevent Function		Slowing down to stop when P-OT、N-OT
		Error Feedback and Record		Red and green double-hole indicator, 9 set of records

1.2 Servo motors and drives model description

1.2.1 Drive Model Description

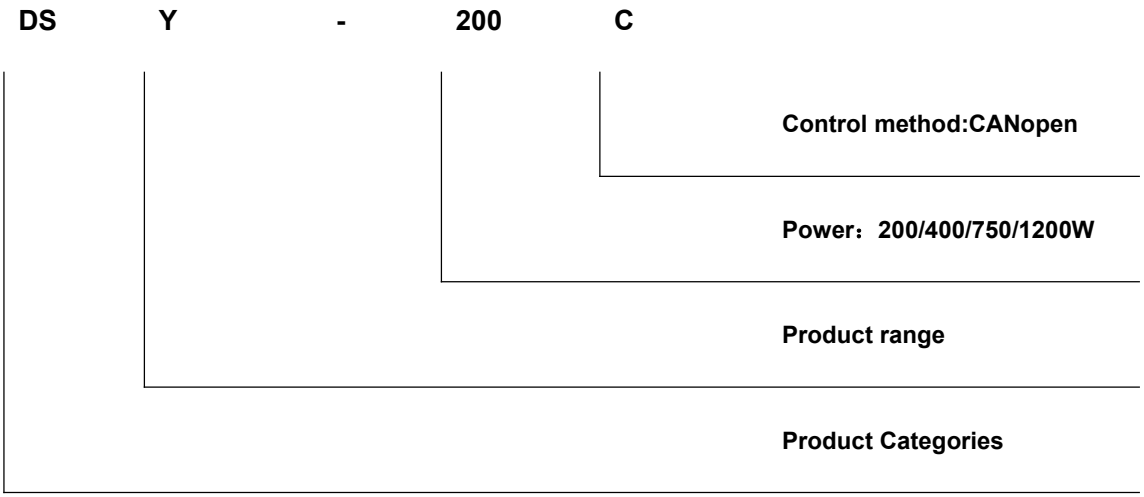


Figure1.1 Drive naming rule

1.2.2 Servo motor model description

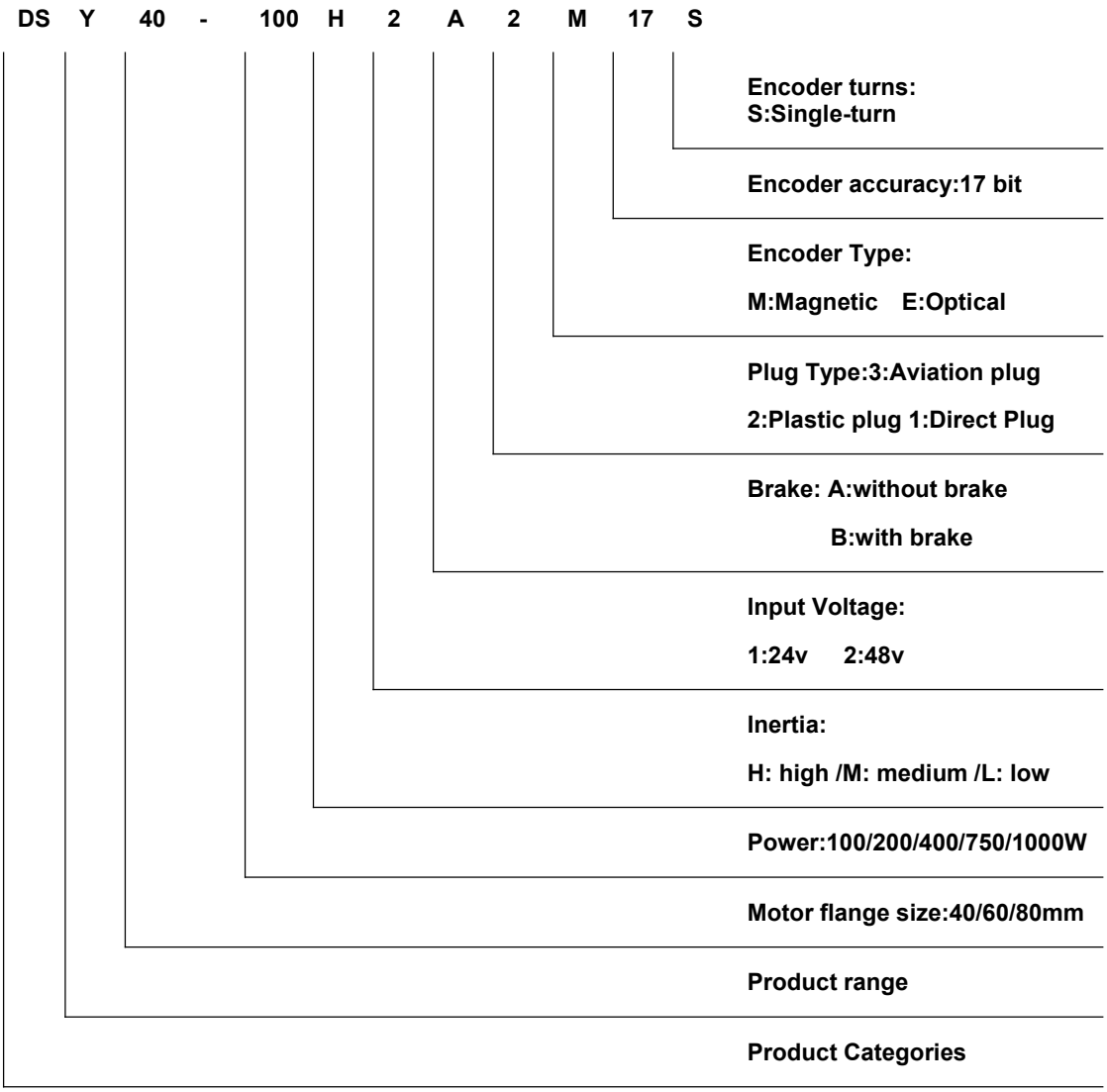


Figure 1.2 Motor naming rules

1.3 Servo drive and servo motor matching list

Table 1.2 Servo drive and motor matching table

Servo Motor Models		Flange	Power	Drive Model	Matching cable
				DSY-xxC	
Small inertia	DSY40-100L2A2M17S	40	100w	200	P1-DS-xx E2-DS-xx
	DSY40-100L2A2M17S	40	100w	200	
	DSY60-200L2A2M17S	60	200W	200	
	DSY60-200L2B2M17S	60	200W	200	
	DSY60-400L2A2M17S	60	400W	400	
	DSY60-400L2B2M17S	60	400W	400	
	DSY80-750L2A2M17S	80	750W	750	P2-DS-xx E2-DS-xx
	DSY80-750L2B2M17S	80	750W	750	
	DSY80-1000L2A2M17S	80	1000W	1200	
	DSY80-1000L2B2M17S	80	1000W	1200	

1.4Matching motor parameters

Table 1.3 Motor parameter table

Motor spec.	Rated output KW	Rated torque Nm	Max.Torque Nm	Rated current Arms	Rated Speed min^{-1}	Rotor Inertia $Kgm^{2}\cdot 10^{-4}$
DSY40-100L2A2M17S	0.1	0.318	0.954	3.1	3000	0.046
DSY40-100L2A2M17S	0.1	0.318	0.954	3.1	3000	0.0483
DSY60-200L2A2M17S	0.2	0.64	1.91	5.2	3000	0.27
DSY60-200L2B2M17S	0.2	0.64	1.91	5.2	3000	0.29
DSY60-400L2A2M17S	0.4	1.27	3.81	10.2	3000	0.49
DSY60-400L2B2M17S	0.4	1.27	3.81	10.2	3000	0.51
DSY80-750L2A2M17S	0.75	2.39	7.17	18.5	3000	1.42
DSY80-750L2B2M17S	0.75	2.39	7.17	18.5	3000	1.62
DSY80-1000L2A2M17S	1.0	3.2	9.6	24.5	3000	1.92
DSY80-1000L2B2M17S	1.0	3.2	9.6	24.5	3000	2.12

1.5Matching cable

The naming rules for motor power lines are as follows：

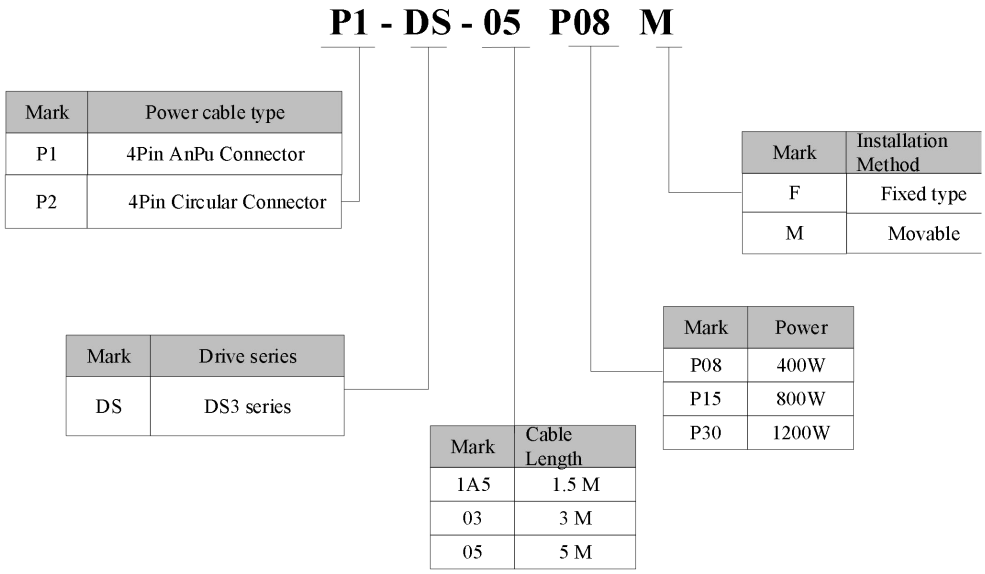
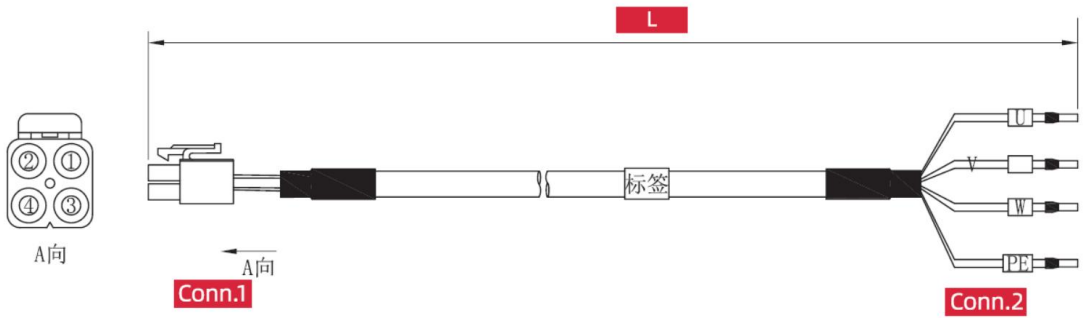
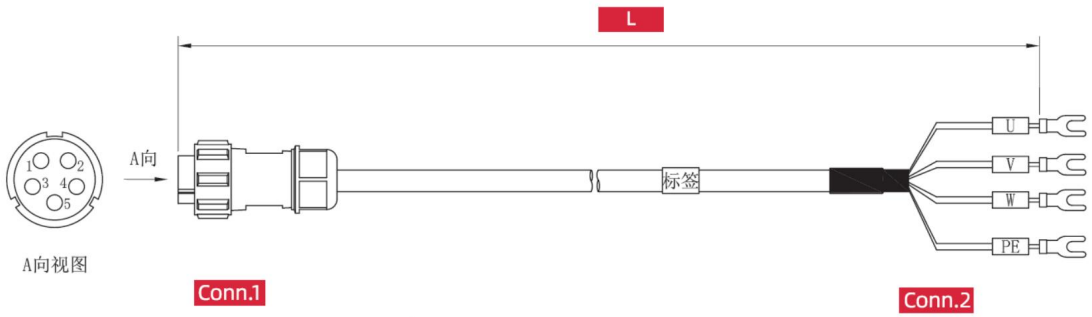


Figure 1.3 Power line naming rules

The structure diagram of the motor power cable is shown in the table below：



1.P1　AnPu Connector Cable(Lead Length L =1.5m、3m、5m)



2.P2.P2　Circular Connector Cable (Lead Length L =1.5m、3m、5m)

Figure 1.4 Power cable structure diagram

The motor power line pins are defined as follows

Conn.1	Pin definition	Conn.2	Pin color
1	U	U	Red
2	V	V	Yellow
3	W	W	Blue
4	PE	PE	Yellow green

The naming rules for motor encoder lines are as follows:

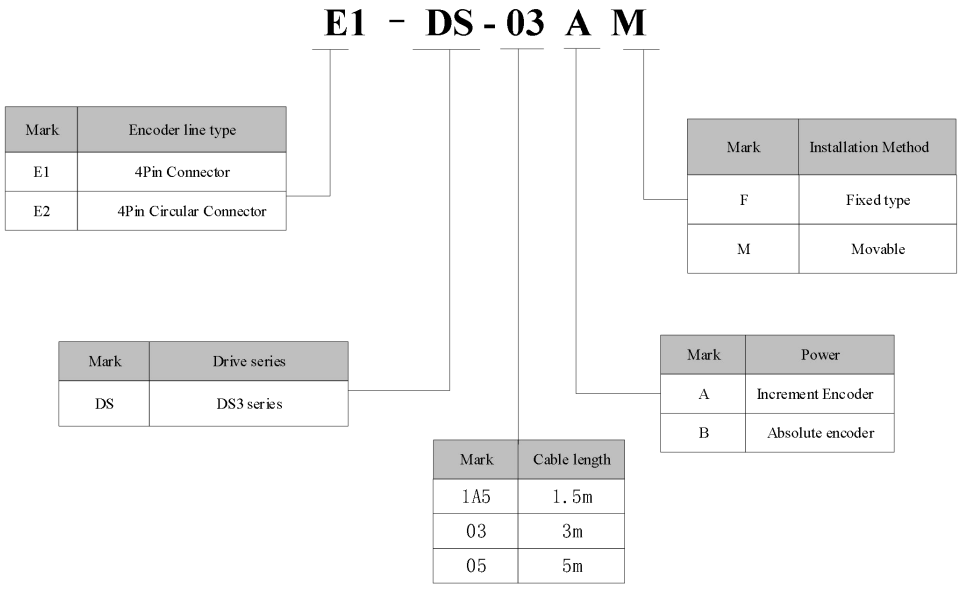


Figure 1.5 Encoder line naming rules

The structure diagram of the encoder cable is shown in the table below:

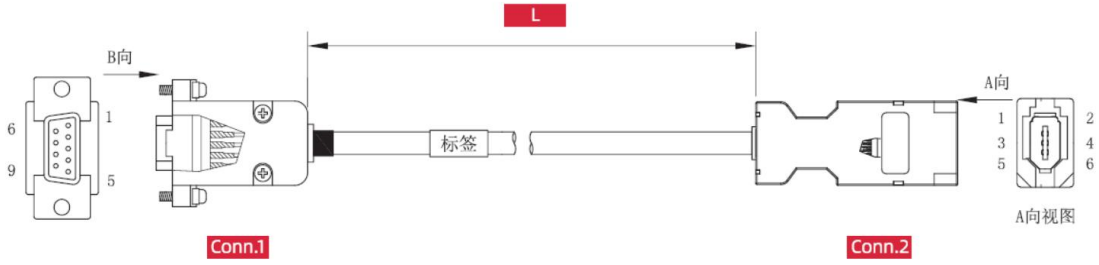


Figure 1.6 Encoder cable structure diagram (L=1.5m、3m、5m)

The motor encoder line pins are defined as follows：

Conn.1	Pin definition	Conn.2	Pin color
7	5V	1	Blue
8	0V	2	Orange
3	SD+	5	Brown
4	SD-	6	Green
Shell	PE	Shell	faraday cage

Chapter 2 Dimensions and Product Installation

2.1 Servo drive dimensions

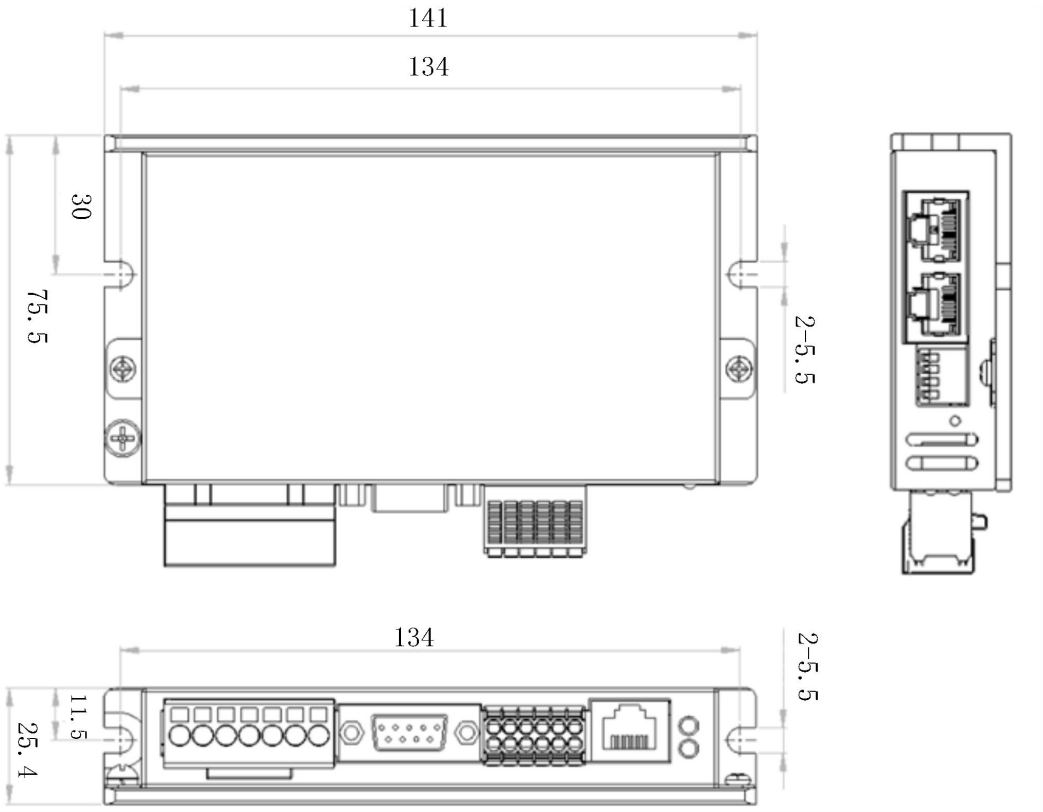


Figure 2.1 The picture of DSY-200C/DSY-400C drive structure

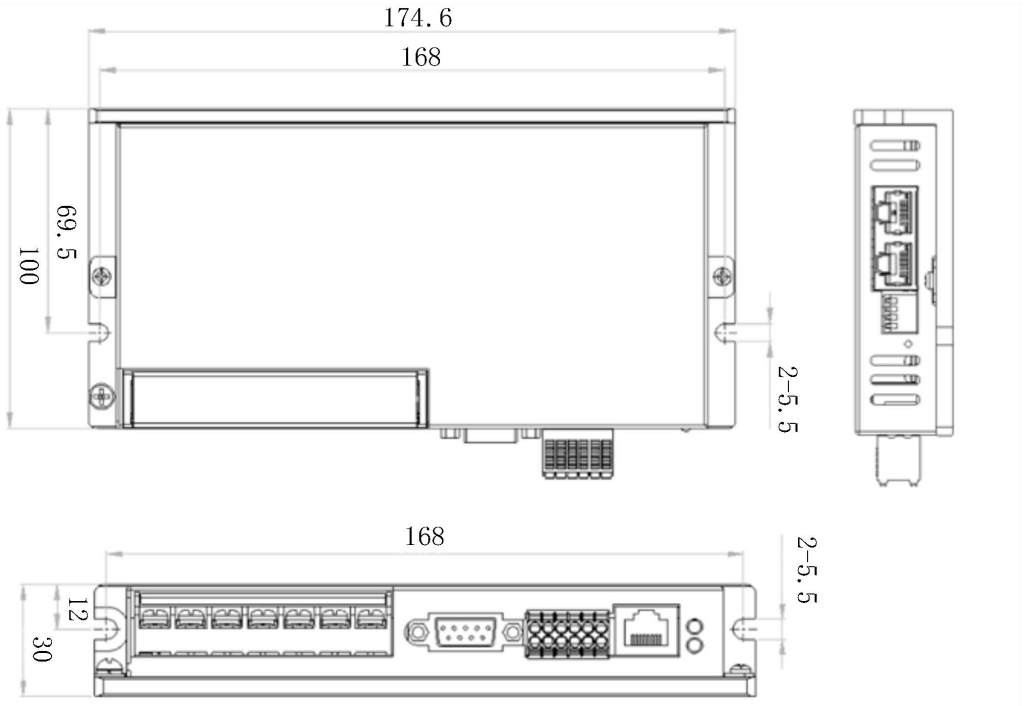


Figure 2.2 The picture of DSY-750C drive structure

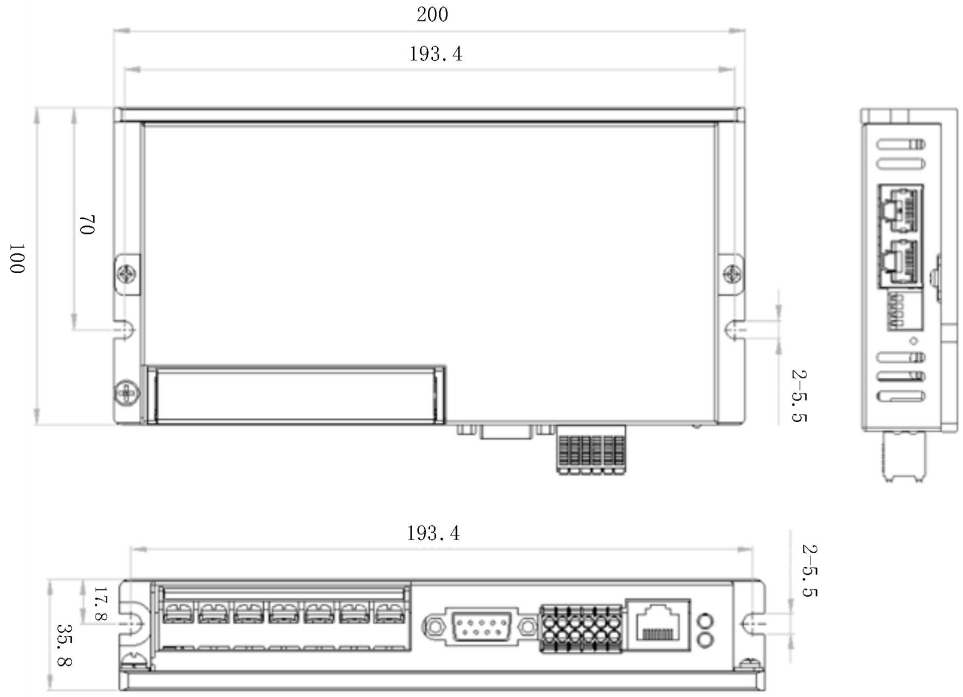


Figure 2.3 The picture of DSY-1200C drive structure

2.2 Servo drive installation

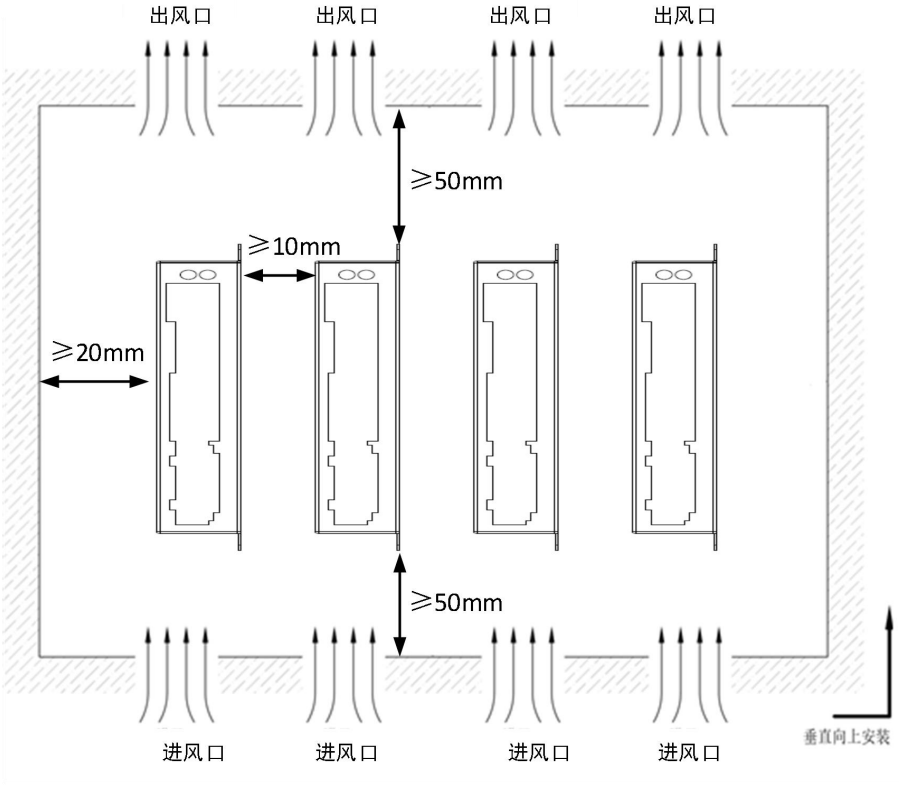
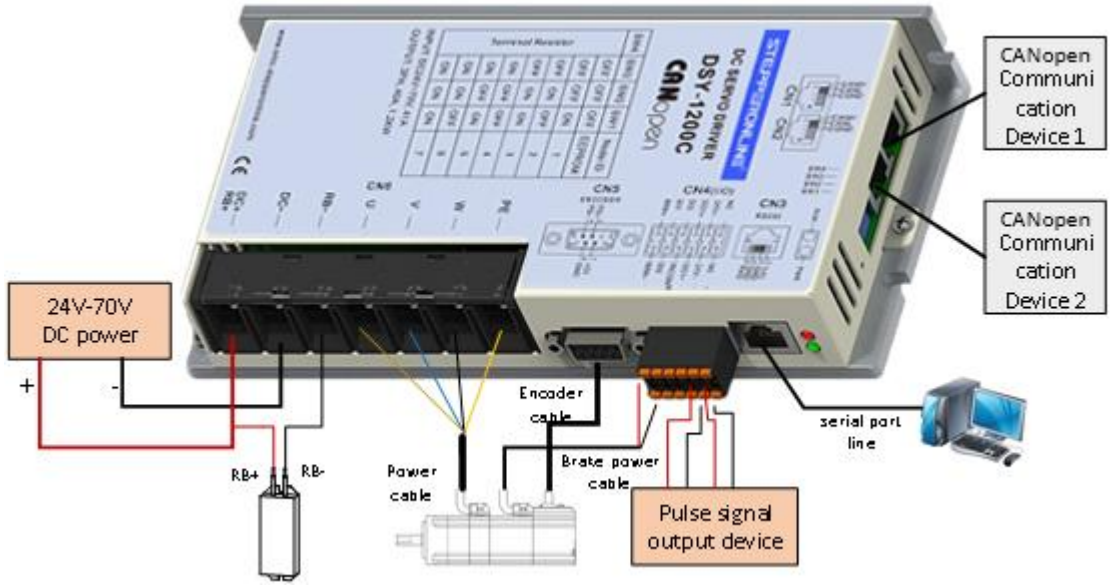


Figure 2.4 Servo drive installation

Chapter 3 Servo Drive and Motor Connection

3.1 System structure diagram



The names, functions and specifications of each part of the drive are as follows.

Table 3.1 Servo drive main circuit terminal name and function

NO.	Name	Describe
1	DC+、DC-	Input control circuit power according to specifications
2	RB+,RB- Regenerative Resistor Terminal	External Brake Resistor Connection Port
3	U、V、W、PE Motor power cable	Connect servo motor three-phase and ground wire
5	Communication port CN3	Connected to PC via RS232 to monitor the drive, test and change parameter, etc
6	Communication control CN2、 CN1	Connected to CANopen communication devic
7	Control port CN4	Connect to host controller for IO signal control
8	Encoder feedack port CN5	Connected to the motor encoder cable

3.2 Main circuit

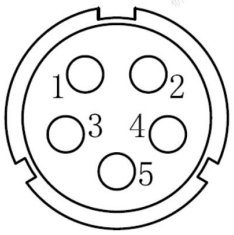
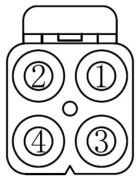
The functions, names and specifications of each part of the main circuit are as follows.：

Table 3.2 Servo drive main circuit terminal function specifications

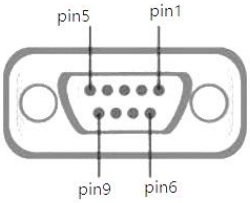
Name	Terminal mark	Function, specification
Power input terminal	DC+、DC-	Power circuit of the power supply according to the input specifications.
Regenerative resistor connection terminal	RB+,RB-	The external regenerative resistor between the RB + with RB -. For the external brake resistor, please buy separately.
Servo motor Power cable connection terminal	U、V、W	Connected to the U, V, W of the motor.
Ground Terminal	PE	Two grounding terminals are connected to the power grounding terminal and the motor grounding terminal. Be sure to ground the entire system.

3.3Power cable connector

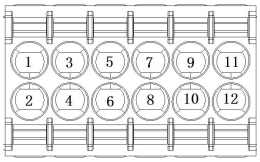
Table 3.3 Power cable connector

Terminal pin		Connector picture
5Pin circular connector		
Pin NO.	Signal name	
1	U	
2	V	
3	W	
4	PE	
4Pin AnPu connector		
in NO.	Signal name	
1	U	
2	V	
3	W	
4	PE	

3.4Encoder connector

Pin definition					Picture
1	2	3	4	5	
		SD+	SD-		
6	7	8	9	PE	
	+5V	GND		PE	

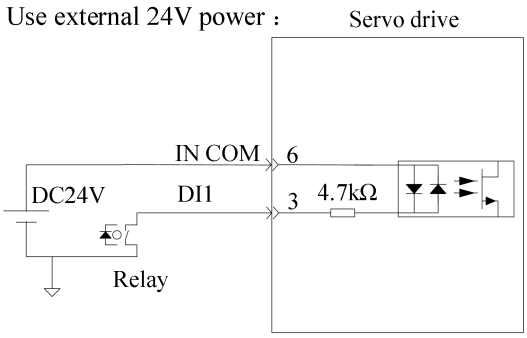
3.5 DIDO Port

Definition		pin	function
General input and output signal 	BRK+	1	Brake output
	BRK-	2	
	DI1	3	Digital input 1
	DI2	4	Digital input 2
	DI3	5	Digital input 3
	INCOM	6	DI power supply input port，24V
	DO+	7	Digital output 1
	DO-	8	
	24V+	9	NC
	24V-	10	
	NC+	11	NC
	NC-	12	

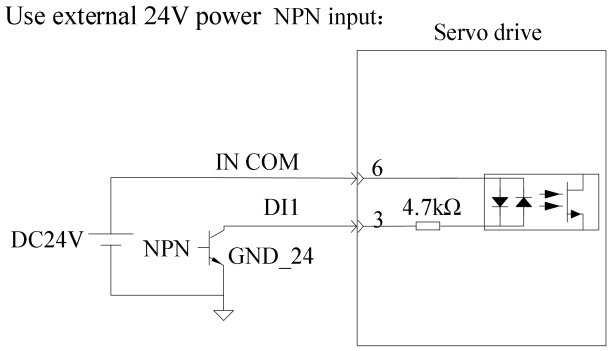
3.5.1 Digital input circuit

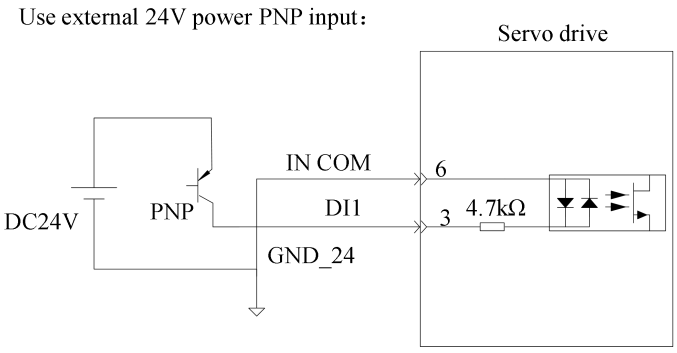
Take DI1 as an example, DI1~DI3 is the same.

a) When the host computer is relay output:



b) When the host computer is open collector output:

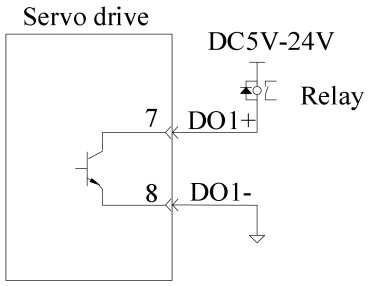




Note：The mixing of NPN and PNP inputs is not supported.

3.5.2Digital output circuit

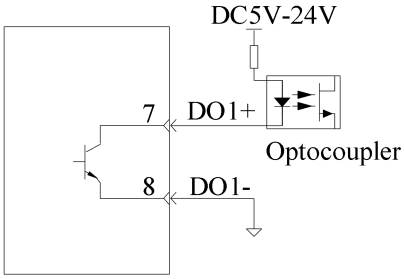
a) When the host computer is a relay input



Note：

When the host computer is relay input, be sure to connect the freewheeling diode in the correct direction, otherwise the DO port may be damaged.

b) When the host computer is optocoupler input



Note：

1. Be sure to connect a current limiting resistor.

2. The maximum allowable voltage and current of the internal optocoupler circuit of the servo driver are as follows:

Voltage: DC30V (maximum)

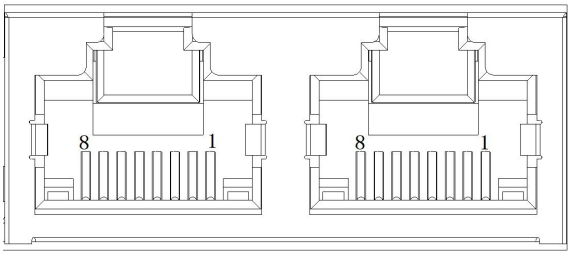
Current: DC50mA (maximum)

3.6 Communication signal port

3.6.1 Bus port CN1/CN2

CN1 and CN2 is the CANopen communication connector：

Table 3.4 communication connector pin definition

Pin	Signal name	Direction	Terminal
1	CAN_H	Output	
2	CAN_L	Output	
3	GND	Output	
4	--	unused	
5	--	unused	
6	--	unused	
7	--	unused	
8	--	unused	

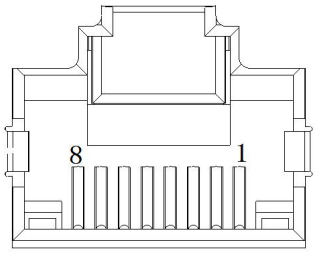
Side-Mounted Dip Switch: The following description:

Dip switch	SW1	SW2	SW3	SW4
1	on	off	off	On: Enable 120Ω termination resistance Off: Disable 120Ω termination resistance
2	off	on	off	
3	on	on	off	
4	off	off	on	
5	on	off	on	
6	off	on	on	
7	on	on	on	

Note: Node_ID definition must be configured via P10.06.

3.6.2 Post computer communication port CN3

Table 3.5 communication connector pin definition

Pin	Definition	Function	Terminal
1	--		
2	--		
3	RS232-TXD	RS232 communication port	
4	GND		
5	GND		
6	RS232-RXD		
7	--		
8	--		
Shell	PE	Shielding	

Chapter 4 CANopen Communication

4.1 Overview of CANopen

CAN stands for Controller Area Network. The CAN communication standard defines only the physical layer and data link layer but does not provide detailed high-layer application protocols for specific application domains. With the rapid development of the industrial control field, the application of CAN bus communication has become increasingly widespread. To promote the standardization and regulation of CAN communication in the same industry, the CIA organization has developed the standardized CANopen communication protocol for various industrial sectors.。

CANopen is a higher-layer protocol based on CAN communication. It specifies the usage guidelines for CAN message identifiers and 8-bit data. This protocol has been widely applied in various industrial sectors. The DS3 drive is a CANopen-compliant slave device that strictly adheres to the DS301 protocol and the DSP402 servo protocol. The device supports standard communication objects such as Process Data Objects (PDO), Service Data Objects (SDO), synchronization messages, emergency messages, and network management messages (NMT messages). In servo control modes, it supports PP (profile position), PV (profile velocity), PT (profile torque), and HM (homing) modes.

4.2 Object Dictionary Structure

The object dictionary is a collection of ordered objects and is the most important component of the CANopen protocol. All servo control objects perform data reading and writing through the object dictionary. The object dictionary uses a 16-bit index and an 8-bit sub-index for addressing. The DS3 drive supports communication protocol indexes in the range 1000h-1FFFh and device protocol indexes in the range 6000-6FFFh.

The object dictionary consists of object name, index, sub-index, data type, and data read/write properties. For example, the object dictionary format for the target speed in PV control mode is as shown in Figure 4.1.

Object Name	Index	Subindex	Data Type	Read/Write Attribute
Target Velocity	0x60FF	0x00	INTEGER32	R/W

Figure 4.1 Object Dictionary Structure

4.3 NMT Object

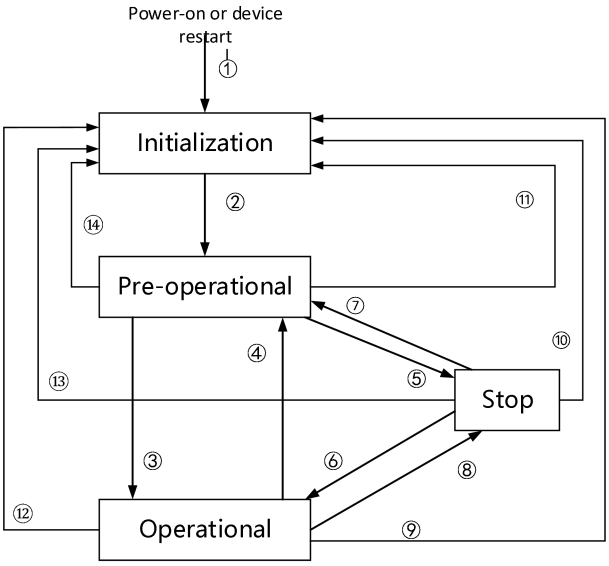


Figure 4.2 CiA 301 Communication State Machine

Figure 4.2 shows the state machine of the CAN bus communication (CiA 301 protocol), comprising four states: Initialization, Pre-operational, Operational, and Stopped. Transitions between states are triggered by sending NMT management messages, with specific switching methods detailed in Table 4.1.

Table 4.1 CiA 402 State Machine Switching Methods

Step(s)	Description
1	System automatically enters NMT Initialization state upon power-on
2	Transitions to Pre-operational state after NMT initialization completes
3	NMT command transitions node to Operational state
4, 7	NMT command forces node into Pre-operational state

5, 8	NMT command halts node communication (Stopped state)
6	NMT command activates node communication to Operational state
9,10,11	NMT command resets node to Initialization state
12,13,14	NMT command resets node communication to Initialization state

The communication functions achievable in various CAN communication operational states are detailed in Table 4.2:

Table 4.2 Communication Functions Supported in NMT States

Communication Object	Pre-operational	Operational	Stopped
PDO (Process Data Object)	×	√	×
SDO (Service Data Object)	√	√	×
SYNC (Synchronization Message)	√	√	×
EMCY (Emergency Message)	√	√	×
NMT (Network Management)	√	√	√
Error Control	√	√	√

Table 4.3 NMT Management Message Frame Format.The CAN ID of NMT management messages is fixed at 0x00. The data field consists of two bytes: Byte 1: CS (Command Specifier) command word (defines the type of command); Byte 2: Node ID. The node ID of the drive device can be set via a DIP switch. If the ID is set to 0, the NMT communication configuration applies to all slave nodes on the bus.

Table 4.3 NMT Message Format

COB-ID	Data 1	Data 2
0x00	CS	Node-ID

The NMT services for each type of CS Command Specifier object are shown in Table 4.4.

Table 4.4 NMT Services

Command Specifier (CS)	NMT Service
0x01	Start remote node
0x02	Stop remote node
0x80	Enter pre-operational state
0x81	Reset node
0x82	Reset communication

The operational state of a node can be configured to implement heartbeat messages. By using SDO communication to configure the object dictionary entry 1017h (Producer Heartbeat Time) in units of 1 ms, the drive will send heartbeat messages indicating the node's operational state at the configured interval. The CAN-ID of the heartbeat message is 0x700 + Node_ID, and the message format contains only one byte of data. A data value of 0x00 indicates the node is in the bootup state, 0x04 indicates the node is in the stopped state, 0x05 indicates the node is in the operational state, and 0x7F indicates the node is in the pre-operational state.

4.4 SDO Objects

SDO (Service Data Object) is primarily used by a CANopen master to configure parameters of slave nodes. It establishes a connection with the object dictionary through indexes and sub-indexes. A defining feature of SDO is its service acknowledgment mechanism, which generates a response for every message to ensure data transmission accuracy. The sender (server) transmits a message with a CAN-ID of 600h + Node-ID, where Node-ID is the address of the receiver (slave node), and the data length is

always 8 bytes. After successful reception, the receiver (client) responds with a message carrying a CAN-ID of 580h + Node-ID, where Node-ID remains the address of the receiver, and the data length is also fixed at 8 bytes. The SDO transmission structure is illustrated in Figure 4.3.

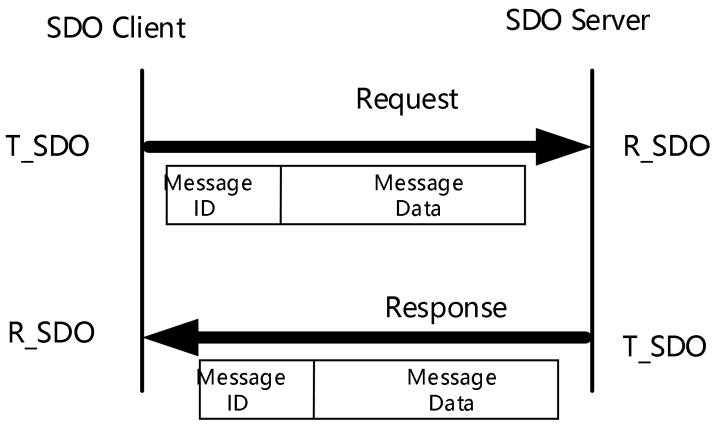


Figure 4.3 SDO Message Structure

4.4.1 Expedited Write SDO Message

The data frame format for SDO expedited read/write messages is shown in Table 4.5. The command code specifies the transmission type and data length of the SDO segment:8-bit data occupies 1 data byte,16-bit data occupies 2 data bytes,32-bit data occupies 4 data bytes.

Both the object dictionary index and the data are formatted in little-endian (LSB first, MSB last).。

Table 4.5

COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
580h+Node_ID/ 600h+Node_ID	Command Code	Index		Subindex	Data			

Table 4.6 describes the format of the expedited write message. The transmission messages for each object dictionary data type differ and require the command code to be determined based on the number of data bytes being transmitted. If the write operation to the object dictionary is successful, a message with CAN-ID 0x580 + Node-ID and a command code of 0x60 will be received. If the write operation fails, a message with CAN-ID 0x580 + Node-ID and a command code of 0x80 (abort code) will be returned.

Table 4.6

<table>
<tr><th></th><th>COB-ID</th><th>data0</th><th>data1</th><th>data2</th><th>data3</th><th>data4</th><th>data5</th><th>data6</th><th>data7</th></tr>
<tr><td rowspan="4">Client</td><td rowspan="4">600h+Node_ID</td><td>23h</td><td colspan="2" rowspan="4">Index</td><td rowspan="4">Subindex</td><td colspan="4">Data</td></tr>
<tr><td>27h</td><td colspan="3">Data</td><td>-</td></tr>
<tr><td>2Bh</td><td colspan="2">Data</td><td>-</td><td>-</td></tr>
<tr><td>2Fh</td><td>Data</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td rowspan="2">Server</td><td rowspan="2">580h+Node_ID</td><td>60h</td><td colspan="2" rowspan="2">Index</td><td rowspan="2">Subindex</td><td>-</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>80h</td><td colspan="4">Abort Code (Exception)</td></tr>
</table>

Example: For Node ID 1, using the SDO expedited write mode to configure the target position object dictionary entry 0x607A to a position value of 10000 (0x2710), the transmitted message data is shown in Table 4.7.

Table 4.7

COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
601	23	7A	60	00	10	27	00	00

If the data is written successfully, the received message format from the drive will be as shown in Table 4.8.

Table 4.8

COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
581	60	7A	60	00	00	00	00	00

If the data write operation fails, the drive will return an abort code, as shown in Table 4.9.

Table 4.9

COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
581h	80	7A	60	00	10	00	07	06

4.4.2 Expedited Read SDO Message

Table 4.10 shows the data format of the expedited read SDO message. The command code in the returned message varies depending on the data type.

Table 4.10 Expedited Read SDO Message Data Format

	COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
Client	600h+Node_ID	40h	Index		Subindex	-	-	-	-
Server	580h+Node_ID	43h	Index		Subindex	Data			
		47h				Data			-
		4Bh				Data		-	-
		4Fh				Data	-	-	-
		80h				Abort Code (Exception)			

Example: For Node ID 1, using the SDO expedited read mode to read the actual position object dictionary entry 0x6064.

Table 4.11

COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
601	43	64	60	00	00	00	00	00

If the data is read successfully, the received message format from the drive will be as shown in Table 4.12.

Table 4.12

COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
581	43	64	60	00	10	27	00	00

If the data read operation fails, the drive will return an abort code, as shown in Table 4.13.

Table 4.13

COB-ID	data0	data1	data2	data3	data4	data5	data6	data7
581h	80	7A	60	00	01	00	04	05

4.5 PDO Communication Objects

PDO (Process Data Object) is used to transmit real-time process data in a unidirectional manner, requiring no acknowledgment from the receiving node. In communication terminology, PDO follows a "producer-consumer" model, and its transmission structure is illustrated in Figure 4.4.

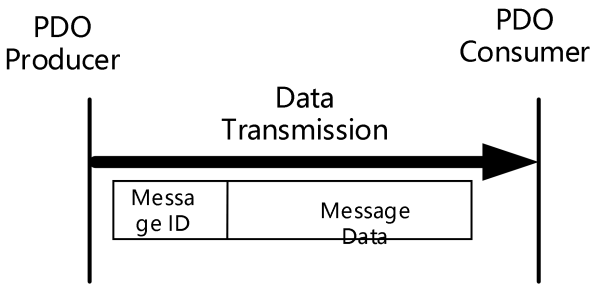


Figure 4.4 PDO Message Transmission Structure

Based on transmission direction, PDO is categorized into TPDO (Transmit PDO) and RPDO (Receive PDO). TPDO refers to data transmitted from the drive, while RPDO refers to data received by the drive. The form and content of data transmitted via PDO are determined by the communication parameter object dictionary and mapping parameter object dictionary. The DS3 drive supports 4 groups of TPDO and RPDO, with related object parameters listed in Table 4.14.

Table 4.14 PDO Object Parameters

Name	COB-ID	Communication Parameter Object Dictionary	Mapping Parameter Object Dictionary
RPDO1	200h+Node_ID	1400h	1600h
RPDO2	300h+Node_ID	1401h	1601h

RPDO3	400h+Node_ID	1402h	1602h
RPDO4	500h+Node_ID	1403h	1603h
TPDO1	180h+Node_ID	1800h	1A00h
TPDO2	280h+Node_ID	1801h	1A01h
TPDO3	380h+Node_ID	1802h	1A02h
TPDO4	480h+Node_ID	1803h	1A03h

4.5.1 Communication Object Parameters

The communication parameter objects (0x1400, 0x1800) share identical configurations, each containing 5 subindices:

Subindex 1: COB-ID for PDO communication.

Subindex 2: Transmission type. The communication functions corresponding to each transmission type parameter are listed in Table 4.15.。

Table 4.15 PDO Transmission Types

ransmission Type	Trigger Condition	Transmission Mode
0	Sync Message and Event	Synchronous Acyclic
1-240	Sync Message	Synchronous Cyclic
241-251	-	-
252	Sync Message and Remote Frame	Asynchronous
253	Remote Frame	Asynchronous
254	Remote Frame or Event	Asynchronous
255	Remote Frame or Event	Asynchronous

When the transmission type of a TPDO is 0, the TPDO is sent if the mapped object dictionary data changes and a SYNC message is received. When the TPDO transmission type is 1–240, the TPDO is sent after receiving the corresponding number of SYNC messages (e.g., type 5 means it sends after 5 SYNC messages). When the TPDO transmission type is 254 or 255, the PDO is sent either when the mapped object dictionary parameters change or when the configured timer expires.

If the transmission type is 0–240, the mapped object dictionary parameters are updated only after receiving one SYNC message. If the transmission type is 254 or 255, the mapped object dictionary parameters are updated immediately (no SYNC required). The drive is preset to 255, and this value is recommended, meaning the drive updates the mapped object dictionary parameters as soon as the RPDO is received.

Sub-index 3 defines the Producer Inhibit Time (unit: 0.1 ms), which constrains the minimum interval between PDO transmissions to prevent excessive bus load. For example, if position status changes too rapidly, this prevents TPDOs from flooding the bus.

Sub-index 4 specifies the Event Timer (unit: 1 ms); the PDO is sent when the timer expires.

Sub-index 5 sets the SYNC Start Value; PDO transmission begins after receiving the specified number of SYNC messages.

4.5.2 Mapping Parameters

The mapping parameters define the object dictionary entries to be mapped into the PDO. These parameters include the object dictionary index, sub-index, and data type, with the data structure shown in Table 4.16. While the drive supports mapping up to 8 object dictionary entries, the PDO data field is limited to 8 bytes (allowing 8 parameters of 8-bit data type). Therefore, the actual mappable parameters depend on the data types of the mapped entries.

Table 4.16 PDO Mapping Parameters

Data Bit	31-16	15-8	7-0
Corresponding Data	Index	Subindex	Data Type

If you need to map the first mapping parameter of TPDO1 to send the status word 6041h data, you can use the fast SDO to send the index 1A00h, sub-index 1, and data 0x60410010 to the drive with node ID 4, as shown in the message of Table 4.17.

Table 4.17

COB_ID	0	1	2	3	4	5	6	7
604	23	00	1A	01	10	00	41	60

If you need to map the first mapping parameter of RPDO1 to receive the control word 6040h data, you can use the fast SDO to send the index 1600h, sub-index 1, and data 60400010 to the drive with node ID 4, as shown in the message of Table 4.18.

Table 4.18

COB_ID	0	1	2	3	4	5	6	7
604	23	00	16	01	10	00	40	60

4.6 Emergency Message

When a CANopen node device experiences a fault, the drive will send an emergency message in accordance with the standards defined by the DS301 and DS402 protocols. DS3 drives will only send emergency messages related to their own device faults and will not process emergency messages from other devices. The format of the emergency message is shown in Table 4.19.

Table 4.19 Emergency Message Format

COB_ID	0	1	2	3	4	5	6	7
80h+Node_ID	Error Code		Error Register	Reserved	Auxiliary Byte			

4.7 Synchronous Messages

When the communication parameter transmission type is 0, the TPDO is sent if a synchronous message is received and the mapped object dictionary parameters have changed.For transmission types 2-240, the TPDO transmission is triggered simply by receiving the corresponding number of synchronous messages.For synchronous RPDO: After receiving RPDO data, the mapped object dictionary data is updated only upon receiving the next synchronous message.

Chapter 5 Control

5.1Servo Operation Status

5.1.1 Servo Operation State Machine

The transition between various operational states of the servo drive under the DS402 protocol is controlled by the control word object dictionary 0x6040h. The data type of 0x6040h is Uint16, and the definitions of each bit are specified in Table 5.1.

Table 5.1 Control Word Data Bit Definitions

Data Bit	Name	Description
0	Servo Ready	0: Inactive / 1: Active
1	Main Circuit Power	0: Inactive / 1: Active
2	Quick Stop	0: Inactive / 1: Active
3	Servo Operation	0: Inactive / 1: Active
4-6	-	Combined with servo operation mode
7	Fault Reset	Effective on rising edge
8	Halt	0: Inactive / 1: Active
9-10	NA	Undefined
11-15	Undefined	Undefined

The operational status of the servo drive is indicated by the status word object dictionary 0x6041h. The data type of 0x6041h is also Uint16, and the definitions of each bit are detailed in Table 5.2.

Table 5.2 Status Word Bit Definitions

Data Bit	Name	Description
0	Servo No Fault	0: Invalid / 1: Valid

1	Waiting to Enable Servo	0: Invalid / 1: Valid
2	Servo Running	0: Invalid / 1: Valid
3	Fault	0: Invalid / 1: Valid
4	Main Loop Energized	0: Invalid / 1: Valid
5	Quick Stop	0: Invalid / 1: Valid
6	Servo Ready	0: Invalid / 1: Valid
7	Warning	0: Invalid / 1: Valid
8	Undefined	Undefined
9	Remote Control	0: No-CANopen mode 1: CANopen mode
10	Target Reached	0: Target Not Reached /1: Target Reached
11	Soft Position Limit	0: Invalid / 1: Valid
12-13	-	Depends on the servo operation mode
14	Undefined	Undefined
15	Homing Completion	0: Invalid / 1: Valid

Individual bits in the Control Word (6040h) and Status Word (6041h) have no standalone meaning; their configuration requires combining with other bits.

The state transition process for the servo drive's operational states, as defined by the CIA 402 protocol, is illustrated in Figure 5.1. To transition the servo drive to a target state, the protocol's prescribed sequence must be strictly followed.

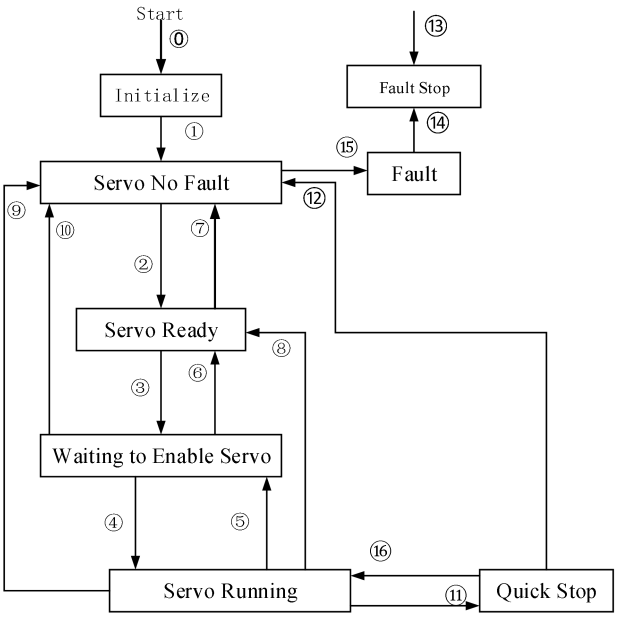


Figure 5.1 CIA 402 State Machine

Table 5.3 lists the data to be sent via the Control Word object dictionary (6040h) for transitioning between servo operational states.。

Table 5.3 CIA 402 State Machine Transition Methods

CIA402 State Transition		6040h	CIA402 State Transition		6040h
0	Power-on → Initialization	Natural transition	9	Servo Running → Servo No Fault	0x00
1	Initialization → Servo No Fault	Natural transition, if error occurs, enter 13	10	Waiting to Enable Servo → Servo No Fault	0x00
2	Servo No Fault → Servo Ready	0x06	11	Servo Running → Quick Stop	0x02
3	Servo Ready → Waiting to Enable Servo	0x07	12	Quick Stop → Servo No Fault	Natural transition
4	Waiting to Enable Servo → Servo Running	0x0F	13	→ Fault Stop	Drive fault, no control needed
5	Servo Running → Waiting to Enable Servo	0x07	14	Fault Stop → Fault	Natural transition

6	Waiting to Enable Servo → Servo Ready	0x06	15	Fault → Servo No Fault	0x80
7	Servo Ready → Servo No Fault	0x00	16	Quick Stop → Servo Running	0x0F
8	Servo Running → Servo Ready	0x06			

Table 5.4 shows the feedback of status bits in the Status Word object dictionary (6041h) during servo operational states, where "x" indicates that the bit is not assigned a specific status.

Table 5.4 Status Word Feedback in Different Servo States

Bit (15-0)	Servo Operation Status
xxxx xxxx x0xx 0000	Initialization
xxxx xxxx x1xx 0000	Servo No Fault
xxxx xxxx x01x 0001	Servo Ready
xxxx xxxx x01x 0011	Waiting to Enable Servo
xxxx xxxx x01x 0111	Servo Running
xxxx xxxx x00x 0111	Quick Stop
xxxx xxxx x0xx 1111	Fault Stop
xxxx xxxx x0xx 1000	Fault

5.1.2 Servo Operating Modes

The servo operating mode can be configured via the object dictionary 6060h, which uses the int8 data type. The DS3 servo drive supports 4 operating modes, with the corresponding data codes for object dictionary 6060h listed in Table 5.5.

Table 5.5 Servo Operating Modes

6060h	Running Mode	6060h	Running Mode
0x1	Profile Position Mode	0x1	Profile Position Mode
0x3	Profile Velocity Mode	0x3	Profile Velocity Mode

The configured servo operating mode can be verified by reading the object dictionary 6061h. After configuring object dictionary 6060h, read object dictionary 6061h to confirm whether the servo is operating in the set mode. The data codes for object dictionary 6061h and their corresponding operating modes are listed in Table 5.6.

Table 5.6 Servo Operating Mode Feedback

6061h	Running Mode	6061h	Running Mode
0x1	Profile Position Mode	0x1	Profile Position Mode
0x3	Profile Velocity Mode	0x7	Profile Velocity Mode
0x4	Profile Torque Mode		

5.2 Stop Methods

The DSY servo drive supports five stop modes: Quick Stop, Shutdown Stop, Disable Enable Stop, Halt, and Fault Stop. These stop modes can be configured via the Control Word object dictionary 0x6040, and the corresponding data to be sent for 0x6040 is listed in Table 5.7.

Table 5.7 Control Commands for Stop Methods

6040h	Running Mode	6040h	Running Mode
0x2	Quick Stop	0x10F	Pause
0x6	Power-off Stop	0x7	Disable Stop

The functional parameters for each stop method can be configured through object dictionaries 0x605A, 0x605B, 0x605C, 0x605D, and 0x605E, respectively.

5.2.1 Quick Stop

The Quick Stop mode is configured via object dictionary 0x605A. 0x605A supports 9 deceleration modes, and the deceleration modes corresponding to the data codes are listed in Table 5.8. The stop deceleration units for both 0x6084 and 0x6085 are r/s².

Table 5.8 Quick Stop Deceleration Modes

605Ah	Stop Mode	605Ah	Stop Mode
0x0	Free Stop	0x5	Ramp Stop (Configured by 6084h)
0x1	Ramp Stop (Configured by 6084h)	0x6	Quick Stop (Configured by 6085h)
0x2	Quick Stop (Configured by 6085h)	0x7	Quick Stop (Configured by 6085h)
0x3	Quick Stop (Configured by 6085h)	0x8	Quick Stop (Configured by 6085h)
0x4	Quick Stop (Configured by 6085h)		

5.2.2 Shutdown Stop

The Shutdown Stop mode is configured via object dictionary 0x605B. 0x605B supports 2 deceleration modes, and the deceleration modes corresponding to the data codes are listed in Table 5.9.

Table 5.9 Shutdown Stop Deceleration Modes

605Bh	Stop Mode	605Bh	Stop Mode
0x0	Free Stop	0x1	Ramp Stop (Configured by 6084h)

5.2.3 Disable Enable Stop

The Disable Enable Stop mode is configured via object dictionary 0x605C. 0x605C supports 2 deceleration modes, and the deceleration modes corresponding to the data codes are listed in Table 5.10.

Table 5.10 Disable Stop Deceleration Methods

605Ch	Stop Mode	605Ch	Stop Mode
0x0	Free Stop	0x1	Ramp Stop (Configured by 6084h)

5.2.4 Pause Mode

The pause mode can be configured through the object dictionary 605Dh, which supports two stop modes. The data bit codes corresponding to the deceleration methods are shown in Table 5.11.

Table 5.11 Pause Deceleration Methods

605Ch	Stop Mode	605Ch	Stop Mode
0x1	Ramp Stop (Configured by 6084h)	0x2	Quick Stop (Configured by 6085h)

5.2.5 Error Stop

The stop mode for Class 2 faults (40 < errorCode < 80) and warnings (errorCode > 80) can be configured through the object dictionary 605Eh, which supports two deceleration methods. The data bit codes corresponding to the deceleration methods are shown in Table 5.12.

Table 5.12 Error Stop Deceleration Methods

605Ch	Stop Mode	605Ch	Stop Mode
0x0	Free Stop	0x1	Ramp Stop (Configured by 6084h)

If it is a Class 1 non-resettable fault (errorCode < 40), the stop mode must be set via the master software.

☆Related Function Codes:

Function Code	Name	Setting Range	Function Description	Setting Type	Effective Time	Factory Default
P00.11	Fault NO.1 Stop Mode Selection	0- Free Stop, maintain free run state 1- Reserved	Sets the stop mode of the motor when the firste class fault occurs	Stop	Immediate	0

5.2.6 Superposition Stop

"Superposition": Refers to a condition where the mechanical movement exceeds the designed safe travel range.

"Superposition Stop": Refers to a safety function where the servo motor is forcibly stopped by the servo drive when the mechanical part exceeds the safe travel range, causing the limit switch to change its output level.

☆Related Function Codes:

Function Code	Name	Setting Range	Function Description	Setting Type	Effective Time	Factory Default
P00.13	Superposition Stop Mode Selection	0- Free Stop, maintain free run state 1- Zero-speed stop (deceleration time set by P05.06), maintain position lock state 2- Zero-speed stop (deceleration time set by P05.06), maintain free run state	Sets the stop mode when superposition occurs.	Stop	Immediate	1

When the servo motor is driving a vertical axis and enters a superposition state, the workpiece may fall. To prevent this, ensure the superposition stop mode (P00.13) is set to "1 - Zero-speed stop, position lock state". In applications involving linear motion, ensure the limit switches are connected to prevent mechanical damage. In a superposition state, the motor (or workpiece) can be reversed by sending a reverse command to return to a safe position.

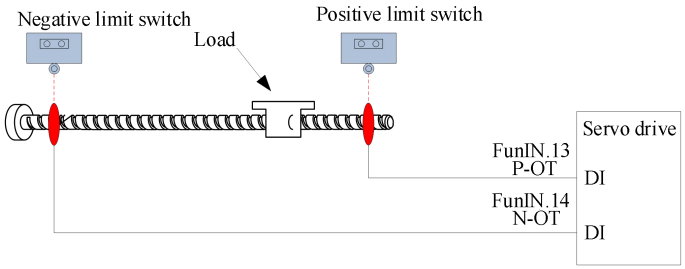


Figure 5.2 Installation Diagram of the Limit Switch

Use the DI (Digital Input) terminals of the servo drive to configure FunctionIN.13 (P-OT: Forward Limit Switch) and FunctionIN.14 (N-OT: Reverse Limit Switch) for superposition stop functionality. These inputs receive the limit switch signals and set the logic level for the DI terminals. The servo drive enables or clears the superposition stop based on whether the DI terminal logic is valid.

☆Related Function Codes:

Encoding	Name	Function Name	Function Description
FunIN.13	P-OT	Forward Limit Switch	When mechanical movement exceeds the movable range, the forward limit switch function is activated Invalid: Allows forward driving Valid: Prohibits forward driving
FunIN.14	N-OT	Reverse Limit Switch	When mechanical movement exceeds the movable range, the reverse limit switch function is activated Invalid: Allows reverse driving Valid: Prohibits reverse driving

5.3 Conversion Factors

5.3.1 Unit Conversion Factor Object Dictionary

The Electronic Gear Ratio Object Dictionary (6091h) essentially signifies that for every displacement unit of the load axis, there is a corresponding displacement of the motor. The Electronic Gear Ratio Object Dictionary (6091h) contains two sub-indices: Sub-index 1: Motor Resolution; Sub-index 2: Load Resolution.

The Velocity Coding Factor Object Dictionary (6094h) includes two sub-indices:

Sub-index 1: Velocity Coding Factor Numerator; Sub-index 2: Velocity Coding Factor Denominator.

The Velocity Coding Factor Object Dictionary 6095h contains two sub-indices: Sub-index 1 is the Velocity Coding Factor Numerator; Sub-index 2 is the Velocity Coding Factor Denominator;

The Acceleration Factor Object Dictionary (6097h) contains two sub-indices: Sub-index 1: Acceleration Factor Numerator; Sub-index 2: Acceleration Factor Denominator;

Related Parameter Conversions:

Actual Motor Position Command（Unit：r）=607A×$\frac{6091\text{Subindex1}}{6091\text{Subindex2}}$

Actual Motor Velocity Command（Unit：r/s）=60FF×$\frac{6091\text{Subindex1}}{6091\text{Subindex2}}$

Actual Motor Acceleration Command（Unit：r/s^2）=6083×$\frac{6091\text{Subindex1}}{6091\text{Subindex2}}$

Actual Motor Deceleration Command（Unit：r/s^2）=6084×$\frac{6091\text{Subindex1}}{6091\text{Subindex2}}$

5.3.2 Command Polarity

The command polarity can be configured via the object dictionary 607Eh, which allows setting the polarity for both velocity and position commands. The data bits of the object dictionary 607Eh are defined in Table 5.13.

Table 5.13 Command Polarity 607E Data Bit Definitions

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Position Command Polarity	Velocity Command Polarity	NA	NA	NA	NA	NA	NA

When bit7 is 0, the position command is forward. When bit7 is 1, the position command is reverse. When bit6 is 0, the velocity command is forward. When bit6 is 1, the velocity command is reverse. NA indicates that the data is undefined.

5.4 Basic control

5.4.1 Check before running

The following checks are required before the servo driver and servo motor run：

Table 5.14 Check steps before operation

Record	No.	Description
Wiring		
□	1	The control circuit power input terminals (DC+, DC-) of the servo drive must be connected correctly.
□	2	The output terminals (U, V, W) of the main circuit of the servo driver and the main circuit cables (U, V, W) of the servo motor must have the same phase and be correctly connected.。
□	3	The wiring of each control signal cable of the servo driver is correct: the external signal cables such as brake and overtravel protection have been reliably connected.
□	4	Servo drives and servo motors must be reliably grounded.
□	5	The stress of all cables is within the specified range.

□	6	The wiring terminals have been insulated.
Condition and machine		
□	1	There are no wire stubs, metal filings and other foreign objects inside and outside the servo drive that may cause short circuits in the signal wires and power wires.
□	2	Servo drives and external braking resistors are not placed on combustible objects.
□	3	The installation of the servo motor, the connection of the shaft and the machine must be reliable.
□	4	The servo motor and connected machinery must be in operable condition.

5.4.2 Power on

1）Turn on the control loop (DC+, DC-)

●After the control circuit power supply and the main circuit power supply are connected, the bus voltage indicator shows no abnormality and wait for the host computer to give the servo enable signal

●If the driver's red alarm light is on, please check the wiring and eliminate the cause of the fault.

5.4.3 JOG operation

Please use the JOG mode in the host computer software to confirm whether the servo motor can rotate normally, and there is no abnormal vibration and abnormal sound when rotating. The JOG mode control panel is shown in Figure 5.3.

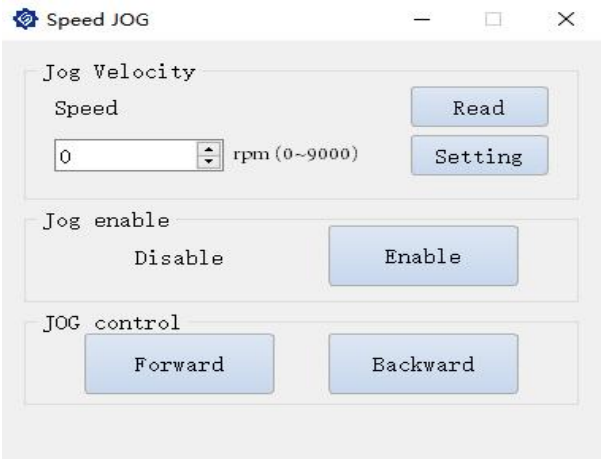


Figure 5.3 JOG mode operation interface

The JOG setting process is shown in the figure below.

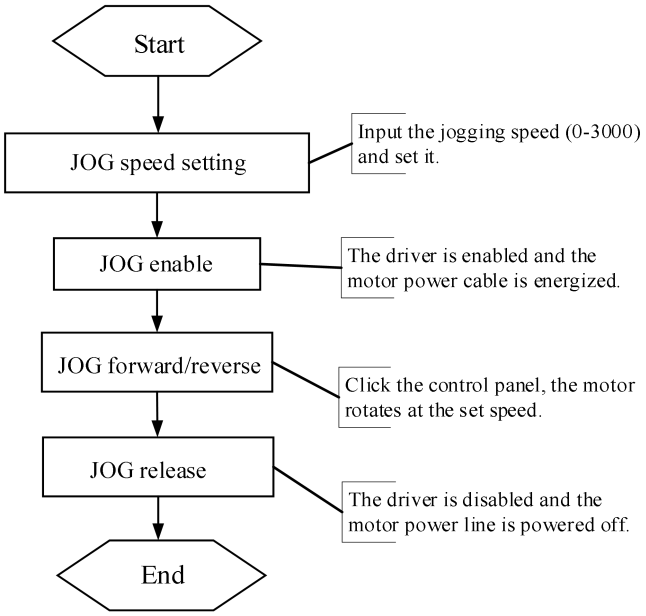


Figure 5.4 JOG control process

5.4.4 Rotation direction selection

By setting "rotation direction selection (P00.01)", the rotation direction of the motor can be changed without changing the polarity of the input command。

☆Associated function code：

Function code	Name	SettingRange	Function	Type	Effective time	Default setting
P00.01	Direction of rotatio	0--CCW direction is the forward direction 1--CW direction is the forward direction	Set the motor forward/reverse direction	Setting after motor stop	Re-power	0

Note：

When the rotation direction selection (P00.01) is changed, the output pulse form of the servo driver and the positive or negative of the monitoring parameters will not change.

5.4.5 Brake Setting

The brake is a mechanism that prevents the servo motor shaft from moving when the servo driver is in a non-running state, to keep the motor locked in position, so that the moving parts of the machine will

not move due to its own weight or external force.

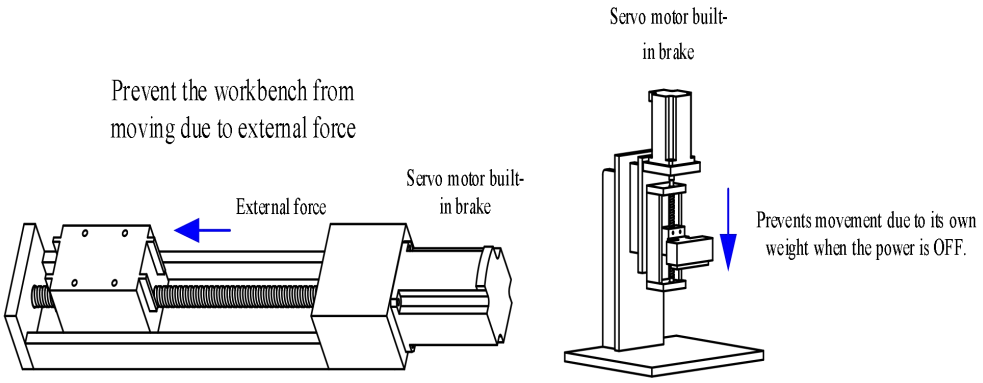


Figure 5.5 Brake application schematic

Note：

• The brake mechanism built into the servo motor is a non-energized fixed special mechanism, which cannot be used for braking purposes, and is only used to keep the servo motor in a stopped state.

• The brake coil has no polarity.

• After the servo motor stops, the servo enable (S-ON) should be turned off.

• When the motor with the built-in brake running, the brake may make a clicking sound, but it has no effect on the function.。

• When the brake coil is energized (the brake is released), magnetic flux leakage may occur at the shaft end and other parts. Be careful when using instruments such as magnetic sensors near the motor.

a）Brake wiring

The connection of the brake input signal has no polarity, and the user needs to prepare a 24V power supply. The standard wiring example of the brake signal BK and the brake power supply is as follows:

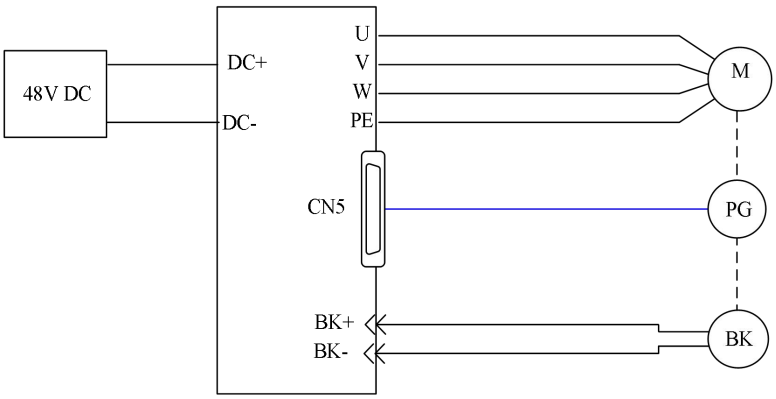


Figure 5.6 Brake Wiring Diagram

Note：

●The length of the motor brake cable needs to fully consider the voltage drop caused by the cable resistance, and the brake needs to ensure that the input voltage is at least 21.6V.

●It is best not to share the power supply with other electrical appliances, so as to prevent the voltage or current from reducing due to the work of other electrical appliances and eventually cause the brake to malfunction.

●It is recommended to use cables above 0.5mm^2.

●Brake is the Homeal power consumption component. Brakes with different powers are installed according to the rated torque of the motor. The output current of 24V power supply inside the servo driver can directly drive the motor lock and drive current 1A.

b）Brake software setting

For a servo motor with a brake, the second DO terminal of the servo drive must be configured as function 11 (FunOUT.11: BK, brake output), and the effective logic of the DO terminal must be determined.

☆Associated function code:

Code	Name	Function Name	Function
FunOUT.11	BK	Brake output	Invalid, the brake power is disconnected, the brake acts, and the motor is in a position-locked state; Valid, the power of the brake is turned on, the brake is released, and the motor can rotate.

5.4.6 Brake setting

When the torque of the motor is in the opposite direction to the rotational speed, the energy is transmitted back to the driver from the motor end, which causes the bus voltage to increase. When it reaches the braking point, the energy can only be consumed through the braking resistor. At this time, the braking energy must be consumed according to the braking requirements, otherwise the servo drive will be damaged. DS3 series drivers need an external braking resistor, and the connection of the braking resistor is shown in Figure5.7.

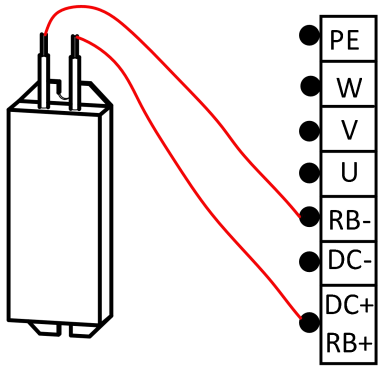


Figure 5.7 Wiring diagram of braking resistor

☆Associated function code:

Function code	Name	Setting Range	Min-Unit	Default setting	Effective time	Type	Mode
P0018	Energy consumption resistance setting	0--Use built-in power consumption resistor. 1--Use external power-consuming resistors and cool naturally. 2--Use external power-consuming resistors and force air-cooled. 3--No energy consumption resistor ,relying on capacitance absorption.	1	0	Effective immediately	setting after motor stop	PST
P0019	External resistor power capacity	1～65535	1W	Machine parameters	Effective immediately	setting after motor stop	PST

P0020	External resistance value	User-defined settings 1～1000	1Ω	Machine parameters	Effective immediately	setting after motor stop	PST
P0021	External resistance heating time constant	User-defined settings 1000~65535	1ms	Machine parameters	Effective immediately	setting after motor stop	PST
P0022	Energy consumption braking start voltage	0~410	1V	Machine parameters	Effective immediately	Set when running	PST

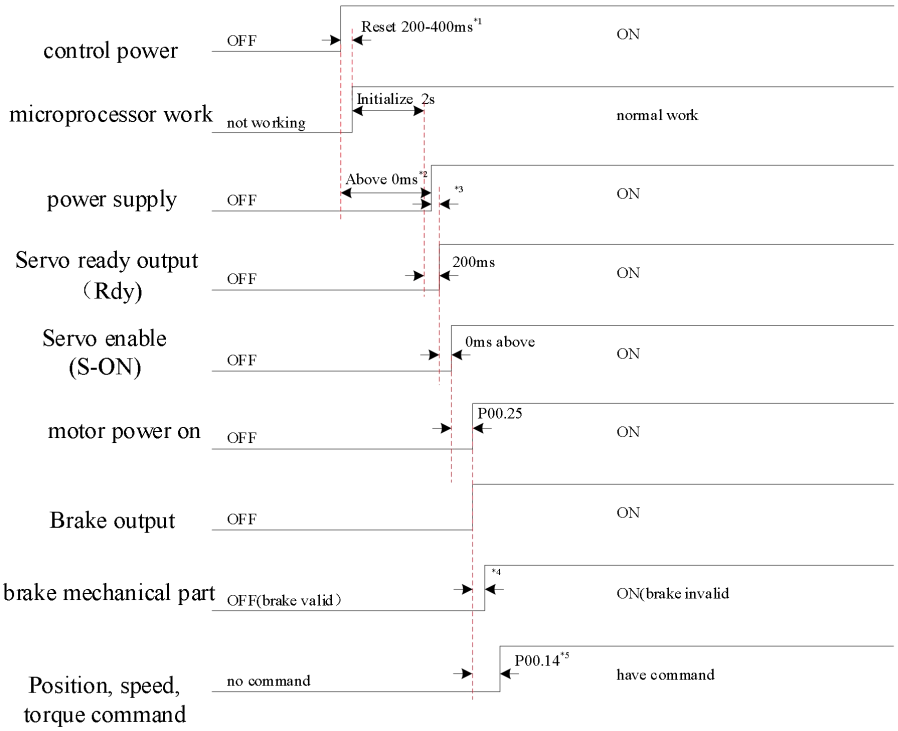
5.4.7 Servo operation

When the servo driver is in the enable state, the servo driver is in the running state at this time; but because there is no command input at this time, the servo motor does not rotate and is in the locked state. After the command is input, the servo motor rotates.

Table 5.15 Servo operation instructions

No.	Description
1	When running for the first time, set appropriate commands to make the motor rotate at a low speed, and confirm whether the motor rotation is correct.
2	Observe whether the direction of rotation of the motor is correct. If it is found that the direction of the motor is opposite to the expected one, please check the input command signal and command direction setting signal.
3	If the rotation direction of the motor is correct, you can use the YAKO host computer software to observe the actual speed P18.01 of the motor, the average load rate P18.02 and other parameters.
4	After checking the operation status of the motor above, you can adjust the relevant parameters to make the motor work in the expected working condition.
5	Refer to Chapter 6 "Operation Performance Adjustment" to debug the servo driver.

1）Power On Timing Chart



Note：

*1：The reset time is determined by the establishment time of the microprocessor +5V power supply;

*2：Above 0ms, it means that the time is determined by the actual moment when the main power supply is switched on

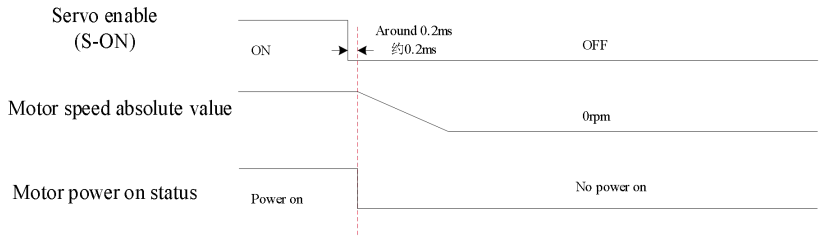
*3：When the control power supply and the power supply are powered on at the same time, this time is the same as the time from the completion of microprocessing initialization to the effective time of Rdy;

*4：For the delay time of the action of the brake mechanical part, please refer to the relevant specifications of the motor;

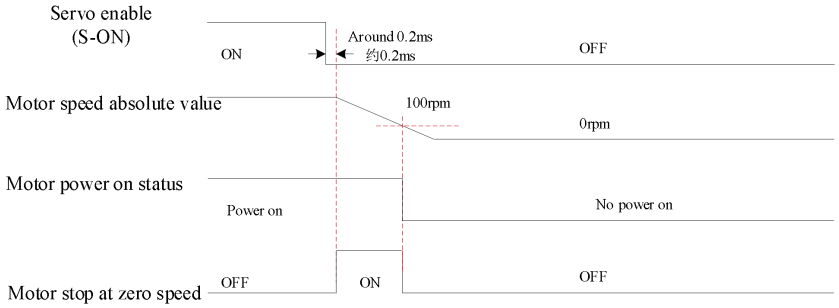
*5：When DO function 11 (FunOUT.11: BK) is not allocated, P00.14 has no effect;

2）Shutdown when servo enable OFF, warning or fault occurs timing diagram

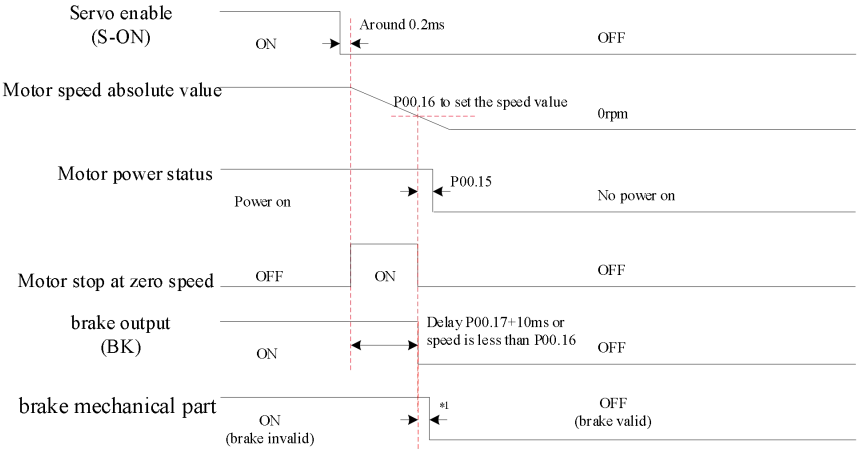
a）Servo enable OFF Non-brake: coast to stop, keep free running state (P00.10=0)



b）Servo enable OFF Non-brake: Stop at zero speed and keep in free running state (P00.10=1)



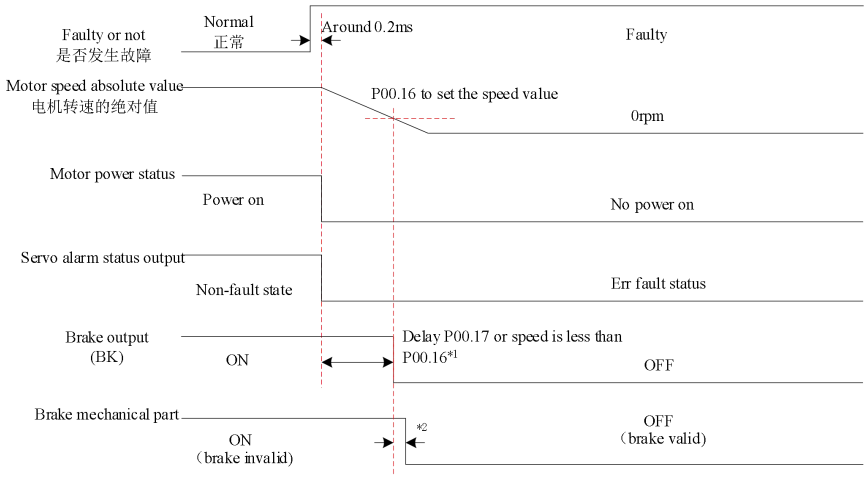
c）Servo enable OFF with brake: forced to stop at zero speed, keep free state



Note:

*1：For the delay time of the action of the brake mechanical part, please refer to the relevant specifications of the motor.

d）Type I faulty : free stop, keep free running state (P00.11=0)

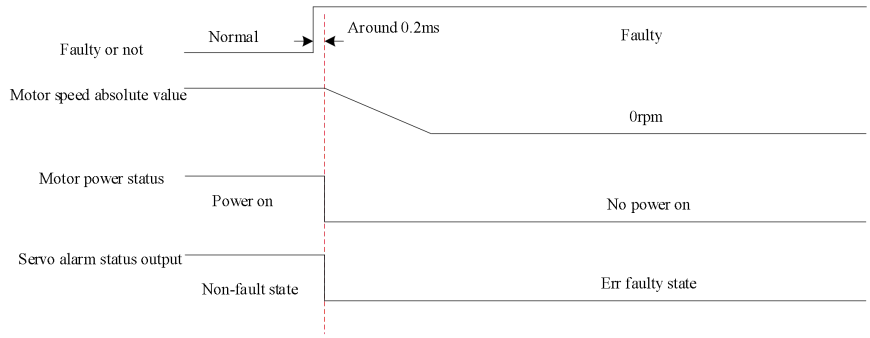


Note:

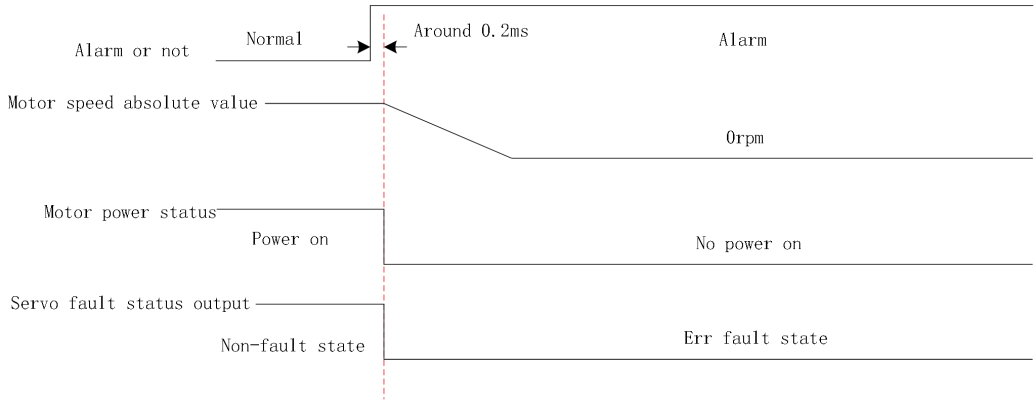
*1：When DO function 11 (FunOUT.11: BK) is not assigned, P00.16 and P00.17 have no effect.

*2：For the delay time of the action of the brake contact part, please refer to the relevant specifications of the motor;

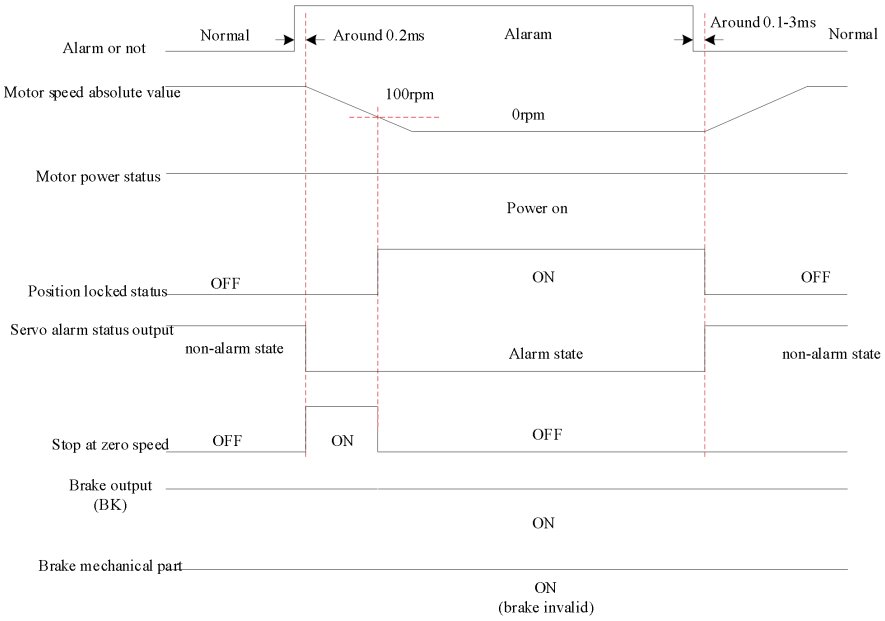
e）Type II faulty



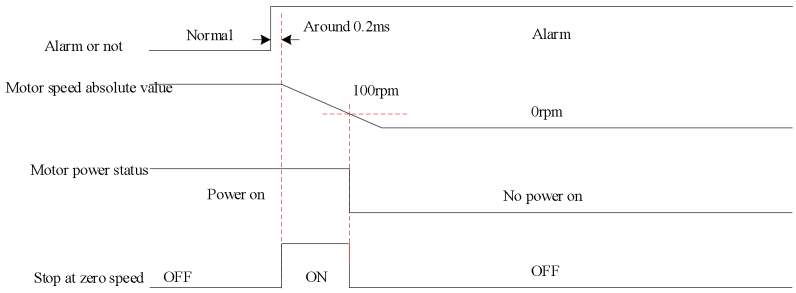
f）Overtravel stop warning non-holding brake: free stop, keep free running state (P00.13=0)



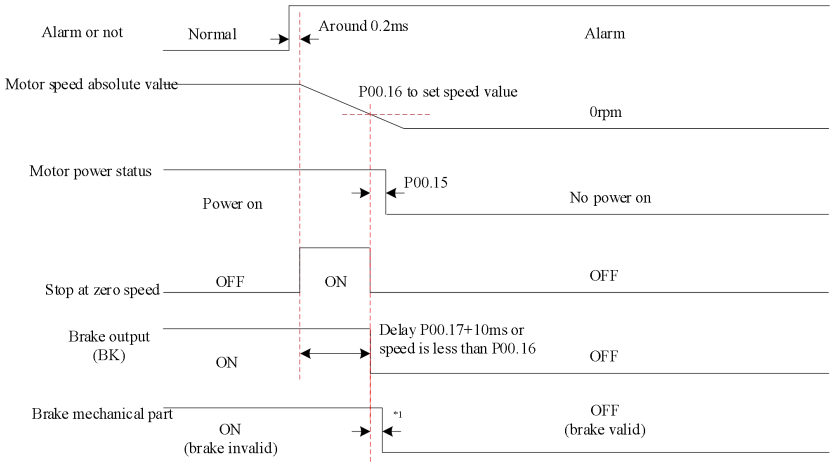
g）Overtravel, brake stop warning: stop at zero speed, keep position locked (P00.13=1)



h）Overtravel stop warning non-brake: stop at zero speed, keep free running state (P00.13=2)



i） Overtravel stop warning with brake: stop at zero speed, keep free running state (P00.13=0 or P00.13=2)

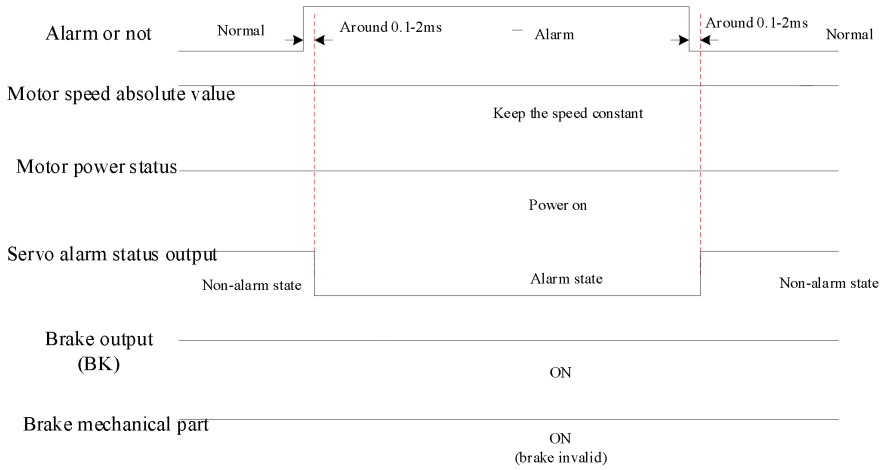


Note:

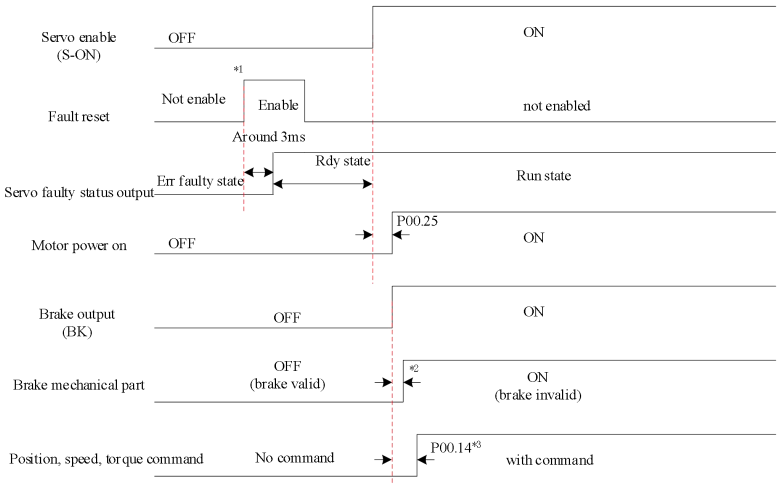
*1：For the delay time of the action of the brake mechanical part, please refer to the relevant specifications of the motor;

j）Non-shutdown warning:

Except for emergency brake, forward overtravel warning, and reverse overtravel warning, other warnings have no effect on the current status of the servo, as shown below.



k）Fault reset:



Note：

*1：DI fault reset signal (FunIN.2: ALM-RST) is valid when edge changes;

*2：For the delay time of the action of the brake contact part, please refer to the relevant specifications of the motor;

*3：When DO function 11 (FunOUT.11: BK) is not assigned, P00.14 has no effect;

5.5 Profile Velocity Mode（PV）

5.5.1 Control Byte and Status Byte

In Profile Velocity Mode, the control word 6040h instructions are the same as the state machine control flow in Section 5.1, and the control instructions are as shown in Table 5.16.

Table 5.16 PV ModeControl Byte Instructions

6040h	Instruction Description
0x06	Servo is ready.
0x07	Servo is ready; servo enable can be activated.
0x0F	The servo is enabled, and the servo operates according to the given instructions.
0x10F	Paused (motor remains in enabled state).

The status byte 6041h is used to feedback the servo running status in Velocity Profile Mode. Among these, bit0 to bit9 are consistent with the state machine function in Section 5.1 and are used to feedback the power-on status of the servo drive. Bit10 is used to feedback whether the instruction has been reached. For specific details, refer to Table 5.17.

Table 5.17

6041 Data Bits	bit13-15	bit12	bit11	bit10	bit0-9
Function Name	NA	Zero Speed Signal	NA	Target Reached	-
Description	-	0: User speed is not 0 1：User speed is 0	-	0: Target speed not reached 1: Target speed reached	Reference 2.1.1

5.5.2 Profile Velocity Mode Related Object Dictionary

The object dictionary related to the velocity profile mainly includes parameter configuration and status feedback. The parameter configuration-related objects, as shown in Table 5.18, are primarily used to set the target speed, target acceleration/deceleration, and limits of target parameters.

Table 5.18 Profile Velocity Mode Control Object Dictionary

Object	Name	Data Type	Description
6060h	Mode Selection	int8	Configure servo operation mode
606Dh	Speed Reach Window	uint16	Target speed threshold (r/s)
606Eh	Speed Reach Window Time	uint16	Target speed threshold time (ms)

606Fh	Zero Speed Threshold	uint16	Zero speed threshold (r/s)
6070h	Zero Speed Threshold Time	uint16	Zero speed threshold time (ms)
607Eh	Command Polarity	uint8	Configuration of speed command polarity
607Fh	Maximum Profile Speed	uint32	Profile speed limitation (r/s)
6083h	Profile Acceleration	uint32	Unit: r/s²
6084h	Profile Deceleration	uint32	Unit: r/s²
6085h	Quick Stop Deceleration	uint32	Unit: r/s
6095h	Speed Factor	uint32	Subindex 1: numerator, Subindex 2: denominator
6097h	Acceleration Factor	uint32	Subindex 1: numerator, Subindex 2: denominator
60C5h	Maximum Acceleration	uint32	Limit maximum profile acceleration (r/s²)
60C6h	Maximum Deceleration	uint32	Limit maximum profile deceleration (r/s²)
60FFh	Target Speed	unint32	Unit: r/s

The status feedback object dictionary is used to provide feedback on the motor's actual speed and position information, as shown in Table 5.19.

Table 5.19 PV mode Status Feedback Object Dictionary

Object	Name	Data Type	Description
6061h	Mode Display	int8	Feedback of servo operation mode
6064h	Actual Position	uint32	Feedback of user's actual position (number of turns)
606Bh	Actual Speed Command	uint32	Feedback of user's actual speed command (rpm)
606Ch	Actual Speed	uint32	Feedback of user's actual speed (rpm)

5.5.3 Profile Velocity Mode Configuration Example

Configure the servo operation mode object dictionary 6060h = 0x03, to operate the driver in profile velocity mode.

Configure the electronic gear ratio object dictionary 6091h, Load axis resolution = 10000，Motor axis resolution = 1.

Configure the profile acceleration a_1=6083h=1000*10000/60（1000rpm/s）。

Configure the profile deceleration d_1=6084h=1000*10000/60 (1000rpm/s)

Configure the command polarity object dictionary 607Eh=0x0，meaning positive direction for speed command.

Configure the target speed V_1=60FFh=1500*10000/60 (1500rpm)

Configure the control word 6040h to enable the driver. the sequence of commands for 6040h is:0x06→0x07→0x0F。

Configure the target speed V_2=60FFh=2000*10000/60 (2000rpm)

Configure the profile acceleration a_2=6083h=600*10000/60 (600rpm/s)

Configure the profile deceleration d_2=6084h=600*10000/60 (600rpm/s)

After reaching the target speed, configure the control word 6040h to stop the driver. The command for 6040h is 0x06.The motor motion process is illustrated in Figure 5.6.

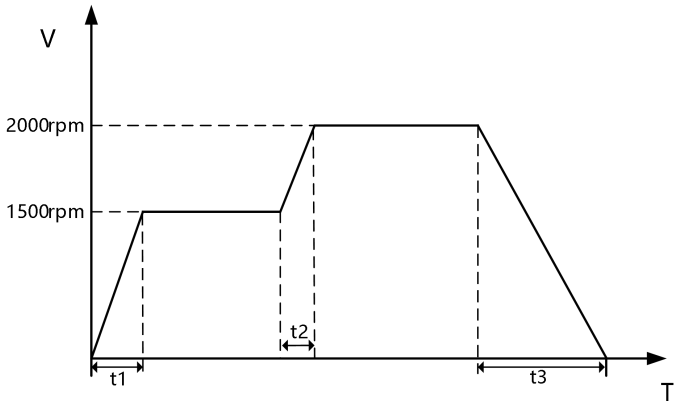


Figure 5.6 PV Mode Motion Process

The acceleration and deceleration times are calculated using the following formulas:

$$t_1 = \frac{V_1}{a_1}s, \qquad t_2 = \frac{V_2}{a_2}s,\ t_3 = \frac{V_2}{d_2}s$$

5.6 Profile Position Mode（PP）

5.6.1 Control Word & Status Word

In Profile Position Mode, the control word 6040h operates similarly to the state machine described in

section 5.1, but with some additional instructions related to position control. The specific control instructions are detailed in Table 5.20.

Table 5.20 PP Mode Control Word Commands

6040h (Hex)	Command Description
0x06	Servo is ready
0x07	Servo is ready; servo enable can be opened
0x0F	Enable servo operation; the servo executes the given command
0x10F	Pause (motor remains in enabled state)
0x0F→0x1F	Absolute position command (not updated immediately)
0x2F→0x3F	Absolute position command (updated immediately)
0x4F→0x1F	Relative position command (not updated immediately)
0x6F→0x7F	Relative position command (updated immediately)

The status word 6041h is used to provide feedback on the servo operation status in Profile Position Mode, Bits 0-9 are consistent with the state machine functionality described in Section 2.1.1, reflecting the power-on status of the servo drive, Bit 10 indicates whether the target position has been reached. The detailed descriptions are provided in Table 2.20.

Table 5.21

6041h	bit14-15	bit13	bit12	bit11	bit10	bit0-9
Function Name	NA	Following Error	New Position Data Reception	NA	Target Reached	-
Description	-	0: No following error 1: Following error detected.	0: New position data can be received 1: Can not receive new position data.	-	0: Target position not reached 1: Target position reached.	Reference 2.1.1

5.6.2 Profile Position Mode Associated Object Dictionary

The object dictionary associated with profile position primarily consists of parameter configuration and status feedback. as illustrated in Table 5.22, the parameter configuration-related object dictionary is designed to set target positions, profile velocity, acceleration/deceleration rates, and limitations for target parameters.

Table 5.22 PP Mode Control Object Dictionary

Object Dictionary	Name	Data Type	Description
6060h	Mode Selection	int8	Configure servo operation mode
6067h	Target Position Window	uint32	Position arrival threshold (r)
6068h	Target Position Window Time	uint16	Position arrival threshold duration (ms)
07Ah	Target Position	int32	Unit: r
607Bh	Position Range Limit	int32	Sub-index 1: Maximum value, Sub-index 2: Minimum value. Unit: r
607Dh	Soft Absolute Position Limit	int32	Sub-index 1: Minimum value, Sub-index 2: Maximum value. Unit: r
607Ch	Home Offset	int32	Origin offset value (r)
607Eh	Command Polarity	unint8	Configure position command polarity
607Fh	Max Profile Velocity	uint32	Profile velocity limit (r/s)
6081h	Profile Velocity	uint32	Unit: r/s
6083h	Profile Acceleration	uint32	Unit: r/s²
6084h	Profile Deceleration	uint32	Unit: r/s²
6085h	Quick Stop Deceleration	uint32	Unit: r/s²
6093h	Position Factor	uint32	Sub-index 1: Numerator, Sub-index 2: Denominator
60C5h	Max Acceleration	uint32	Limit maximum profile acceleration (r/s²)
60C6h	Max Deceleration	uint32	Limit maximum profile deceleration (r/s²)

The Status Feedback Object Dictionary is used to provide feedback on the motor's actual speed, position, and related parameters, as detailed in Table 5.23.

Table 5.23 PP Mode data feedback object dictionary

Object Dictionary	Name	Data Type	Description
6061h	Mode Display	int8	Feedback servo operation mode
6062h	User Actual Position Command	uint32	Feedback user actual position command (r)
6064h	User Actual Position	uint32	Feedback user actual position (revolutions)
606Ch	User Actual Velocity	uint32	Feedback user actual velocity (r/s)

5.6.3 Profile position mode instruction update instructions

The profile position mode command updates comprise four types: non-immediate update of absolute position commands, immediate update of absolute position commands, non-immediate update of relative position commands, and immediate update of relative position commands.种。

For non-immediate position update instructions, the speed is reduced to 0 after the current position instruction arrives, and the next position instruction is executed from zero speed, as shown in Figure 5.7. The first position command is executed at t0, the current first position command is reached at t1 and slowed down to 0, and the second position command is executed from t2.

For the absolute position command, the user's final absolute position = the second absolute position 607Ah. For the relative position instruction, the user's final position increase = the first relative position 607Ah + the second relative position 607Ah.

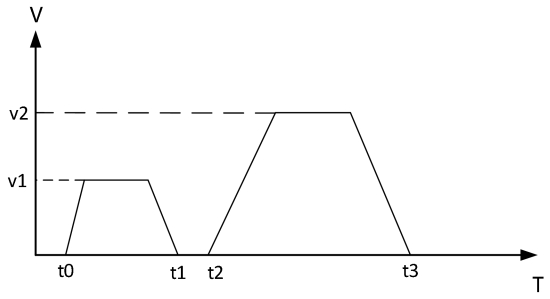


Figure 5.7 PP Non-immediate positioning motion process

For immediate position update commands, upon receiving a new position command, the system will immediately execute the new command at its specified speed, as shown in Figure 5.8. the first position command begins execution at t0. During its execution, a new position command is received at t1, prompting the motor to instantly switch to the new command’s speed while discarding the unfinished portion of the first command. The motor then completes deceleration to zero at t3 after executing both position command segments.

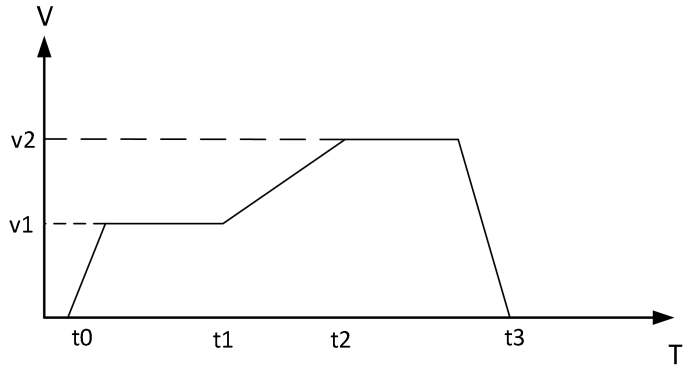


Figure 5.8 PP Immediate Positioning Motion Process

5.6.4 Profile Position Configuration Example

Configure the servo operation mode object dictionary 6060h=0x01 to let the driver work in profile position mode.

The 6091h load axis resolution of the electronic gear ratio object dictionary is 10000, and the resolution of the motor shaft is 1.

Configure the target position P_1=607Ah=10*10000=10r.

Configure the profile velocity V_1=6081h=1000*10000/60（1000rpm）。

Configure the profile acceleration a_1=6083h=100*10000/60（100rpm/s）。

Configure the profile deceleration d_1=6084h=100*10000/60（100rpm/s）。

Configure the command polarity object dictionary 607Eh=0x0， meaning positive direction for position command.

Configure the control word 6040h to enable the driver. The position update mode is absolutely immediately updated, and the 6040h instruction is:0x06→0x07→0x0F→0x2F→0x3F。

Configure the target position P_2=607Ah=20*10000 (20r)。

Configure the profile velocity V_2=6081h=2000*10000/60（2000rpm）

Configure the profile acceleration a_2=6083h=100*10000/60（100rpm/s）

Configure the profile deceleration d_2=6084h=200*10000/60（200rpm/s）

Configure the command polarity object dictionary 607Eh=0x0， meaning positive direction for position command.

Configure the control word 6040h, so that the position update method is absolutely updated immediately, and the 6040h instruction is: 0x2F→0x3F. after the motor reaches the target position command, its speed decelerates to zero, with the motion process illustrated in Figure 5.9.

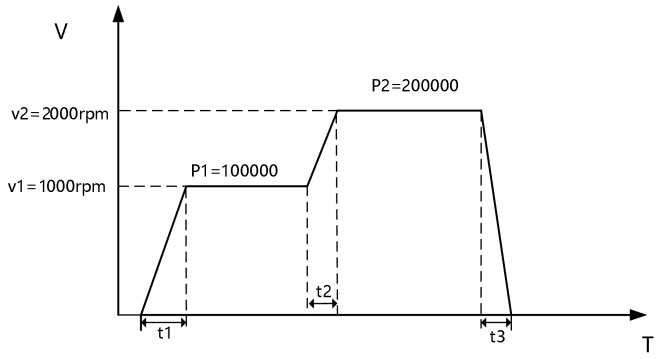


图 5.9

The acceleration and deceleration times are calculated using the following formulas:

$$t_1 = \frac{V_1}{a_1}s, \qquad t_2 = \frac{V_2}{a_2}s,\ t_3 = \frac{V_2}{d_2}s$$

5.7 Profile Torque Mode（PT）

5.7.1 Control Word & Status Word

In Profile Torque Mode, the Control Word (Object Dictionary 6040h) commands align identically with the state machine defined in Section 5.1, and the control commands are detailed in Table 5.24.

Table 5.24 PT Mode Control Word Commands

6040h	Command Description
0x06	Servo is ready
0x07	Servo is ready; servo enable can be opened
0x0F	Enable servo operation; the servo executes the given command
0x10F	Pause (motor remains in enabled state)

The Status Word 6041h is used to feedback the servo operational status in Profile Torque Mode, where bits 0-9 align with the state functions defined in Section 5.1 to reflect the servo drive's power-on status, and bit 10 indicates command completion status, as detailed in Table 5.25.

Table 5.25

6041 data bits	bit12-15	bit11	bit10	bit0-9
Function Name	NA	NA	Target Reached	-
Description	-	-	0: Target position not reached 1: Target position reached.	Reference 2.1.1

5.7.2 Profile Torque Mode Associated Object Dictionary

The profile torque mode associated Object Dictionary include parameter configuration and status feedback. The parameter configuration entries, listed in Table 5.26, are used to set the target speed, target acceleration/deceleration, and limits of the target parameters.

Table 5.26 PT Mode Control Object Dictionary

Object Dictionary	Name	Data Type	Description
6060h	Mode Selection	int8	Configure servo operation mode
6071h	Target Torque	int16	Unit: 0.1% of rated torque
6072h	Maximum Torque	int16	Unit: 0.1% of rated torque
6073h	Maximum Current	unint16	Unit: 0.1% of rated torque
607Fh	Maximum Profile Speed	uint32	Profile speed limit (r/s)
6087h	Torque Slope	uint32	Unit: 0.1% of rated torque/s

The state feedback object dictionary is used to feedback information such as the actual torque, current, speed and position of the motor, as detailed in Table 5.27.

Table 5.27 PT Mode data feedback object dictionary

Object Dictionary	Name	Data Type	Description
6061h	Mode Display	int8	Feedback servo operation mode
6063h	Motor Actual Position	uint32	Feedback motor actual position (revolutions)
6064h	User Actual Position	uint32	Feedback user actual position (revolutions)
606Ch	User Actual Speed	uint32	Feedback user actual speed (rev/s)
6074h	User Torque Command	int16	Unit: 0.1% of rated torque
6075h	Motor Rated Current	uint32	Unit: mA
6076h	Motor Rated Torque	uint32	Unit: mN·m
6077h	Torque Actual Value	int16	Unit: 0.1% of rated torque
6078h	Current Actual Value	int16	Unit: 0.1% of rated current
6079h	DC Link Voltage	uint32	Unit: mV

5.7.3 Profile Torque Mode Configuration Example

Configure the servo operation mode object dictionary 6060h=0x04 to let the driver work in profile torque mode.

Configure the target torque =6071h=100 (10% of rated torque)

Configure the torque slope =6087h=10 (1%/s of rated torque)

Configure the control word 6040h to enable the driver. the sequence of commands for 6040h is: 0x06→0x07→0x0F。

5.8 Homing Mode（HM）

5.8.1 Control Word & Status Word

In Homing Mode, the Control Word (Object Dictionary 6040h) commands are as detailed in Table 5.28.

Table 5.28 Homing Mode Control Word Commands

6040h	Command Description
0x06	Servo is ready
0x07	Servo is ready; servo enable can be opened
0x0F	Enable servo operation; the servo executes the given command
0x1F	Homing Operation (Execute Homing Command)
0x10F	Pause (motor remains in enabled state)

The status word 6041h is used to feedback the servo operation status in homing mode. Bits 0-9 correspond to the state machine functionality described in section 2.1.1, reflecting the power-on status of the servo driver. Bit 10 indicates whether the target position has been reached, Bit 12 reports the completion status of the homing mode, and Bit 13 detects the presence of homing errors. Detailed descriptions are provided in Table 5.29.

Table 5.29

6041 data bits	bit14-15	bit13	bit12	bit11	bit10	bit0-9
Function name	NA	Homing Error	Homing Completed	NA	Target Reached	-
Description	-	0: No homing error 1: Homing error detected	0: Homing mode incomplete 1: Homing mode completed	-	0: Homing position not reached 1: Homing position reached	Reference 2.1.1

5.8.2 Homing Mode Associated Object Dictionary

The homing mode associated object dictionary mainly includes parameter configuration and status feedback. The parameter configuration-related objects, as shown in Table 5.30, are primarily used to set the target positions, profile velocity, target acceleration/deceleration, and limits of target parameters.

Table 5.30 HM Control Object Dictionary

Object Dictionary	Name	Data Type	Description
6060h	Mode Selection	int8	Configure servo operation mode
607Dh	Soft Position Limits	int32	Sub-index 1: Maximum value Sub-index 2: Minimum value Unit: rev
607Ch	Origin Offset	int32	Homing origin offset (rev)
6085h	Quick Stop Deceleration	uint32	Unit: rev/s²
6098h	Homing Type	int8	Configure homing search method
6099h	Homing Speed	int32	Sub-index 1: Speed for approaching; Sub-index 2: Origin search speed; Unit: rpm
609Ah	Homing Acceleration	uint32	Unit: rpm/s

The Status Feedback Object Dictionary is used to provide feedback on the motor's actual speed, position, and related parameters, as detailed in Table 5.31.

Table 5.31 HM data feedback object dictionary

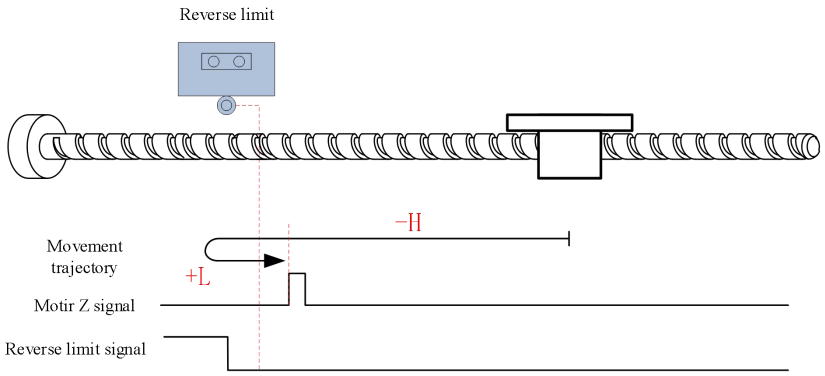
Object Dictionary	Name	Data Type	Description
6061h	Mode Display	int8	Feedback servo operation mode
6064h	User Actual Position	uint32	Feedback user actual position (revolutions)
606Ch	User Actual Speed	uint32	Feedback user actual speed (rev/s)

5.8.3 Explanation of homing mode origin search method

In the legend, H represents the higher speed set by parameter 6099-01, while L represents the lower speed set by parameter 6099-02.

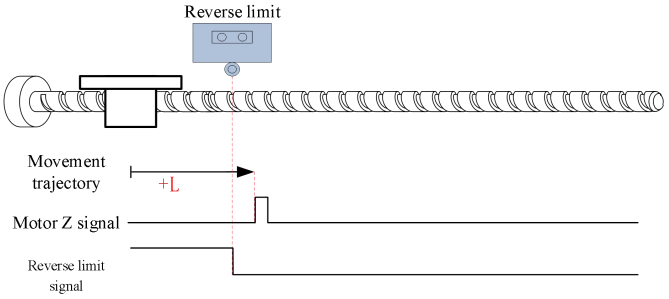
Homing Mode 1: Homing by reverse limit switch and zero position pulse signal（0x6098=1）

1） During homing, the reverse limit switch signal is inactive.



Description: When homing, the reverse limit switch signal is invalid. The motor searches for the reverse limit switch signal at a higher speed until the rising edge of the limit switch signal is detected, then decelerates, reverses, and runs forward at a lower speed. When the falling edge of the reverse limit signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

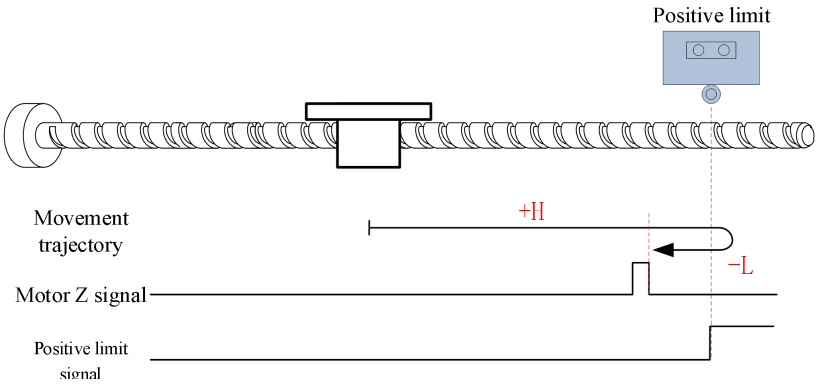
2）When homing, the reverse limit switch signal is valid



Description: When homing, the reverse limit switch signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the reverse limit signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

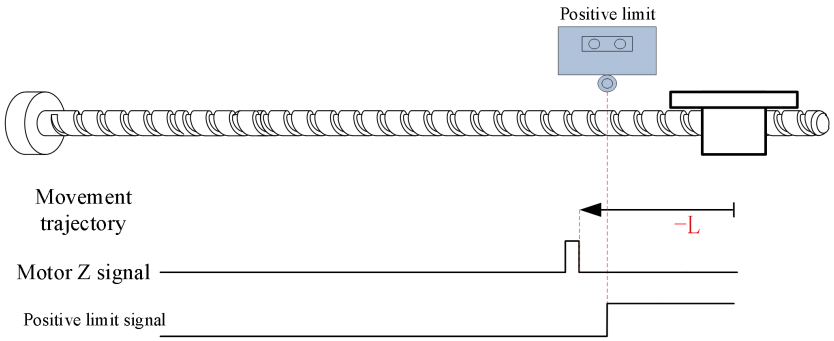
Homing mode 2: hoimg with positive limit switch and zero pulse signal (0x6098 = 2)

1）When homing, the positive limit switch signal is invalid



Description: When homing, the positive limit switch signal is invalid. The motor searches for the positive limit switch signal at a higher speed in the forward direction until the rising edge of the limit switch signal is detected, then decelerates, reverses, and runs in the reverse direction at a lower speed. When the falling edge of the positive limit signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

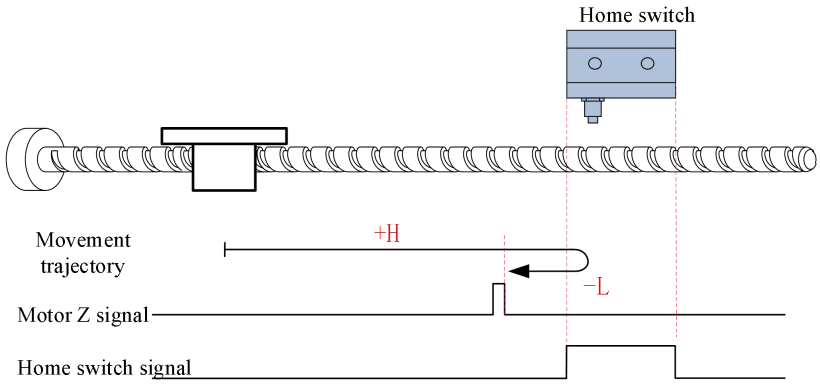
2）When homing, the positive limit switch signal is valid



Description: When homing, the positive limit switch signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the positive limit signal is detected, the motor starts to search for the Z signal of the motor, and stops running after the Z signal is found.

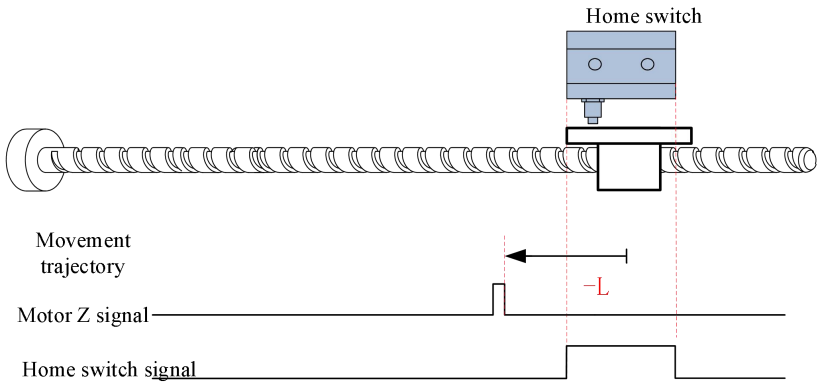
Homing mode 3: homing with Home switch and zero pulse signal (0x6098 = 3)

1）When homing, the Home switch signal is invalid



Description: When homing, the home switch signal is invalid. The motor searches for the home switch signal at a higher speed in the forward direction until the rising edge of the Home switch signal is detected, then decelerates, reverses, and runs in the reverse direction at a lower speed. When the falling edge of the home signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

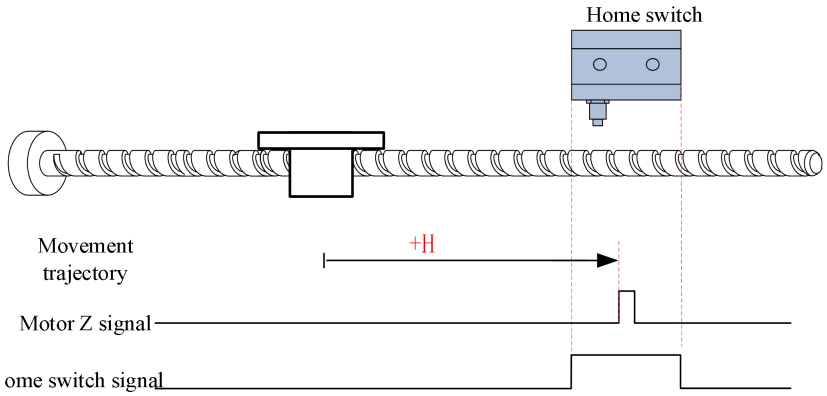
2）When homing,the home switch signal is valid



Description: When homing, the home switch signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home switch signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

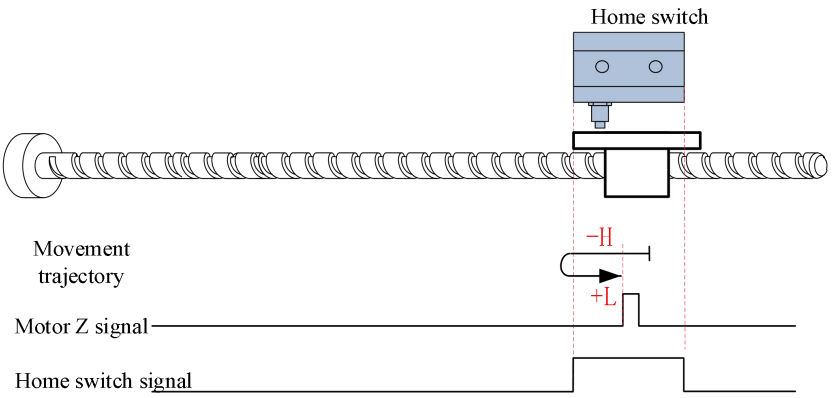
Homing mode 4: homing with Home switch and zero pulse signal（0x6098 =4）

1）When homing, the Home switch signal is invalid



Description: When homing, the home switch signal is invalid. The motor searches for the home switch signal in the forward direction at a lower speed until the rising edge of the Home switch signal is detected. Then it starts to search for the motor Z signal and stops running after the Z signal is found.

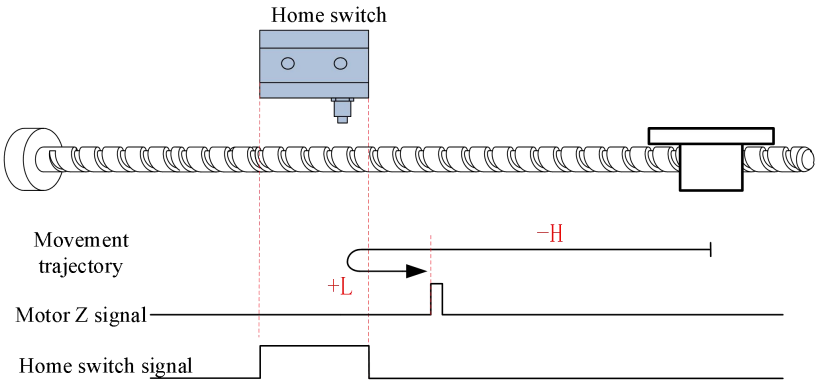
2）When homing, the Home switch signal is valid



Description: When homing, the home switch signal is valid. The motor directly runs in the reverse direction at a higher set speed. When the falling edge of the home switch signal is detected, it decelerates, reverses, and runs forward at a lower speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

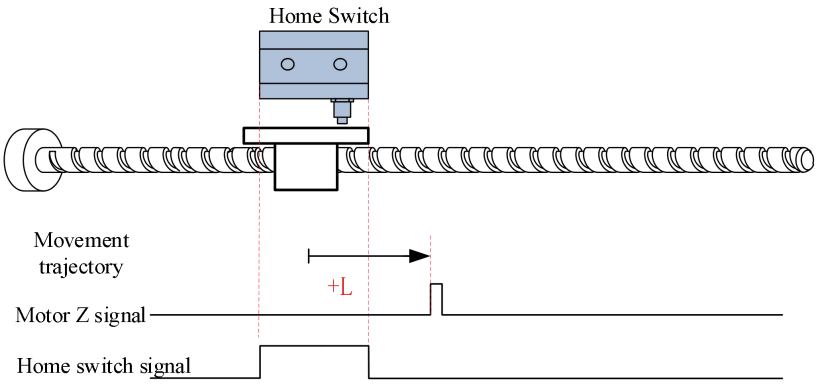
Homing mode 5: homing with Home switch and zero pulse signal（0x6098 =5）

1）When homing, the Home switch signal is invalid



Description: When homing, the Home Switch signal is invalid. The motor searches for the Home Switch signal in reverse at a higher speed. When the rising edge of the Home Switch signal is detected, it decelerates, reverses, and runs forward at a lower set speed. When the falling edge of the Home Switch signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

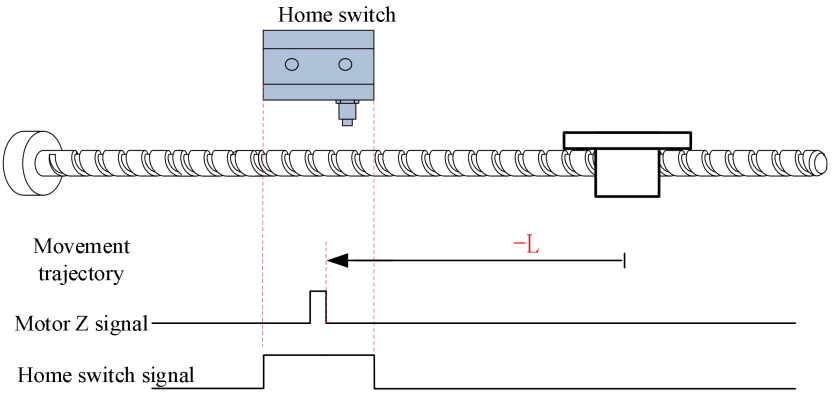
2）When homing, the Home switch signal is valid



Description: When homing, the Home Switch signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the Home Switch signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

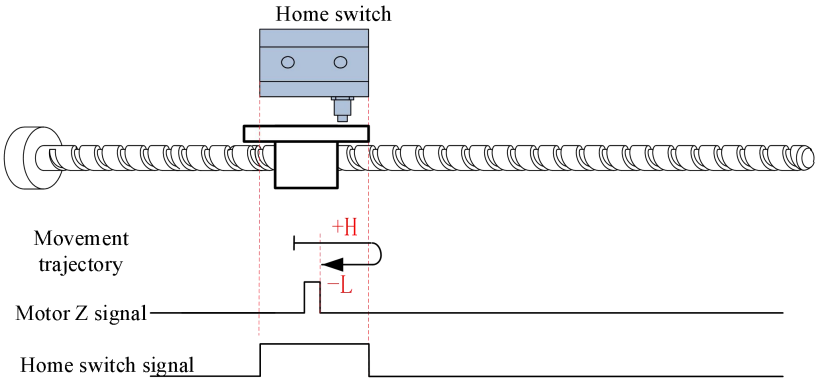
Homing mode 6: homing with Home switch and zero pulse signal（0x6098 =6）

1）When homing, the Home switch signal is invalid



Description: When homing, the Home Switch signal is invalid. The motor searches for the Home Switch signal in reverse at a lower speed until the rising edge of the Home switch signal is detected. Then it starts searching for the motor Z signal and stops running after the Z signal is found.

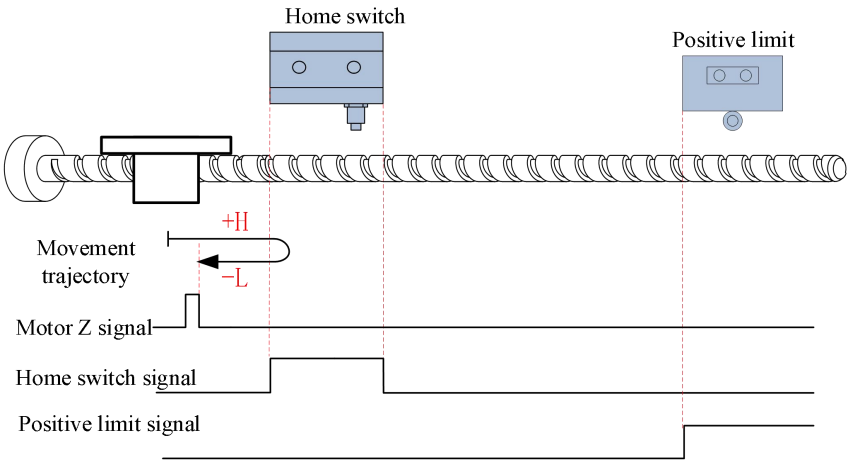
2）When homing, the Home switch signal is valid



Description: When homing, the Home Switch signal is valid. The motor directly runs forward at a higher set speed. When the falling edge of the Home Switch signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower speed. When the rising edge of the Home Signal is detected, it starts to search for the motor Z signal (falling edge), and stops running after the Z signal is found.

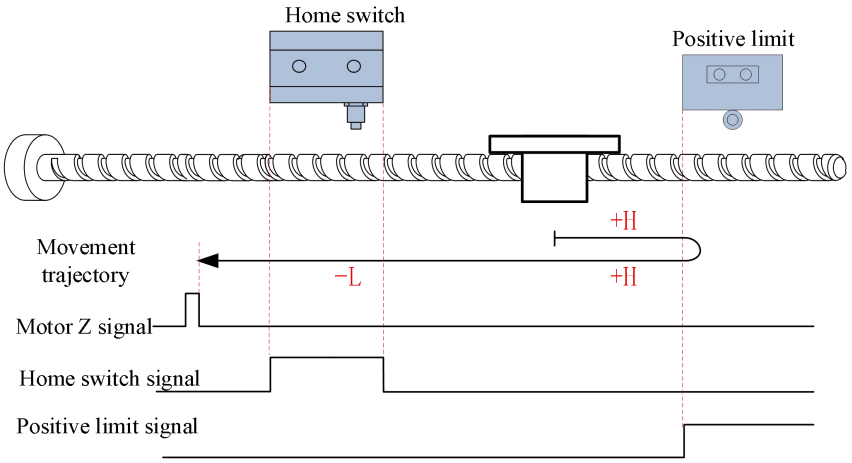
Homing mode 7: homing with Home switch, positive limit signal and zero pulse signal （0x6098 =7）

1）The Home signal is invalid when starting the homing, and the positive limit switch is not encountered.



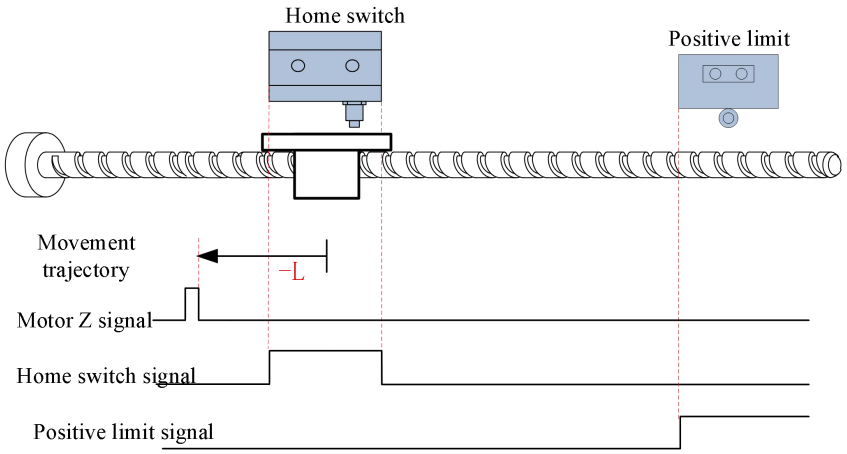
Description: When starting homing, the Home Signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the Home Signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the Home Signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The Home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a high speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates and continues to run in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

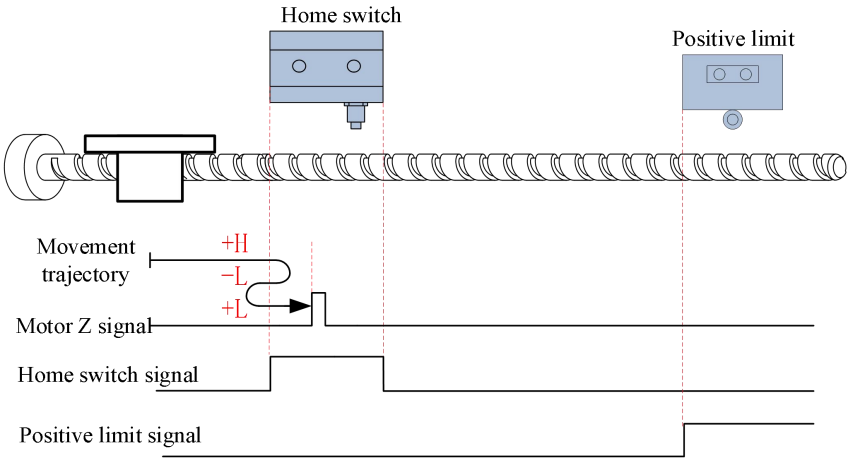
3）The Home Signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

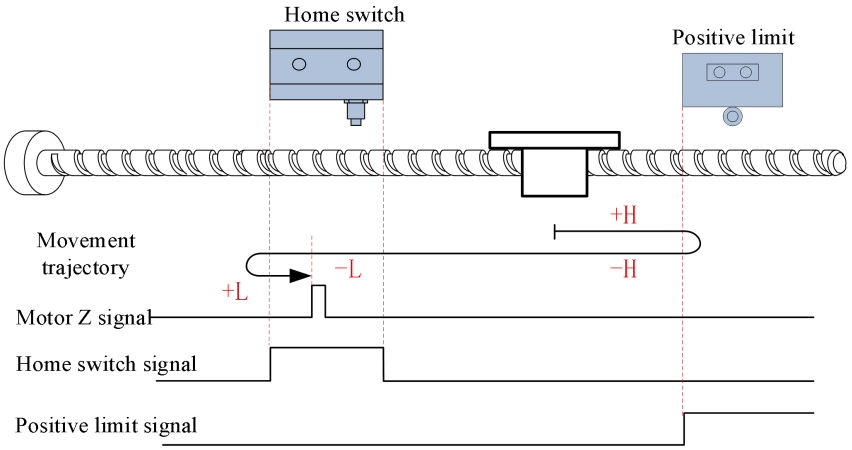
Homing mode 8: homing with home switch, positive limit signal and zero pulse signal （0x6098 =8）

1）The home signal is invalid when starting the homing, and the positive limit switch is not encountered.



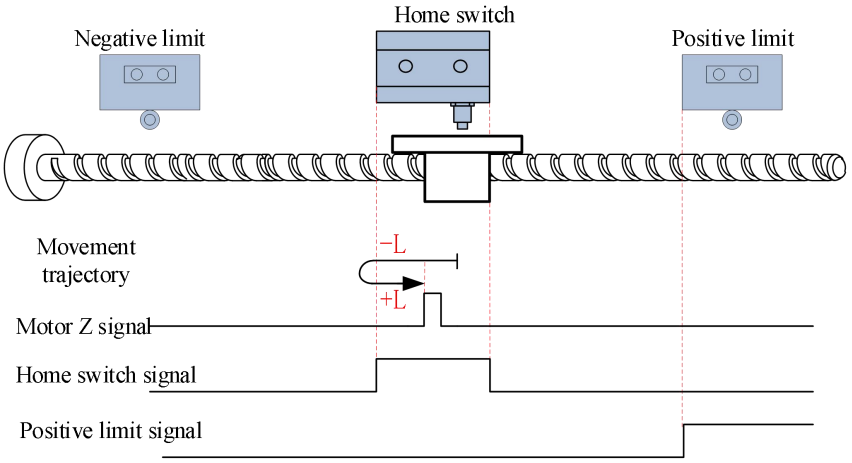
Description: When homing, the home signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the home signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is encountered, it reverses and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a high speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates and continues to run in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it reverses and runs forward at a lower set speed. When detecting the rising edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

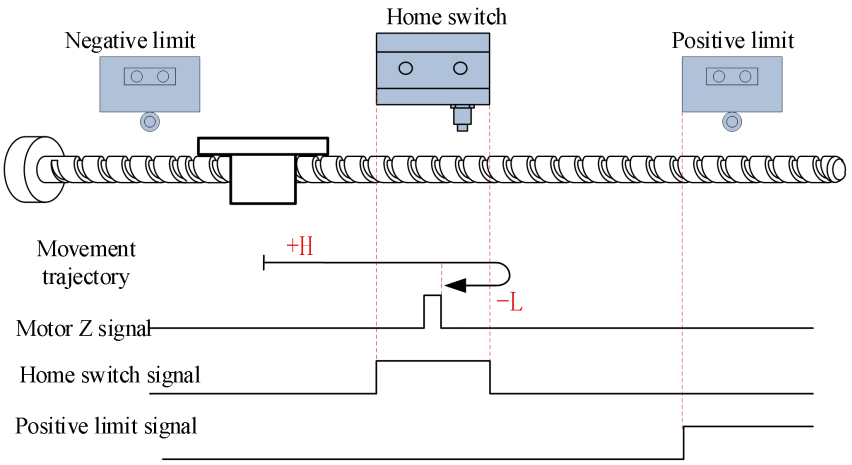
3）The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it reverses and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

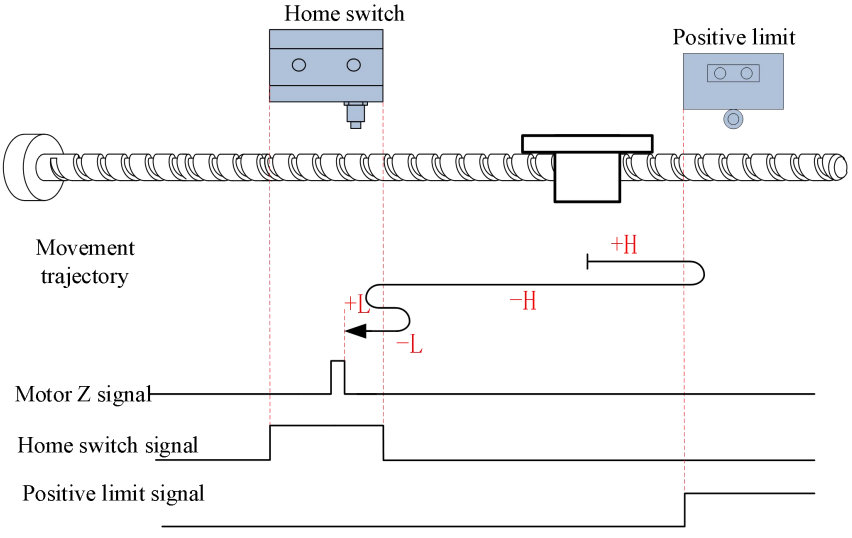
Homing mode 9: homing with home switch, positive limit signal and zero pulse signal （0x6098 =9）

1）The home signal is invalid when starting the homing, and the positive limit switch is not encountered.



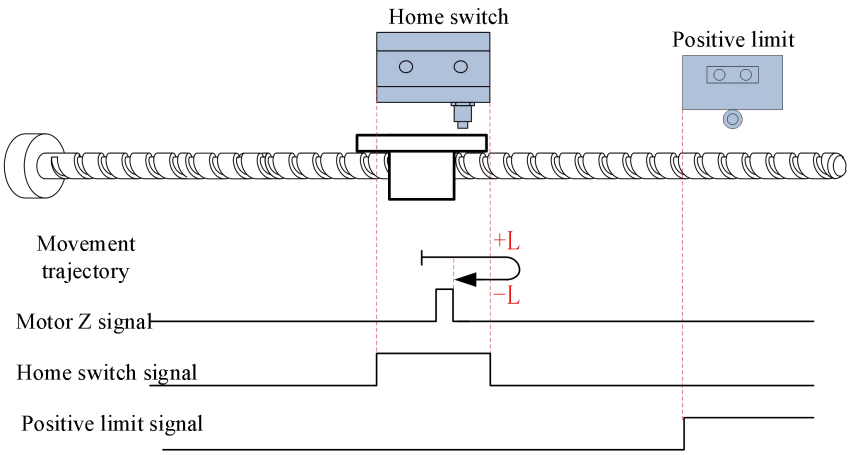
Description: When homing, the home signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is encountered, it reverses and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a higher speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates, reverses, and runs forward at a lower set speed. When encountering the falling edge of the home signal, it reverses and continues to run in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after finding the Z signal.

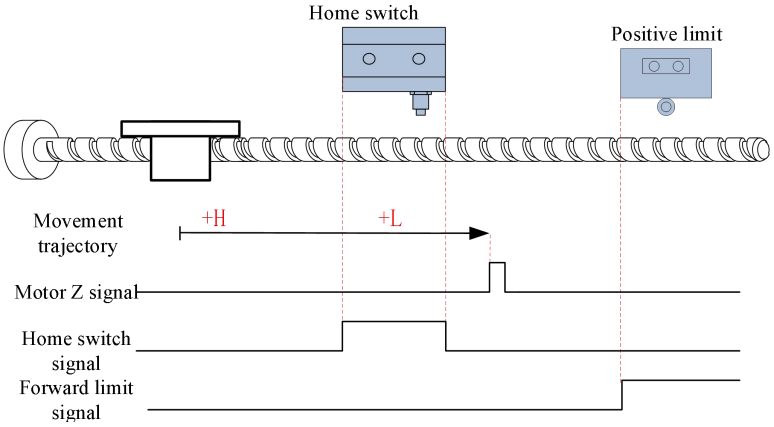
3）The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it reverses and continues to run in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

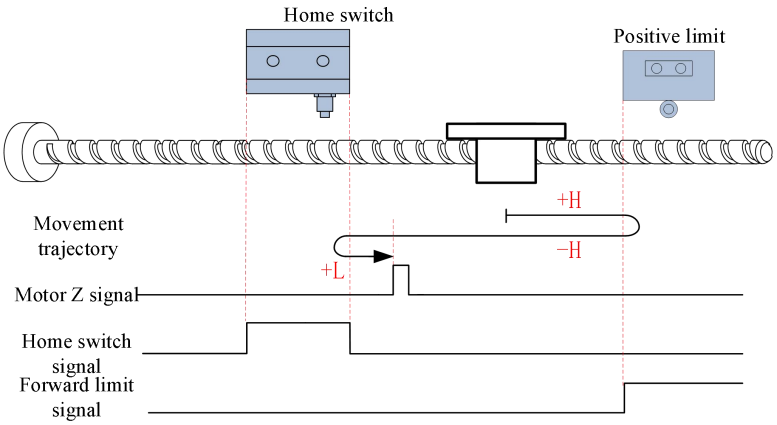
Homing mode 10: homing with home switch, positive limit signal and zero pulse signal （0x6098 =10）

1）The home signal is invalid when starting the homing, and the positive limit switch is not encountered.



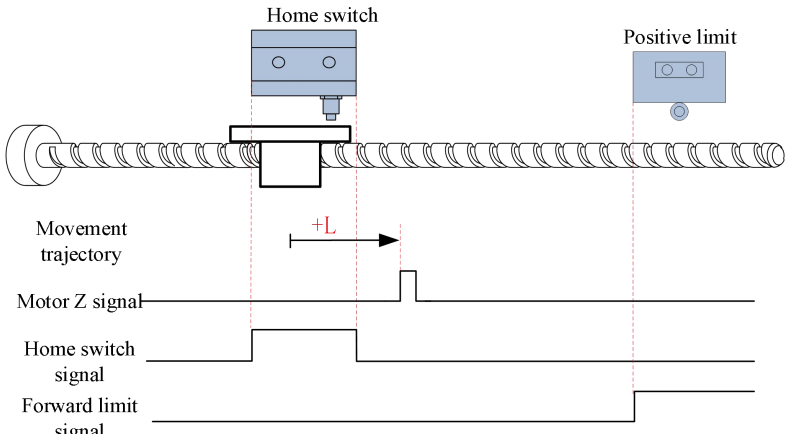
Description: When homing, the home signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a high speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates, reverses, and continues to run forward at a lower set speed. When encountering the falling edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

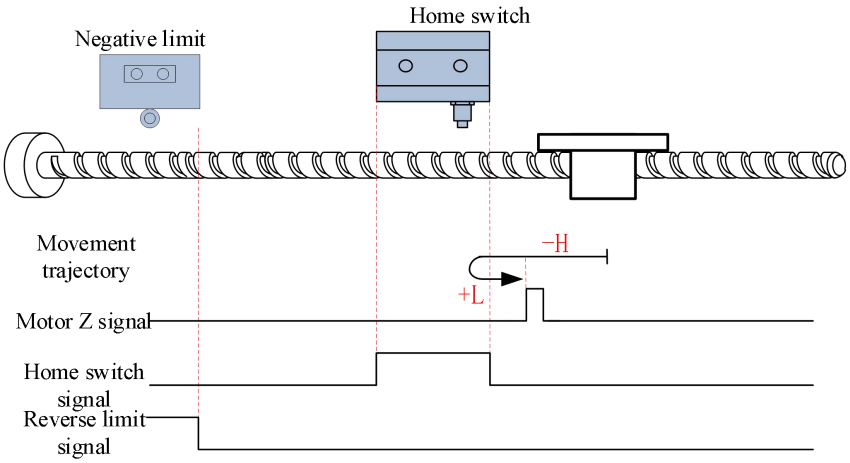
3）The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

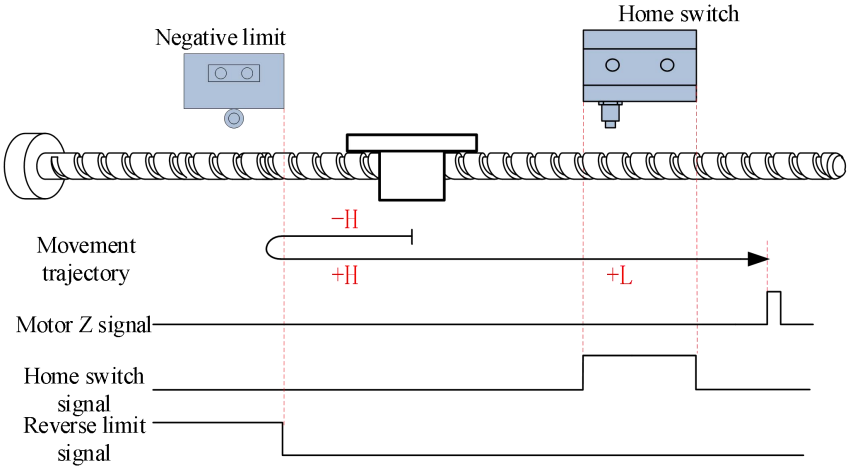
Homing mode 11: homing with home switch, reverse limit signal and zero pulse signal （0x6098 =11）

1）The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



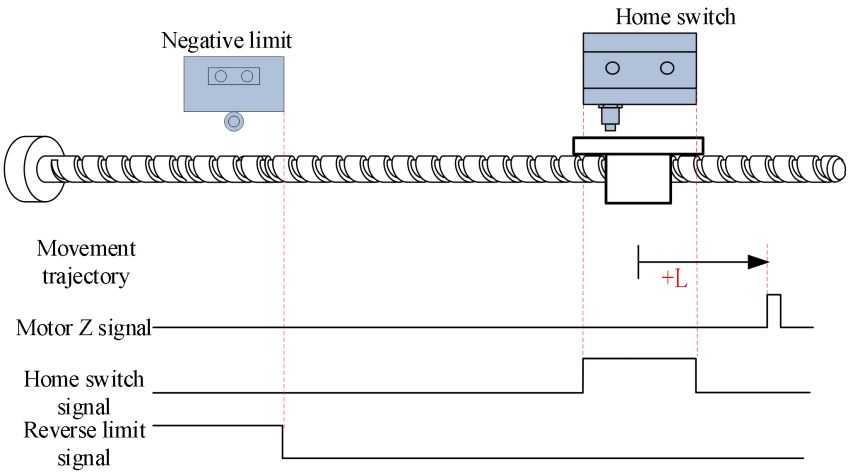
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is detected, it decelerates, reverses, and runs forward at a lower set speed. When the falling edge of the home signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates and reverses, and runs in the forward direction at a higher set speed. After detecting the rising edge of the home signal, it decelerates and continues to run in the forward direction at a lower set speed. When encountering the falling edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

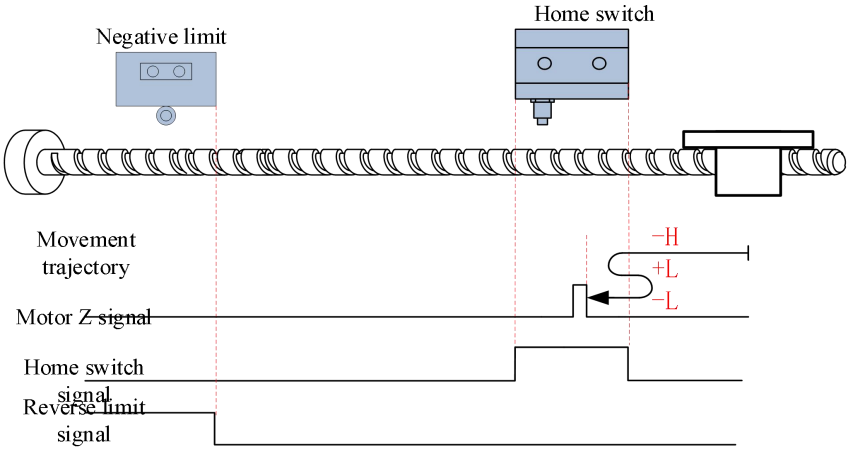
3）The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

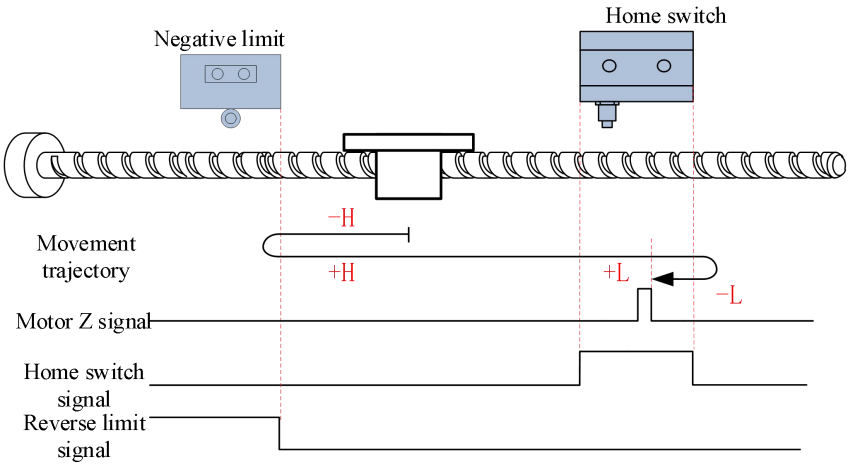
Homing mode 12: homing with home switch, reverse limit signal and zero pulse signal （0x6098 =12）

1）The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



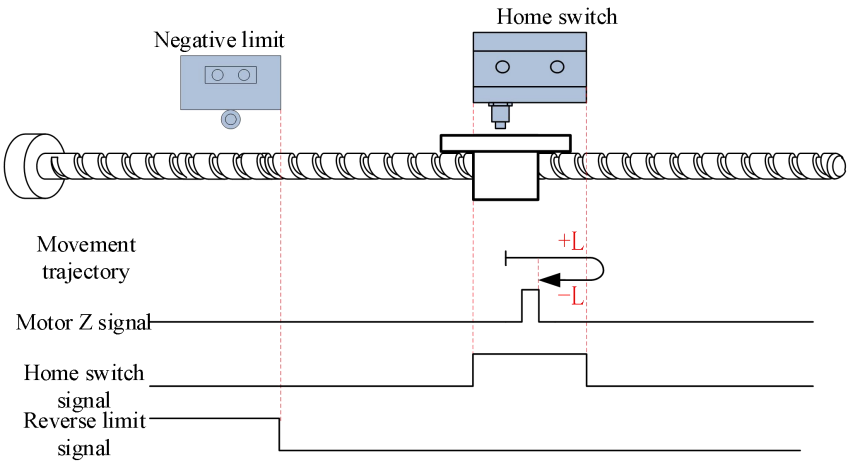
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is detected, it decelerates, reverses, and runs forward at a lower set speed. When the falling edge of the home signal is encountered, it reverses again and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates, reverses, and runs forward at a higher speed. After detecting the rising edge of the home signal, it decelerates and continues to run forward at a lower set speed. When encountering the falling edge of the home signal, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When detecting the rising edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

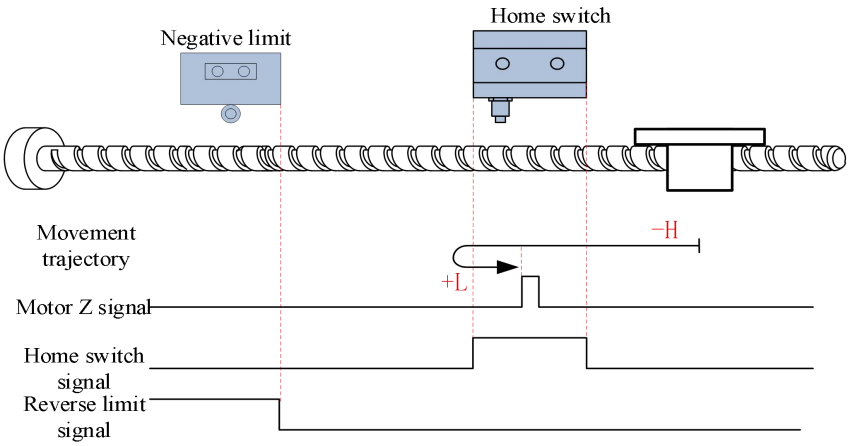
3）The home signal is valid when homing



Description: When starting from zero, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

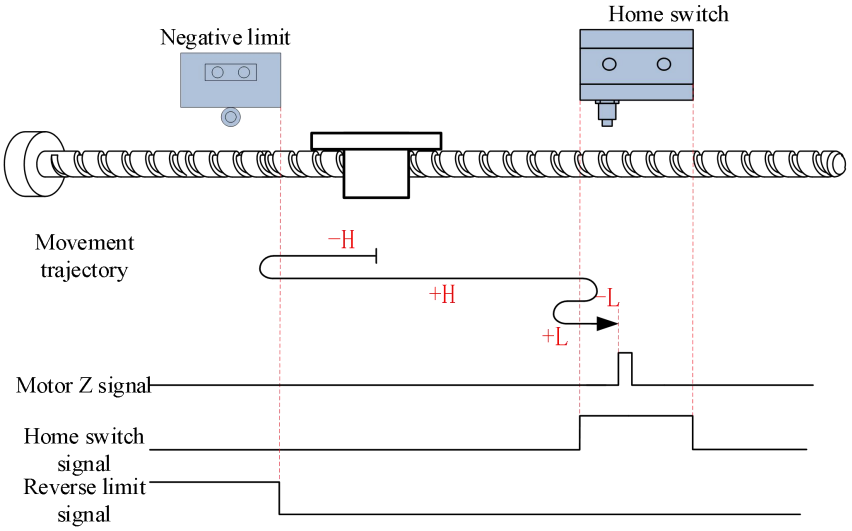
Homing mode 13: homing with home switch, reverse limit signal and zero pulse signal （0x6098 =13）

1）The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is encountered, it reverses and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

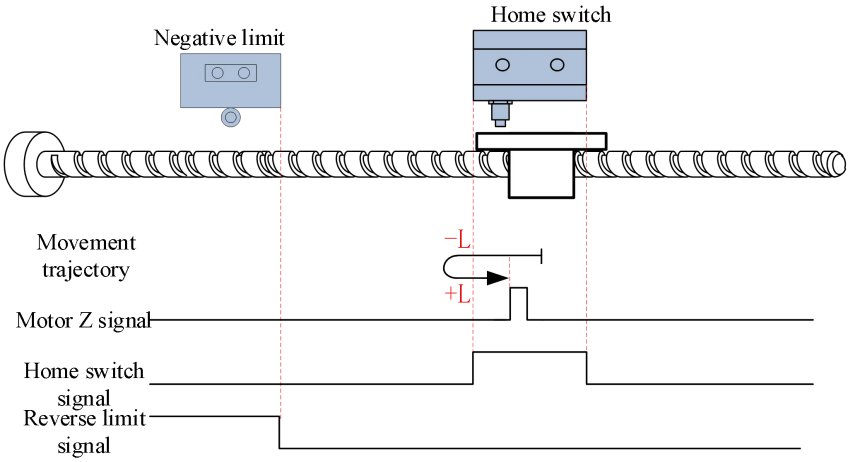
2）The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates, reverses, and runs forward at a higher set speed. After detecting the rising edge of the home signal, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it decelerates again and reverses, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is

found.

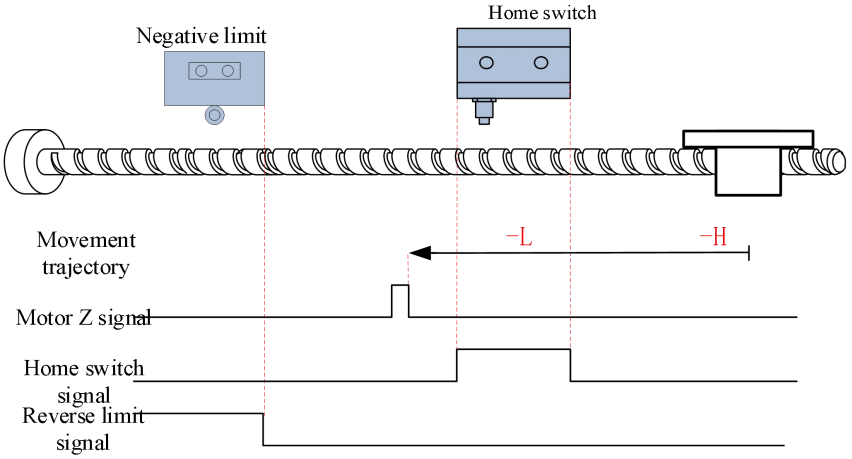
3）The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal, and stops running after the Z signal is found.

Homing mode 14: homing with home switch, reverse limit signal and zero pulse signal （0x6098 =14）

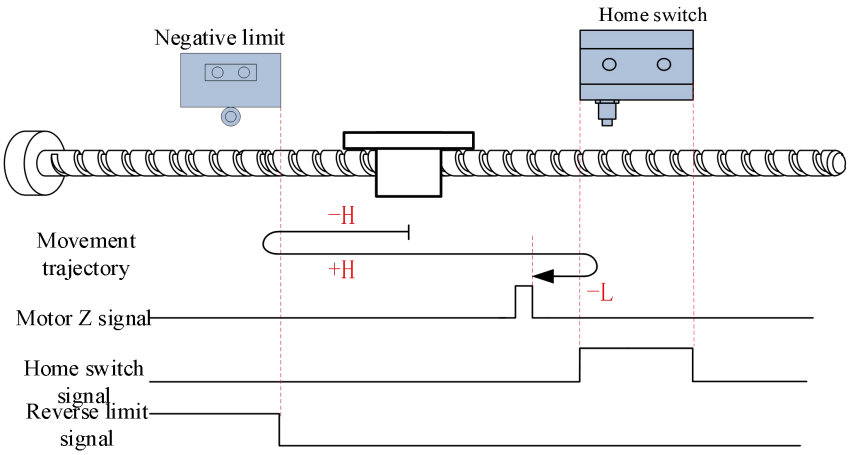
1）The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is

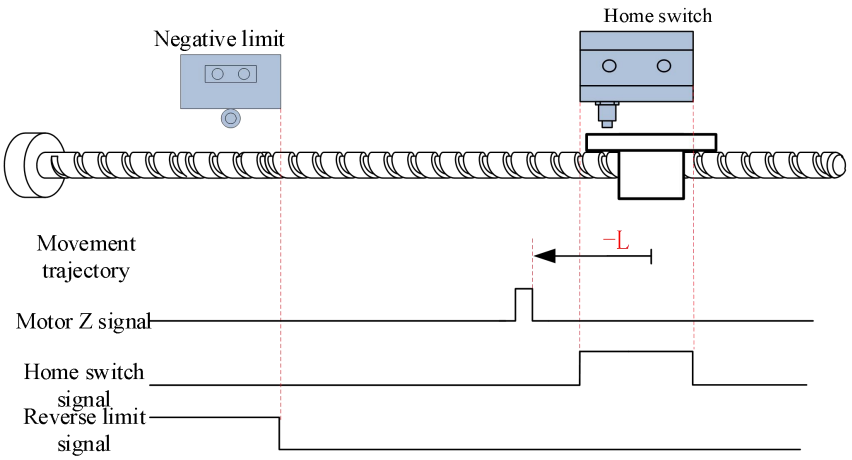
detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates, reverses, and runs forward at a higher set speed. After detecting the rising edge of the home signal, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

3）The home signal is valid when homing



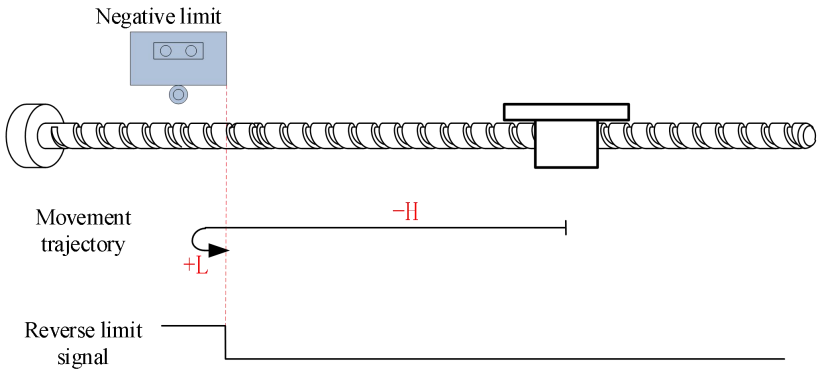
Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

Homing method 15,1615,16：

Reserve

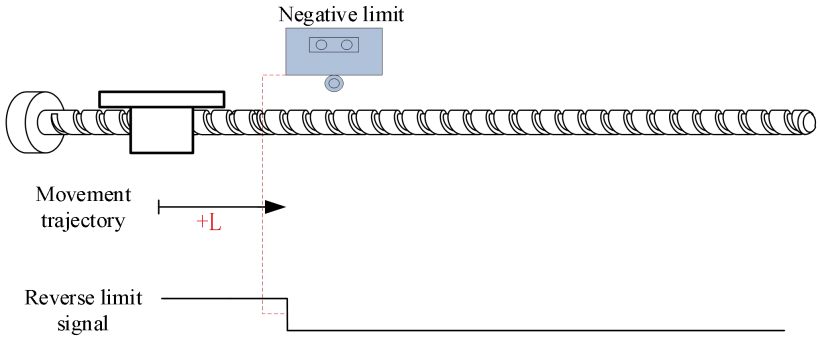
Homing mode 17: Homing with reverse limit signal (6098=17)

1）When homing, the reverse limit signal is invalid.



Description: When starting from zero, the reverse limit signal is invalid. The motor runs in the reverse direction at a higher set speed. When the rising edge of the reverse limit signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the reverse limit signal is detected, it stops running.

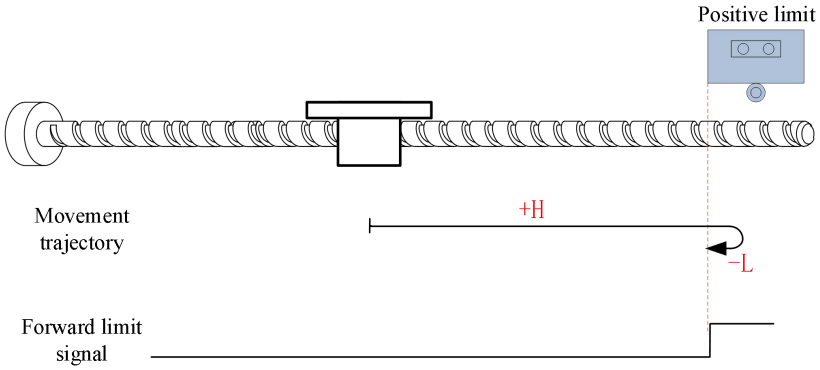
2）When homing and starting, the reverse limit signal is valid



Description: When starting from zero, the reverse limit signal is valid. The motor directly runs forward at a lower set speed and stops when the falling edge of the reverse limit signal is detected.

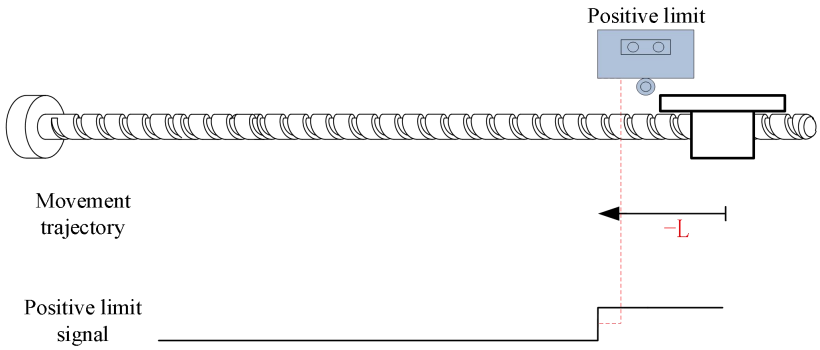
Homing mode 18: Homing with positive limit signal (6098=18)

1）When homing, the positive limit signal is invalid



Description: When starting from zero, the positive limit signal is invalid. The motor runs forward at a higher set speed. When the rising edge of the positive limit signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the positive limit signal is detected, it stops running.

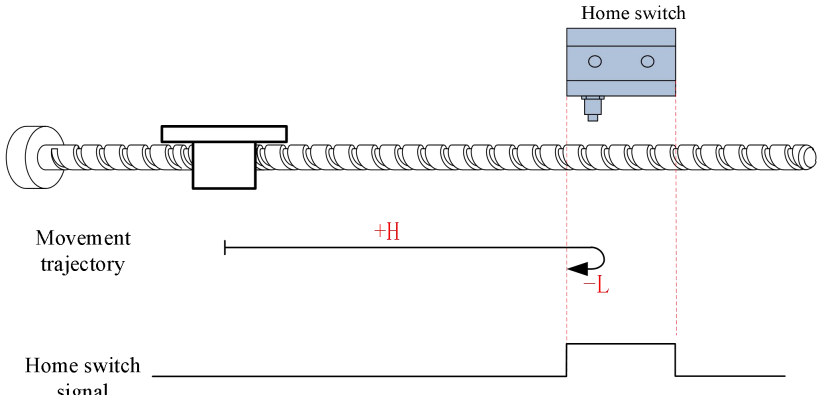
2）When homing, the positive limit signal is valid



Description: When starting from zero, the positive limit signal is valid. The motor directly runs in the reverse direction at a lower set speed, and stops when the falling edge of the positive limit signal is detected.

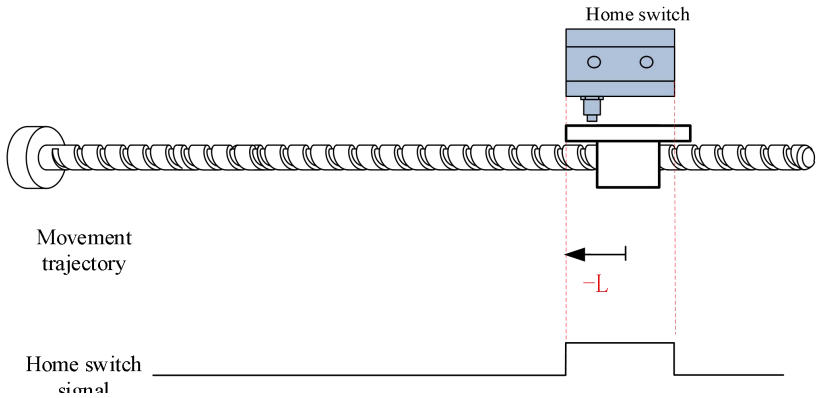
Homing method 19: homing with home signal (6098=19) (6098=19)

1）When homing, the home signal is invalid



Description: When starting from homing, the home signal is invalid. The motor runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

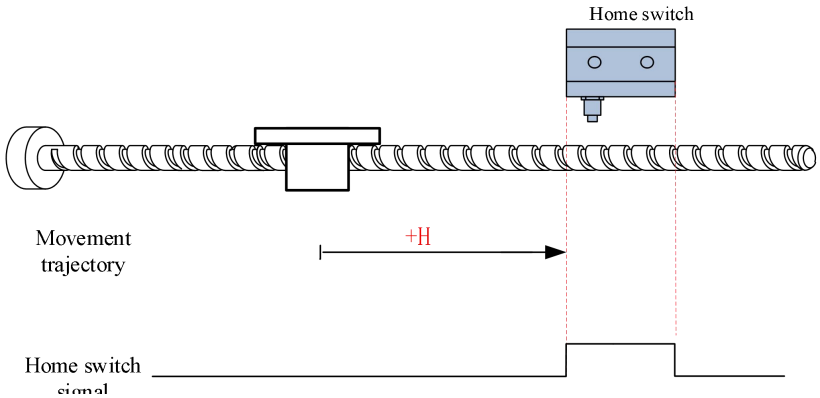
2）When homing, the home signal is valid



Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed, and stops when the falling edge of the home signal is detected.

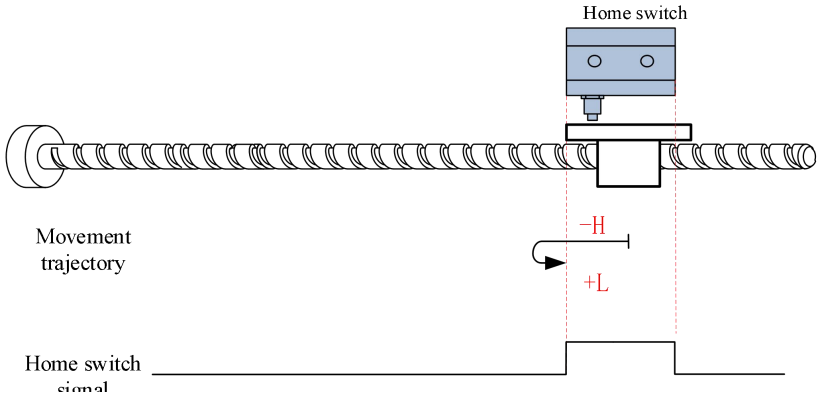
Homing method 20: homing with home signal (6098=20)

1）When homing, the home signal is invalid



Description: When starting the homing, the home signal is invalid. The motor runs forward at a lower speed and stops when the rising edge of the home signal is detected.

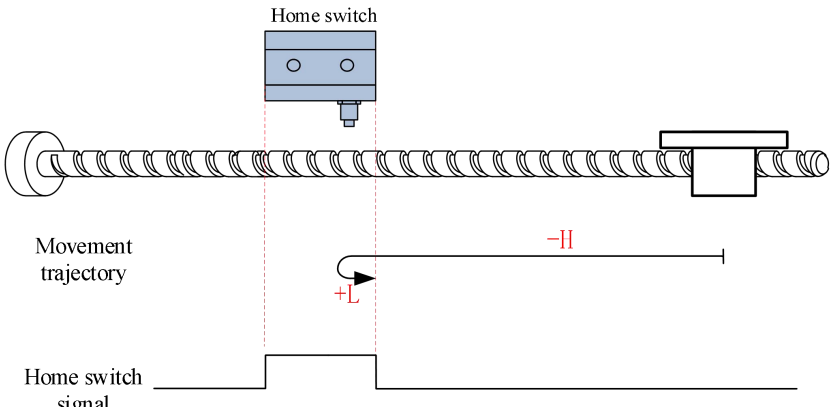
2）When homing, the home signal is valid



Description: When starting from zero, the home signal is valid. The motor first runs in the reverse direction at a higher set speed. When the falling edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops.

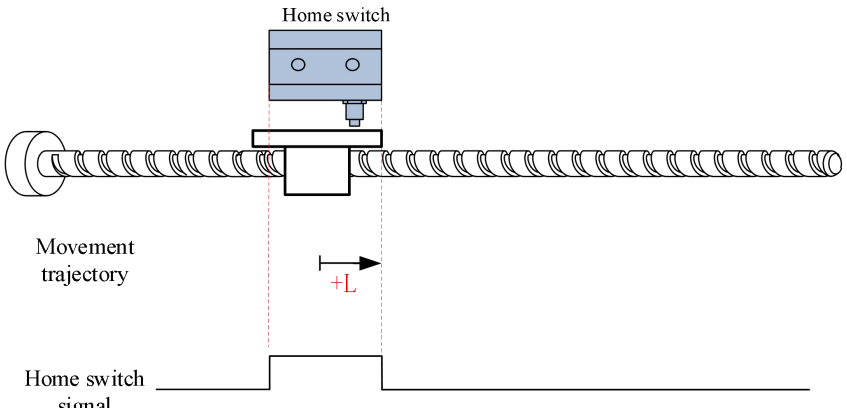
Homing method 21: homing with home signal (6098=21)

1）When homing, the home signal is invalid



Description: When starting from homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it stops.

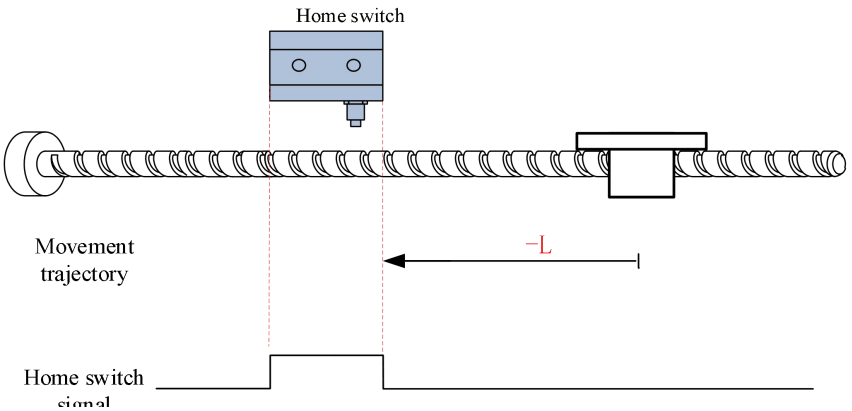
2）When homing, the home signal is valid



描述：Description: When starting from zero, the home signal is valid. The motor directly runs forward at a lower set speed and stops when the falling edge of the home signal is detected.

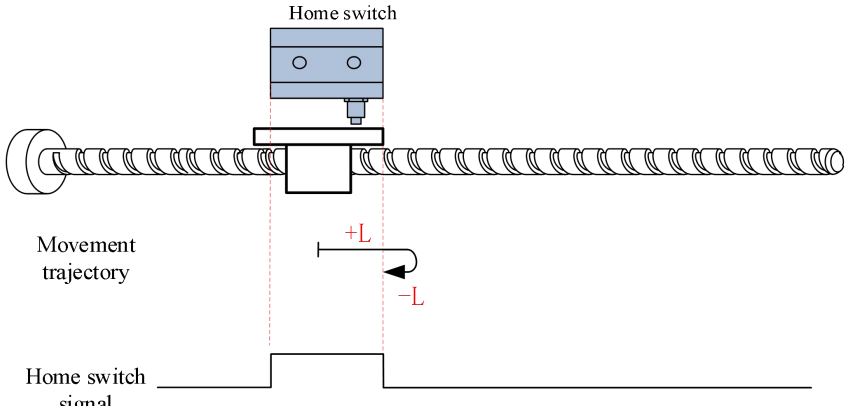
Homing method 22: Homing with home signal (6098=22)

1）When homing, the home signal is invalid



Description: When starting the homing, the home signal is invalid. The motor runs in the reverse direction at a lower speed and stops when the rising edge of the home signal is detected.

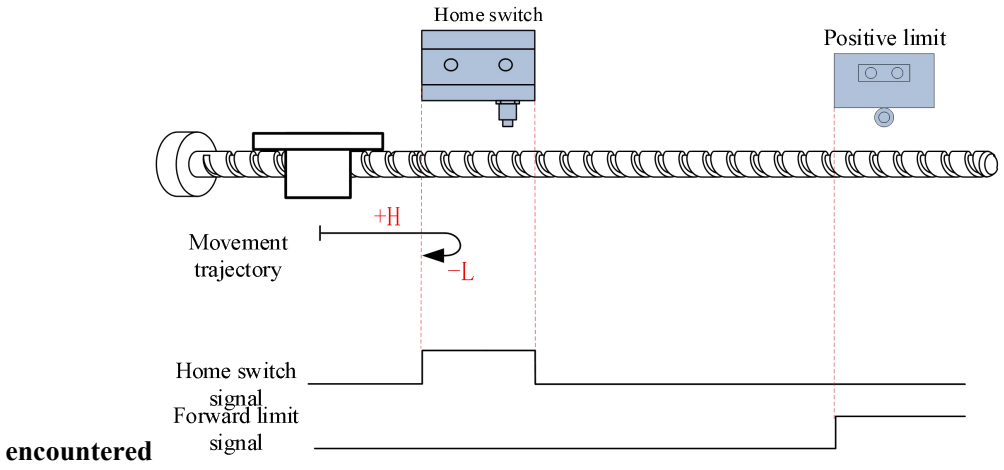
2）When homing, the home signal is valid



Description: When starting from homing, the home signal is valid. The motor first runs forward at a higher set speed. When the falling edge of the home signal is detected, it decelerates and reverses, running in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

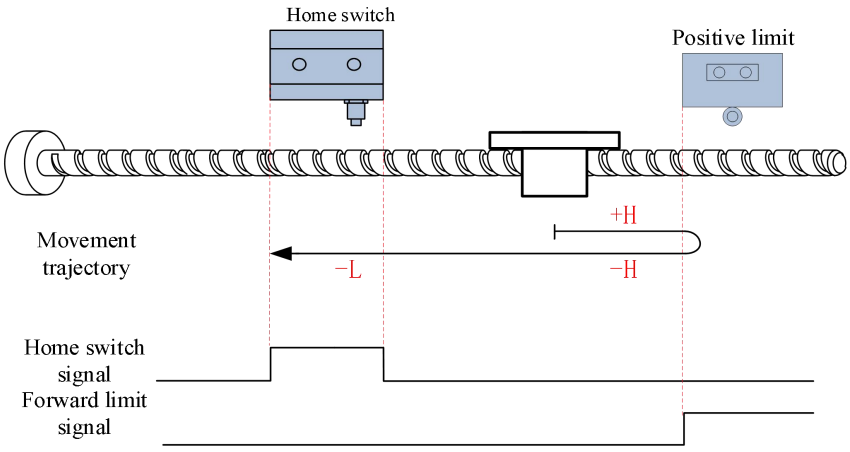
Homing mode 23: homing with home signal and positive limit signal (6098=23)

1）When homing, the home signal is invalid and no positive limit switch is



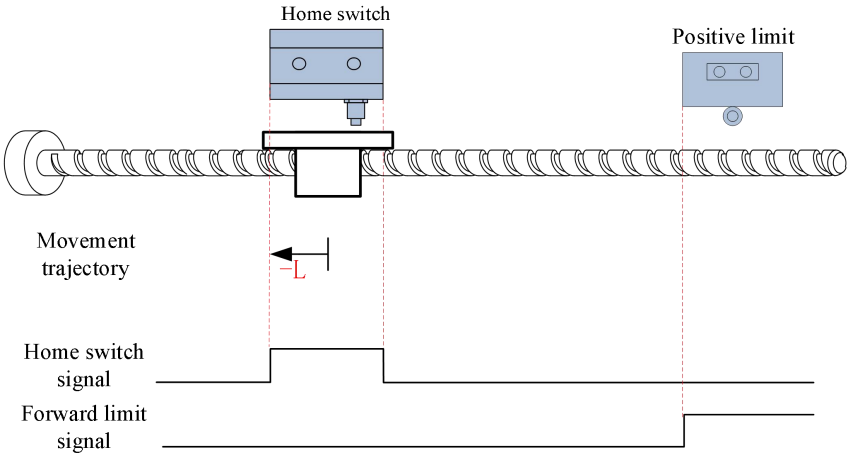
Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, runs in the reverse direction at a lower set speed, and stops when the falling edge of the home signal is detected.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When it encounters the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

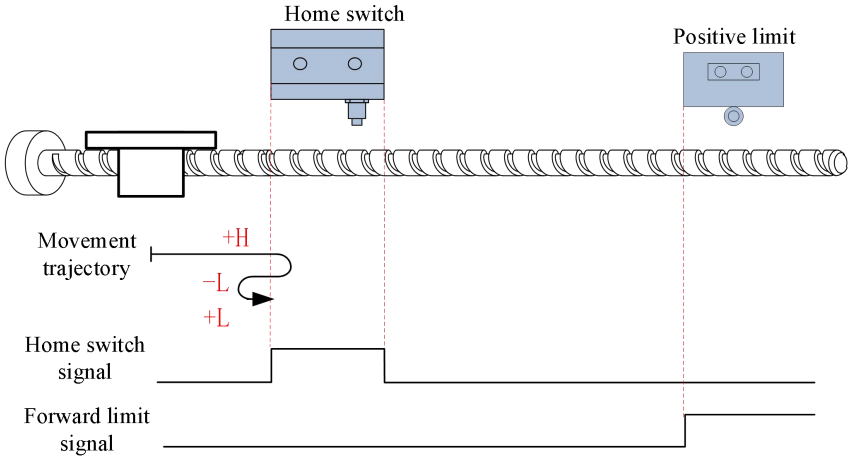
3）When homing, the home signal is valid



Description: When starting the homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed and stops when the falling edge of the home signal is detected.

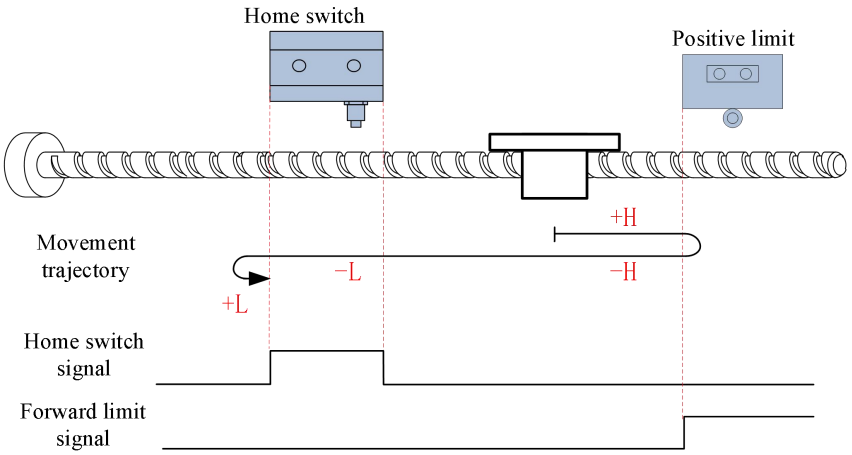
Homing mode 24: homing with home signal and positive limit signal (6098=24)

1）When homing, the home signal is invalid and no positive limit switch is encountered



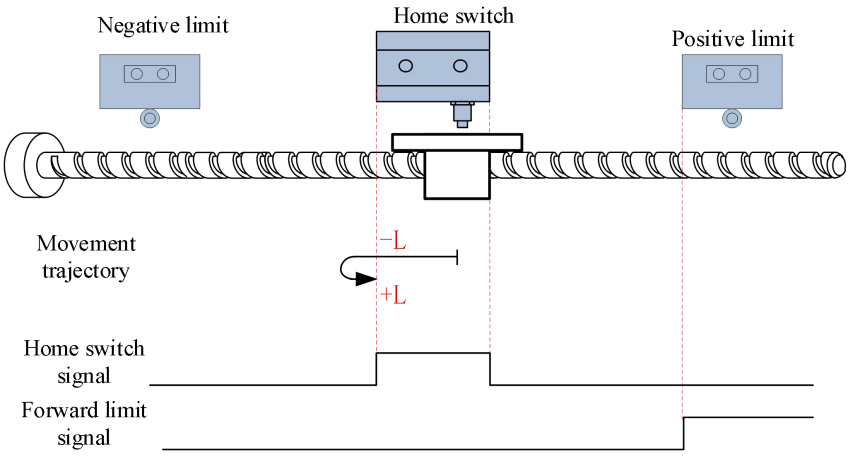
Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When encountering the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

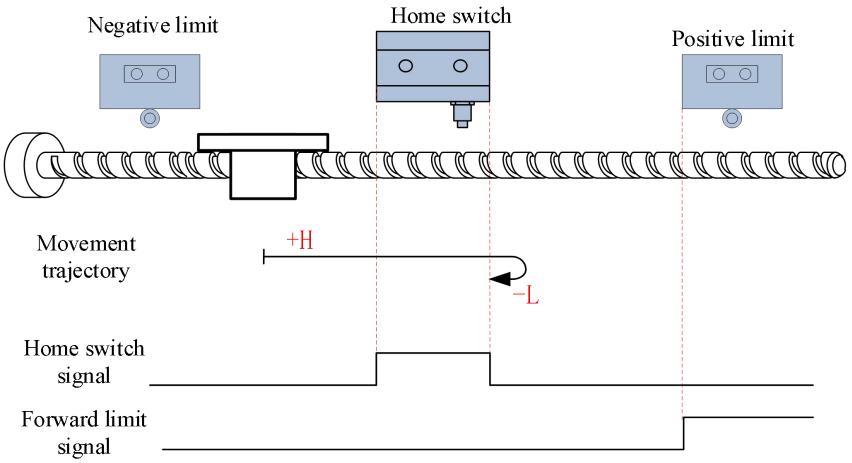
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

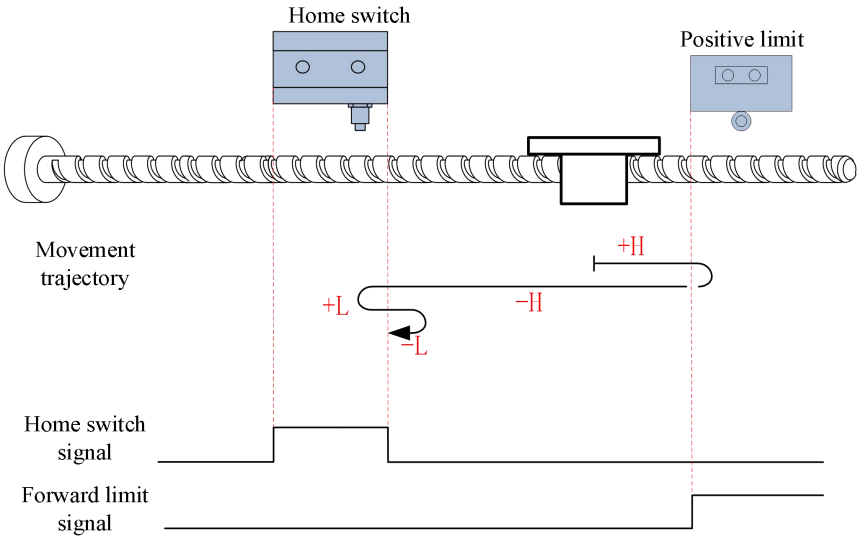
Homing mode 25: homing with home signal and positive limit signal (6098=25)

1）When homing, the home signal is invalid and no positive limit switch is encountered



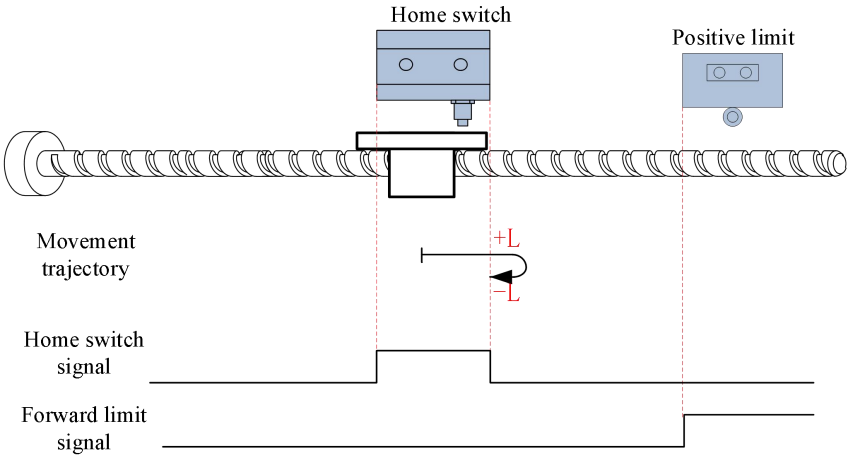
Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates again and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When encountering the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

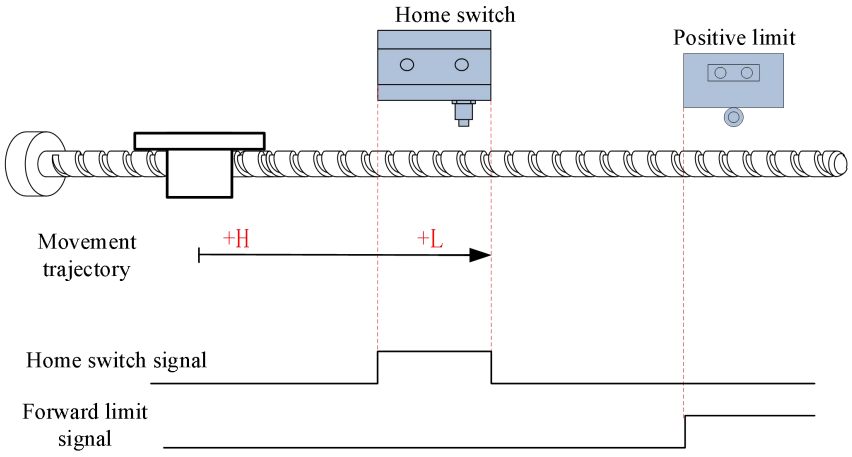
3）When homing, the home signal is valid



描述：Description: When homing, the home signal is valid. The motor runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, running in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

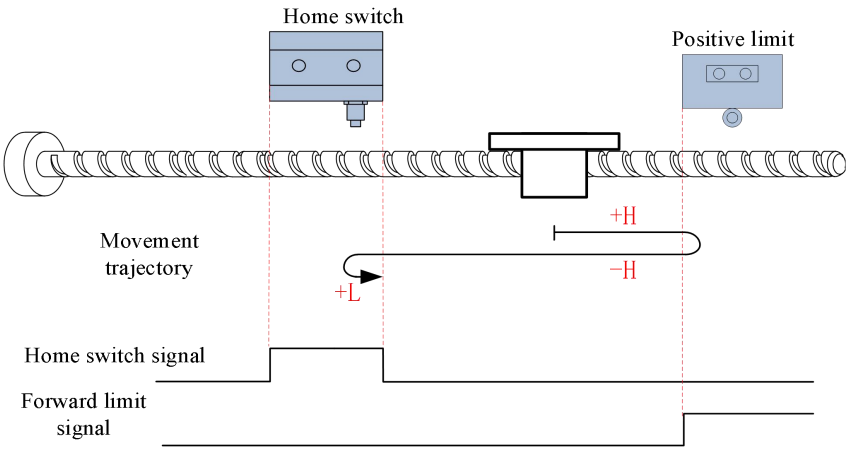
Homing mode 26: homing with home signal and positive limit signal (6098=26)

1）When homing, the home signal is invalid and no positive limit switch is encountered



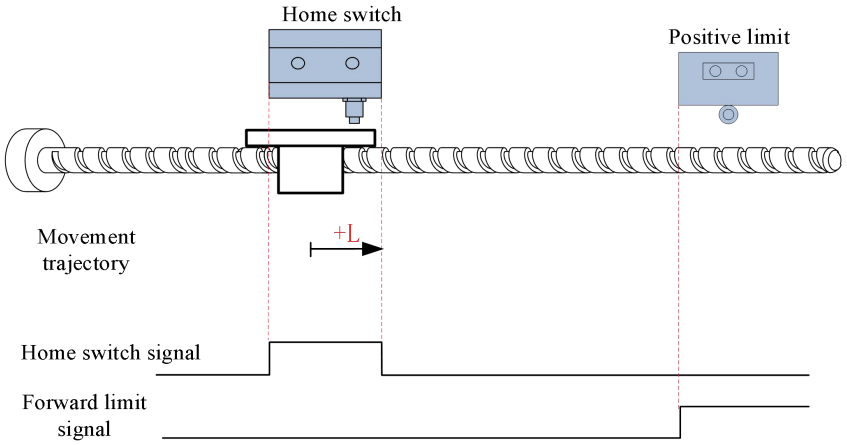
Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When encountering the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it stops running.

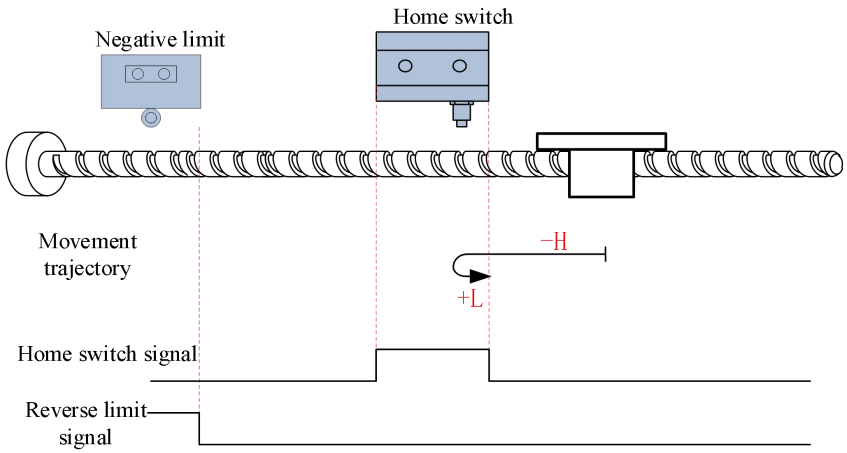
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs forward at a lower set speed and stops when the falling edge of the home signal is detected.

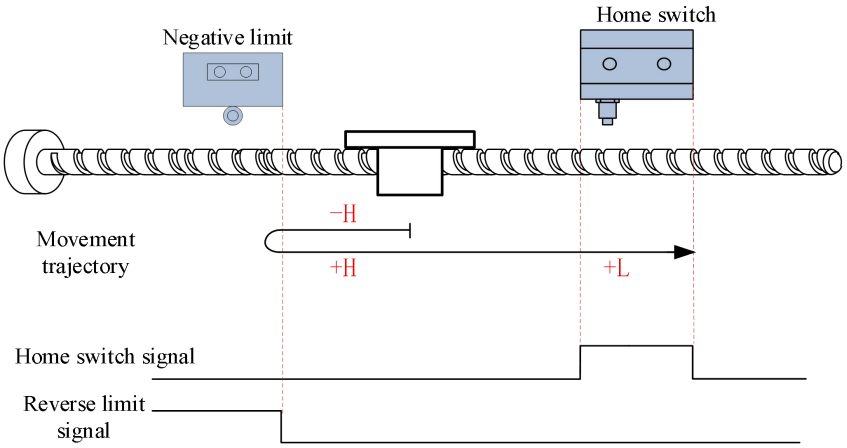
Homing mode 27: homing with home signal and reverse limit signal (6098=27)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



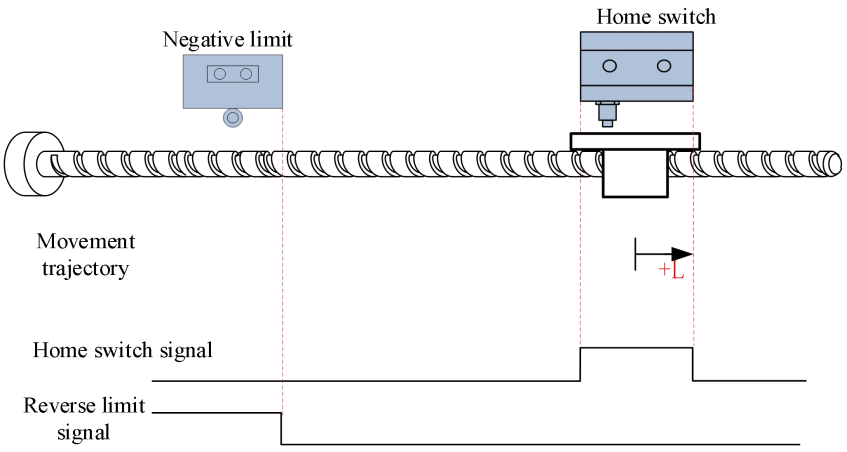
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters a reverse limit switch



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When it encounters the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it stops running.

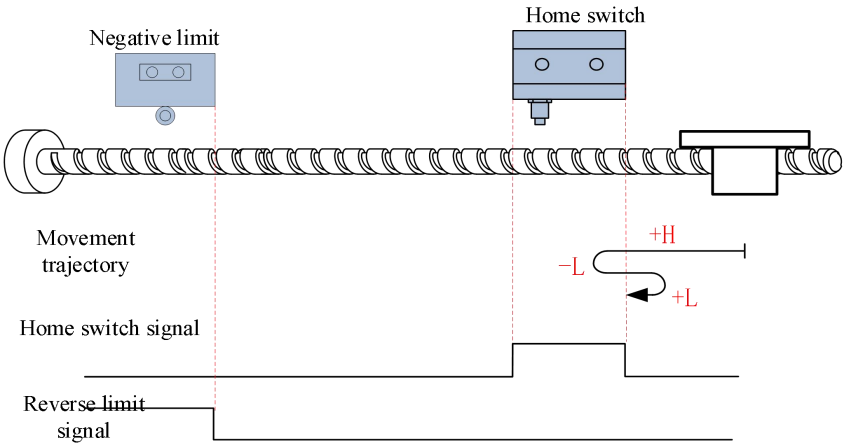
3）When homing, the home signal is valid



Description: During homing initialization, the origin signal is active. The motor operates in the positive di rection at a preset low speed and stops upon detecting the falling edge of the origin signal.

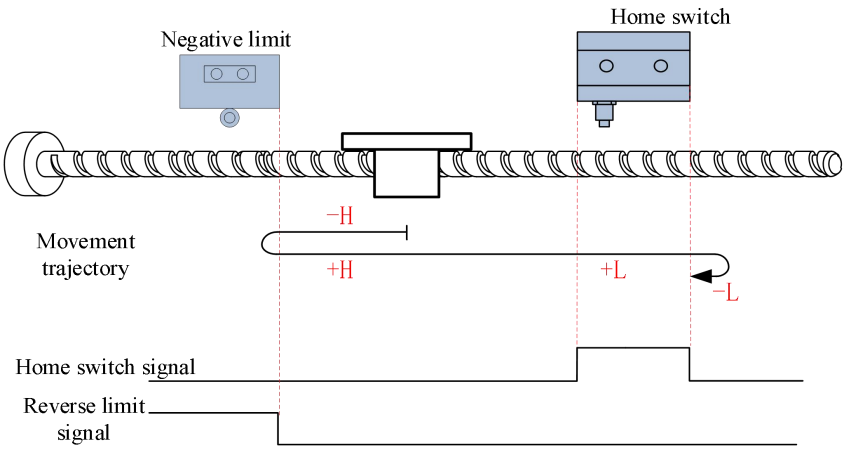
Homing mode 28: homing with home signal and reverse limit signal (6098=28)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



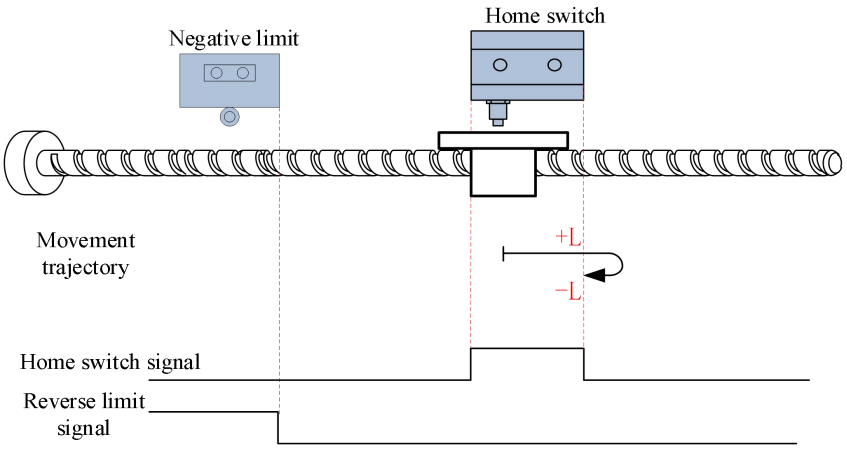
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters a reverse limit switch



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When encountering the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

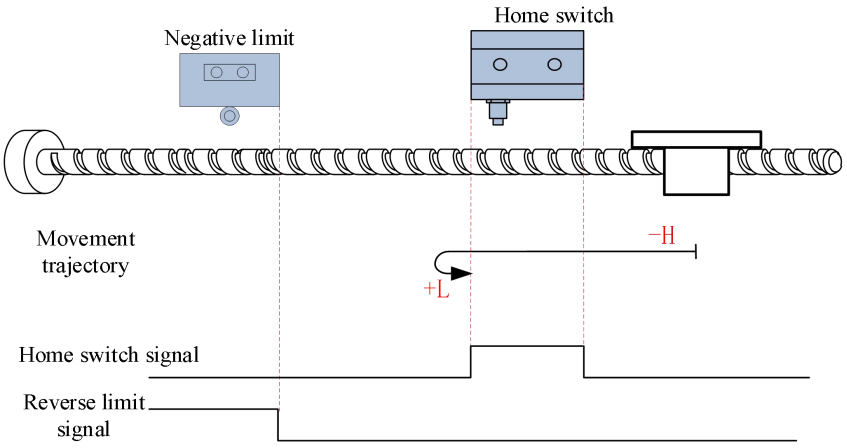
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, running in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

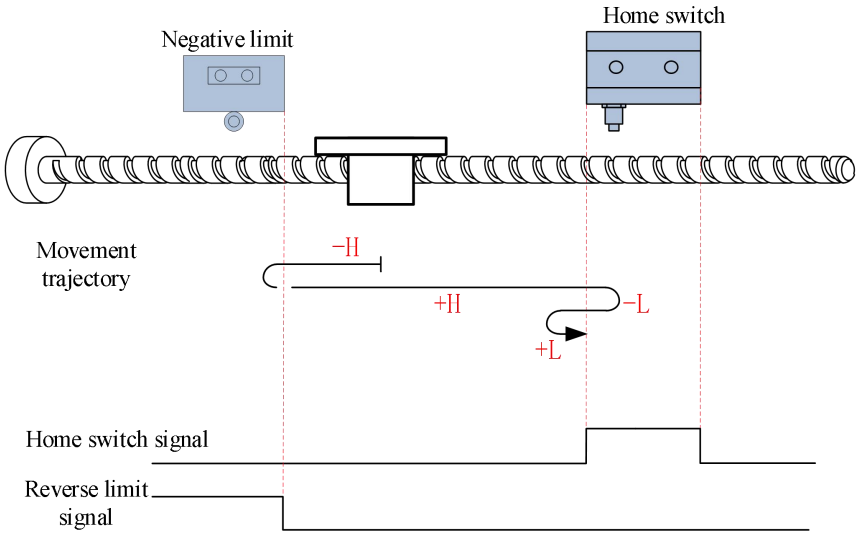
Homing mode 29: homing with home signal and reverse limit signal (6098=29)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



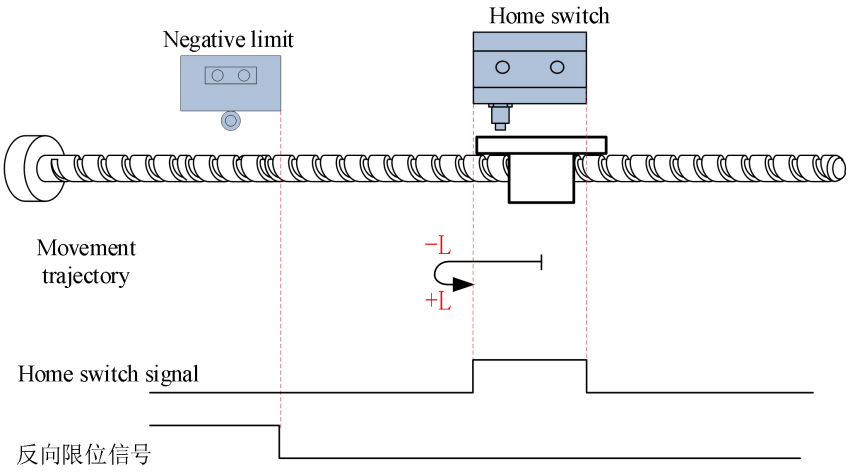
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, runs forward at a lower set speed, and stops when the rising edge of the home signal is detected.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When encountering the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

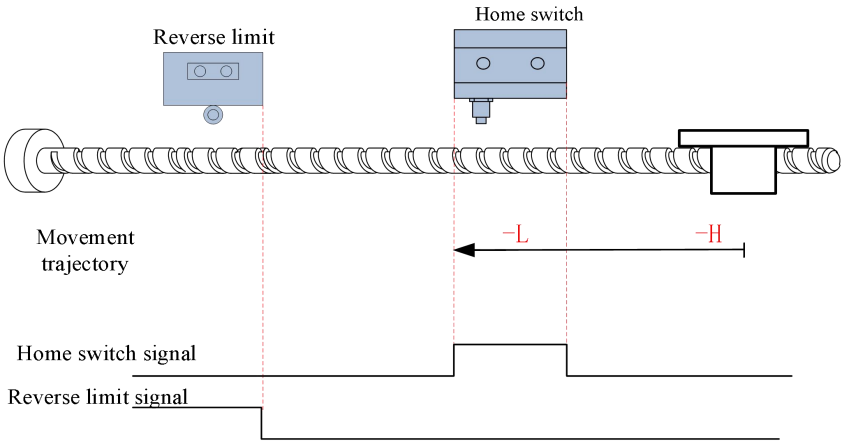
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops.

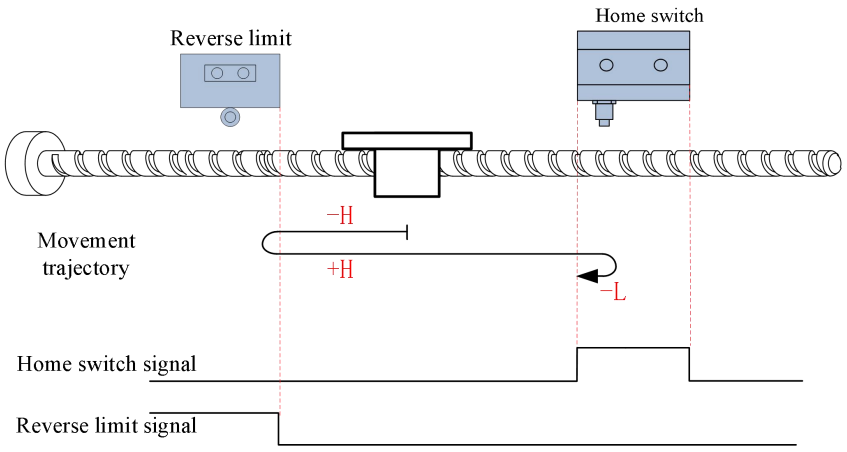
Homing mode 30: homing with home signal and reverse limit signal (6098=30)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



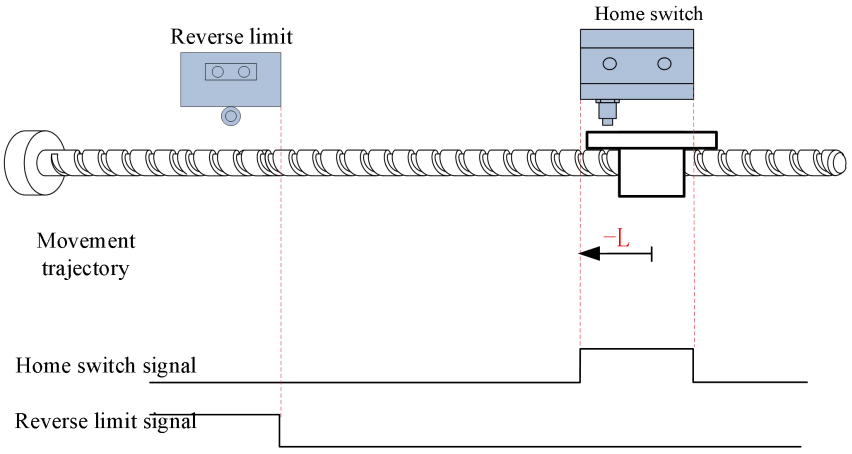
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters a reverse limit switch



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When it encounters the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

3）When homing, the home signal is valid



Description: When starting the homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed and stops when the falling edge of the home signal is detected.

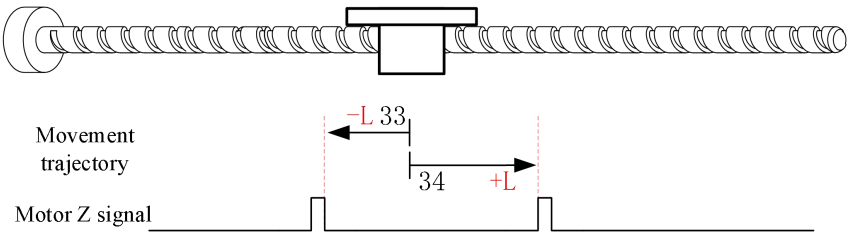
Homing method 31: (used in conjunction with homing method 32)

Description: When the motor uses an absolute position encoder, perform absolute position homing (the absolute position recorded by homing method 32).

Homing method 32: (used in conjunction with homing method 31)

Description: When the motor uses an absolute position encoder, the current absolute position is recorded as the zero point for absolute position return.

Homing mode 33,34: Homing with motor zero position pulse signal（6098=33,34）



Description:

Homing mode 33: The motor runs in the reverse direction at a set lower speed and stops when it encounters the motor Z signal;

Homing mode 34: The motor runs in the forward direction at a set lower speed and stops when it encounters the motor Z signal;

Homing method 35: current point homing

Description: With bit 4 of control word 6040 as the trigger signal, the motor does not move and directly sets the homing completion state.

5.8.4 Homing Mode Configuration Example

Configure the servo operating mode object dictionary 6060h=0x06 to operate in homing mode.

Configure the homing speed deceleration point speed =6091 subindex 1=100（100rpm）

Configure the homing speed zero point speed=6091 subindex 2=50（50rpm）

Configure the homing acceleration 609Ah=100（100ms）。

Configure the homing type=6098h=1（homing using the reverse limit switch and zero pulse signal）.

Configure the control word 6040h to enable the drive in the enabled state. The command sequence for 6040h is: 0x06→0x07→0x0F→0x1F。

Chapter 6 Operation performance adjustment

6.1 Overvie

The Servo drive need to drive the motor as quickly and accurately as possible to track the instructions from host or internal settings. In order to meet the requirement, the servo gain must be reasonably adjustment.

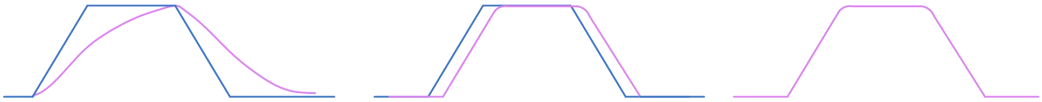


Figure 6.1 Gain setting example

Position loop gain: 40.0Hz	Position loop gain: 200.0Hz	Position loop gain: 200.0Hz
Speed loop gain: 200.0Hz	Speed loop gain: 25.0Hz	Speed loop gain: 25.0Hz
Speed loop integral time constant:100.00ms	Speed loop integral time constant: 50.00ms	Speed loop integral time constant: 50.00ms
Speed feedforward gain: 0	Speed feedforward gain: 0	Speed feedforward gain: 50%
Load inertia ratio: 30	Load inertia ratio: 30	Load inertia ratio: 30

The servo gain is setting by the combination of multiple parameters (Position loop, speed loop gain, filter, load rotate inertia ratio, etc.), which influence each other. Therefore, the setting of servo gain must considerate the balance between each parameter setting value.

Note:

Before performing gain adjustment, it is recommended to perform a jog test run to confirm that the motor can operate normally.

The normal process of gain adjustment is shown as the figure below:

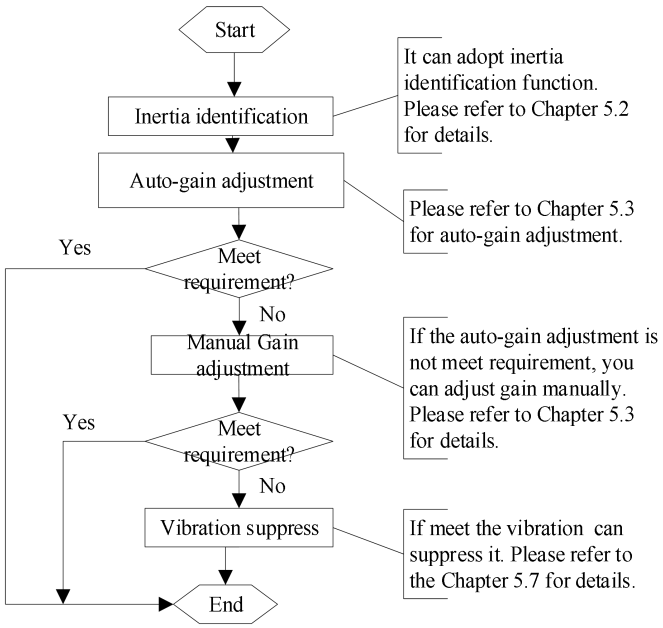


Figure 5.2 Gain adjustment process

6.2 Inertia Identification

$$\text{Load inertia ratio} = \frac{\text{mechanical load ttotal rotation inertia}}{\text{Motor}-\text{self rotation inertia}}$$

Load inertia ratio is the important parameter of servo system, correctly set load inertia ratio is benefit for quickly complete test. The load inertia ratio can be set manually or automatically identified through the inertia identification function of the servo drive.

Use the “Rotating inertia identification function (P11.03)” to realize inertia identification by operating the function code of servo drive host computer to rotate the motor.

Note：

Using the inertia identification function, in order to accurately calculate the load inertia ratio, the following conditions must be met:

• The actual motor maximum rotation speed is up to 200rpm;

• When the actual motor accelerates and decelerates, the acceleration is above 3000rpm/s.;

• The load torque is relatively stable and cannot change strenuously;

• The actual load inertia ratio doea not exceed 120 times;

If the actual load inertia ratio is large and the drive gain is low, the motor will move slowly and cannot meet the motor's maximum speed and accerlation requirements. In this case, the speed loop gain (P07.01) can be increased and the inertia identification can be performed again.

If vibration occurs during the identification process, the inertia identification should be stopped immediately and the gain should be reduced. In addition, when the backlash of the transmission mechanism is large, the inertia identification may fail. Before performing the offline inertia identification, first comfirm the following:

1) The motor movable process should meet the requirement bellowing

There is a movable stroke of more than 1 turn in the forward and reverse directions between the mechanical limit switches: Before performing the offline inertia identification, please make sure that a limit switch is installed on the machine and sure the motor has a movable stroke of more than 1 turn in the forward and reverse directions to prevent overtravel during the inertia identification process and cause an accident!

2) Estimated load inertia ratio P00.05 value

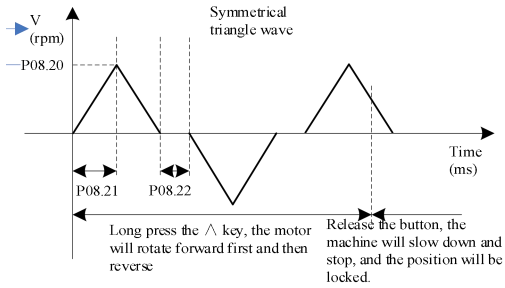
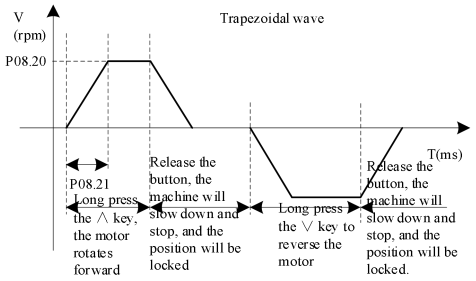
If P00.05 is the default value (1.00) and the actual load inertia ratio is greater than 30.00, the motor may move slowly causing identification failure. In this case, the following two measures can be taken:

a) The preset P00.05 is a larger initial value: it is recommended to use 5.00 times as the starting value for the preset value, and gradually increase it until the panel display value is updated during the identification process.

b) Appropriately increase the drive rigidity level (P00.04) so that the actual motor speed can reach the maximum speed of inertia identification (P08.20).

Inertia identification is divided into positive and negative triangle wave mode and JOG mode, and the motor rotation needs to be realized through the host computer function code.

Table 6.1 Inertia identification method

Project	Positive and negative triangle wave form (P08.23=0)	JOG mode (P08.23=1)
Instruction form		
Max speed	P08.20	P08.20
Acceleration and deceleration time	P08.21	P08.21
Interval time	P08.22	The time interval between two key operations
Motor rotation times	View P08.24	Human Control
Button Description	Long press the inertia identification forward button: the motor first rotates forward and then reverses Long press the inertia identification reverse button: the motor will first reverse and then forward Release the button: stop at zero speed and keep the position locked	Long press the inertia identification forward button: the motor runs forward Long press the inertia identification reverse button: the motor reverses Release the button: zero speed stop, keep the position locked
Applicable occasions	When the motor stroke is short	The motor has a long stroke and can be manually controlled

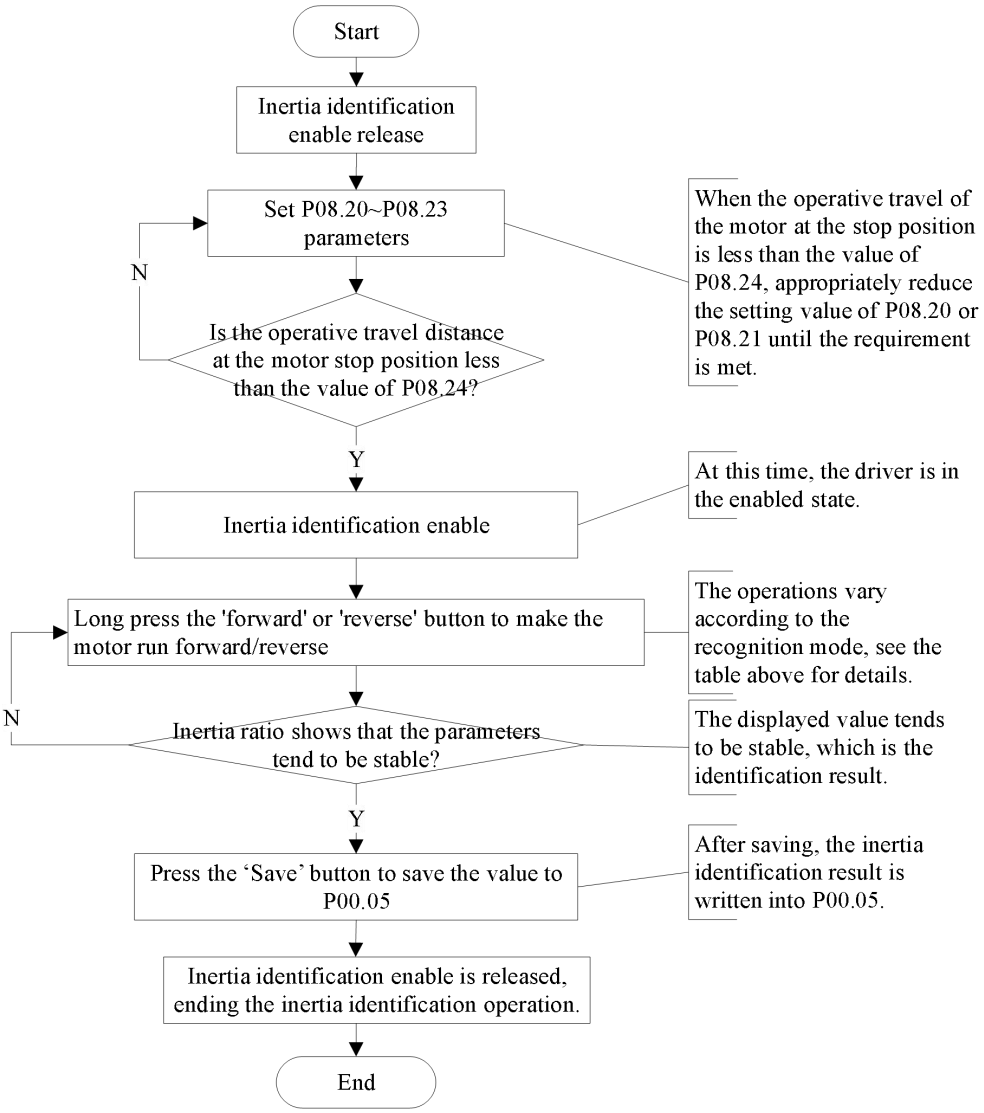


Figure 6.3 Inertia identification process

☆Related function code:

Function code	Name	Setting range	Minimum unit	Function	Factory setting	Effective time	Setting method
P08.20	Inertia identification maximum speed	200~1000	1rpm	Set the maximum speed command for inertia identification	500	Effective immediately	Shutdown setting
P08.21	Inertia identification acceleration and deceleration time	50~800	1 ms	Set the time for the motor to accelerate from 0rpm to the maximum speed of inertia identification (P08.20) during inertia identification.	100	Effective immediately	Shutdown setting

6.3 Gain adjustment

6.3.1 PI Parameters Adjustment

When the automatic gain can not achieve the expected result, you can manually minute adjust the gain. To optimize the result through more detailed adjustment. The servo system is composed of three control loops, from outside to inside, they are position loop, speed loop and current loop, the basic control block diagram is shown in figure 6.4.

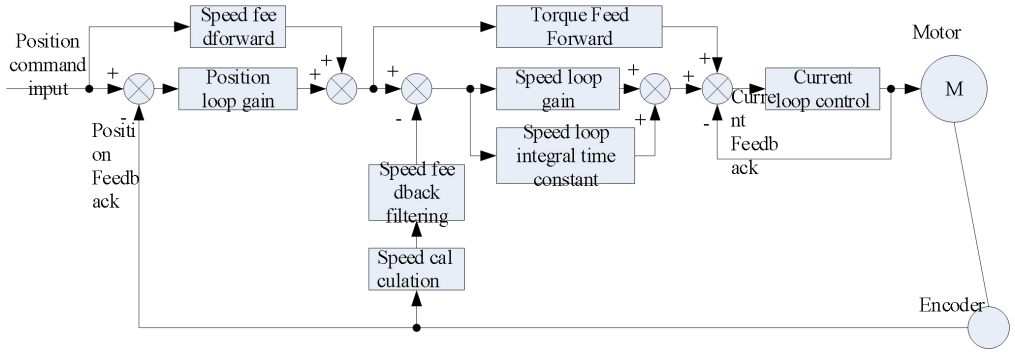
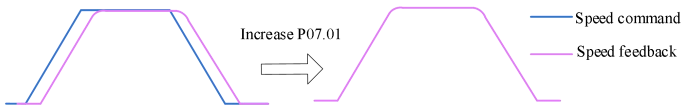
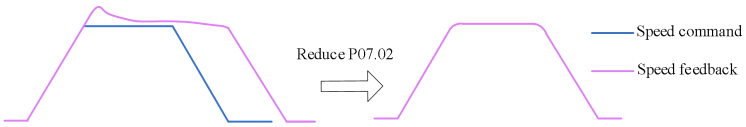
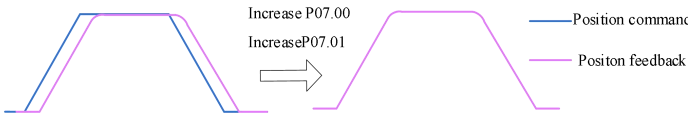


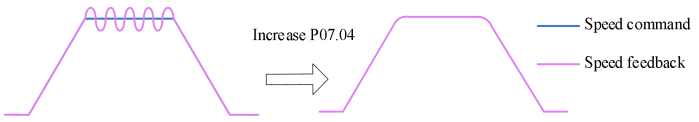
Figure 6.4 Manual Gain Basic Description Diagram

The default current loop gain of servo drive is already ensured the sufficient responsiveness, generally does not need to be adjusted, only position loop gain, speed loop gain and other auxiliary gain need to be adjusted. Therefore, when adjust gain in position control mode, in order to ensure system stability, while increasing the position loop gain, it is necessary to improve speed gain, and ensure that the response of position loop is lower than speed loop, the basic gain parameters adjustment method are as follows.

Table 5.2 Gain adjustment method

Steps	Functio n code	Name	Adjustment Description
1	P07.01	Speed loop gain	◆Parameters function: Determines the maximum frequency of changing speed commands that the speed loop can follow. Under the premise that the load inertia ratio (P00.05) is set correctly,it can be considered that the maximum follow frequencyof speed loop=P07.01

			 ◆Adjustment method: Within the range where noise and vibration do not occurs, increasing this parameter can accelerate positioning time and bring better speed stability and followability. If noise occurs, reduce the parameter setting value.
2	P07.02	Speed loop integral time constant	◆Parameters function: Eliminate speed loop deviation.  ◆Adjustment method: It is recommend to take the value from the following relationship: 500 ≤P07.01×P07.02 ≤ 1000 For example:when the speed loop gain P07.01=40.0Hz, the speed loop integrated time constant should meet: 12.50ms ≤P07.02 ≤ 25.00ms。 Decreasing the set value can strengthen the integral effect and speed up the positioning time, but a set value that is too small can easily cause mechanical vibration. If the setting value is too high, the speed loop deviation will never be returned to zero. When P07.02=512.00ms，integral is invalid.
3	P07.00	Position loop gain	◆Parameters function: Determines the maximum frequency of position instruction that the position loop can follow. Position loop maximum following frequency = P07.00  ◆Adjustment method: In order to ensure system stability, it should be ensured that the maximum following frequency of speed loop is 3-5 times the maximum following frequency of position loop. Therefore: $3 \le \frac{2\pi \cdot \mathrm{P07.01}}{\mathrm{P07.00}} \le 5$ For example, when the speed loop gain: P07.01=40HZ, positon loop gain should meet: 50.2Hz ≤ P07.00≤ 83.7Hz。 According the position time to adjust, increasing this parameter can accelerate positioning time and improve the ability of motor to resist external disturbance when it is stationary. If the setting value is too high may causing systemunstable and occurs vibration.
4	P07.04	Torque instructio n filter time	◆Parameters function: Eliminate high-frequency noise and suppress mechanical resonance.

		constant	 ◆Adjustment method: It should be ensured that the cutoff frequency of the torque instruction filter is higher than 4 times highest following frequency of speed loop, therefore: $$\frac{1000}{2\pi \cdot \text{P07.04}} \geq (\text{P07.01}) \cdot 4$$ For example, when the speed loop gain P07.01=40.0Hz, the torque instruction filter c=time constant should meet: P07.04 ≤ 1.00ms。 When vibration occurs when P07.01 is increased, the vibaration can be suppressed by P07.04; If the setting value is too high, it will cause the responsivemess decrease of current loop. The vibration in motor shutdown status need to be suppressed, you can try to increase P07.01, decrese P07.04; The vibration of the motor is too large when it is stopped. You can try to reduce the setting value of P07.04.

6.3.2 Speed Feedforward Gain

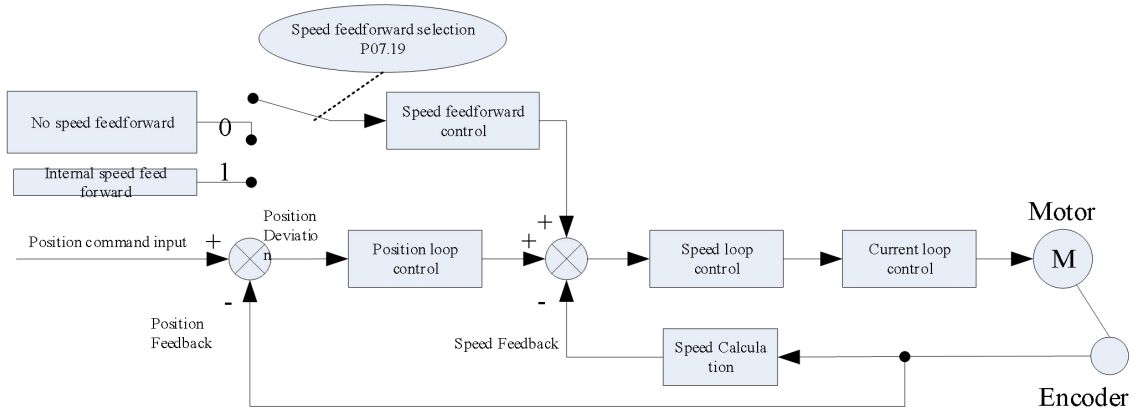


Figure 6.5 Speed feedforward control operation diagram

The velocity forward is only applicable to position control mode. Using velocity feedforward function can improve response of speed command, reduce the position deviation at a fixed speed.

The operation steps of speed feedforward:

a）Setting speed feedforward signal source

Set P07.19 (Speed feedforward control selection) to a non-0 value, the speed feedforward function

takes effect and the corresponding signal source is selected.

Function code	Name	Setting value	Remark
P07.19	Velocity feedforward control selection	0- No speed feedforward	-
		1-Internal speed feedforward	Use the speed information corresponding to the position command (encoder unit) as the source of the speed feedforward signal.
		4- EtherCTASpeed feedforward EtherCTA	Use the 60B1 input from the master station as speed feedforward.

b）Set the speed feedforward parameters

Including the speed feedforward gain (P07.20) and speed feedforward filter time constant (P07.21).

Table 6.3 Speed feedforward adjustment

Function code	Name	Adjustment description
P07.20	Speed feefforward gain	 ◆Parameters function: Increasing P07.20 can improve response, but may generate speed overshoot during acceleration and deceleration;
P07.21	Speed feedforward filter time constant	Reducing P07.21, can suppress the speed overshoot occur in acceleration and deceleration; Increasing P07.21 can suppress the noise caused by the position command update cycle being longer than the drive control cycle, the pulse frequency of the position command being uneven, and suppressing the jitter of the positioning completion signal; ◆Adjustment method: First, setting P07.21 to a fixed value; then gradually increase P07.20 from zero start, until the speed feedforward takes effect at a certain setting value. When adjusting, P07.20 and P07.21 should be adjusted repeatedly the to find a setting with good balance.

6.4 Torque feedforward

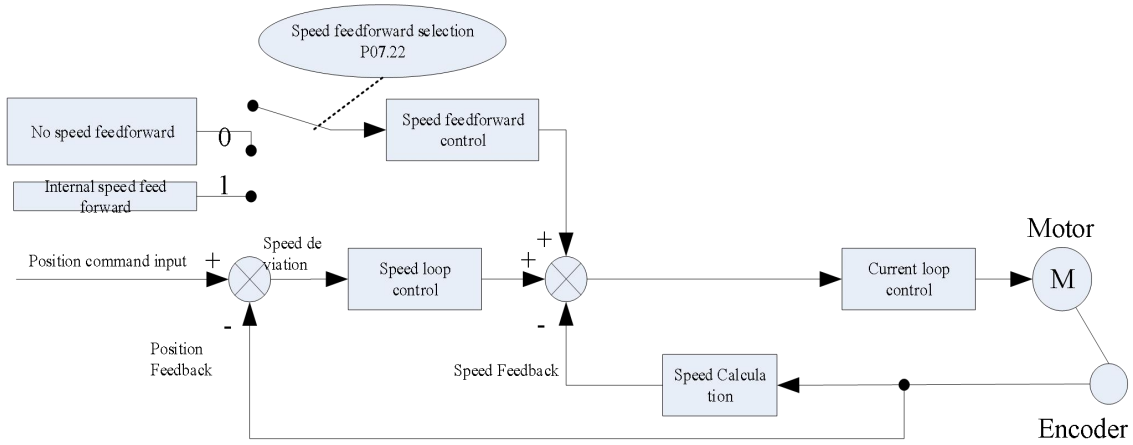


Figure 6.6 Torque feedforward control operation diagram

Only applicable to non-torque control mode situations. The position control mode uses torque feedforward, which can improve the torque command response and reduce the position deviation at fixed acceleration and deceleration; the speed control mode uses torque feedforward, which can improve the torque command response and reduce the speed deviation at fixed speed.

Torque feedforward function operation steps:

a）Setting torque feeforward signal source

Set P07.22 (Torque feedforward control selection) to a non-0 value, the torque feedforward function takes effect and the corresponding signal source is selected.

Function code	Name	Setting value	Remark
P07.22	Torque feedforward control selection	0- No torque feedforward	-
		1-Internal torque feedforward	Using the acceleration information correspondingto speed instruction as the source of the torque feedforward signal. In position control mode, the speed instructioncomes from the output of the position controller.

b）Setting torque feedforward parameters

Function code	Name	Adjustment description
P07.23	Torque feedforward gain	◆Parameters function: Increasing the P07.23 can improve the response, but may occurs overshoot when accelerate and decelerate； Reducing the P07.24 can suppress the overshoot occurs in accelerate and decelerate; Increasing P07.24 can suppress the noise; ◆Adjustment method: When adjusting, first keep P07.24 as the default value; then gradually increase the setting value of P07.23 from 0 until the torque feedforward takes effect at a certain setting value. When adjusting, P07.23 and P07.24 should be adjusted repeatedly to find a setting with good balance.
P07.24	Torque feedforwardfilter time constant	

6.5 Command filter adjustment

6.5.1 Position command smoothing filter

Setting the time constant of the once delay filter for the position command.

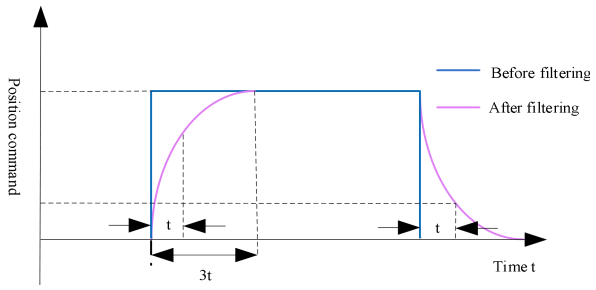


Figure 6.7 Schematic of the position command smoothing filter

☆Related function code:

Function code		Name	Setting range	Minimum unit	Factory setting	Effective time	Category	Mode
P04	03	Position command smoothing filter	0～65535	0.1ms	0	Effective immediately	Shutdown setting	P

6.5.2 Position command FIR filter

Setting the time constant of FIR filter for position command.

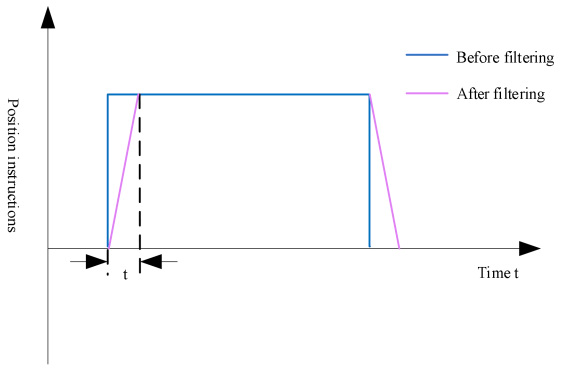


Figure 6.8 Schematic of the position command FIR filter

Function code		Name	Setting range	Minimum unit	Factory setting	Effective time	Category	Mode
P04	04	Position command FIR filter	0.～1280	0.1ms	0	Effective immediately	Shutdown setting	P

6.6 Adjustment parameters in different mode

Parameters adjustment under different control mode must follow the consequence of “Inertia identification” ==> “Automatic Gain Adjustment” ==> “Manual Gain Adjustment”.

6.6.1 Adjustment parameters in position mode

1）Through inertia identification, obtain load inertia ratio P00.05:

2）The gain parameter in position mode:

①The first gain:

Function code	Name	Function	Default value
P07.00	Position loop gain	Set position loop proportional gain	48.0Hz
P07.01	Speed loop gain	Set speed loop proportional gain	50.0Hz
P07.02	Speed loop integral time constant	Set the integral time constant of the speed loop	12.00ms
P07.04	Torque command filter time constant	Set the torque command filter time constant	1.26ms

②The second gain:

Function code	Name	Function	Default value
P07.05	Set the integral time constant of the speed loop	Set position loop proportional gain	38.0Hz
P07.06	The second speed loop gain	Set speed loop proportional gain	18.0Hz
P07.07	The second speed loop integral time constant	Set the integral time constant of the speed loop	512.00ms
P07.09	The second torque command filter time constant	Set the torque command filter time constant	1.26ms
P07.10	DI function GAIN-SWITCH switching action selection	Set the GAIN-SWITCH switching action selection	0
P07.11	Gain switching mode	Set the gain switching condition	0
P07.12	Gain switching delay	Set the delay time for gain switching	5.0ms
P07.13	Gain switching level	Sets the level at which gain switches	50
P07.14	Gain switching hysteresis	Set the hysteresis of gain switching	30
P07.15	Position gain switching time	Set the switching time of the position loop gain	3.0ms

Public Gain:

Function code	Name	Function	Default value
P07.03	Speed feedback filtering	Set the speed feedback filter time	0.00ms
P07.16	Pseudo-differential feedforward control coefficient	Setting the coefficients of the PDFF controller	100.0%
P07.20	Speed feed forward gain	Set the speed feedforward gain	0.0%
P07.21	Speed feedforward filter time constant	Set the filter time constant of the speed feedforward signal	0.50ms

P07.23	Torque feed forward gain	Setting the torque feedforward gain	0.0%
P07.24	Torque feedforward filter time constant	Set the filter time constant of the torque feedforward signal	0.50ms

3) By setting the rigidity level of P00.04 and automatic adjustment gain, obtain the first gain (or second gain). If the rigid connection equipment of the lead screw and rack encounters whistling when the rigidity level is increasing, can open the instantaneous speed observer (P08.39), in order to improve rigidity level.

4）Mannually fine-tune the following gains, if there are problems such as mechanical whistling, mechanical shaking, and tip jitter, you can use notch and model compensation control to optimize:

Function code	Name	Function
P07.00	Position loop gain	Set the position loop proportional gain
P07.01	Speed loop gain	Set the speed loop proportional gain
P07.02	Speed loop integral time constant	Set the integral time constant of the speed loop
P07.04	Torque command filter time constant	Set the torque command filter time constant
P07.20	Speed feedforward gain	Set the speed feedforward gain
P07.21	Speed feedforward filter time	Set the speed feedforward command filter

6.6.2 Parameters adjustment in the speed mode

The parameters adjustment in speed control mode is same as that in position mode, except for the position loop gain (P07.00, P07.05), please adjust according to 5.6.1 “parameters adjustment in position mode”.

6.6.3 Parameters adjustment in torque control mode

parameters adjustment in torque control mode:

When the actual speed reaches the speed limit value, the adjustment method is the same as 6.7.2 "Parameter adjustment in speed mode";

When the actual speed do not reaches the speed limit value, except for speed loop gain(P07.01, P07.06) and speed loop integral time constant (P07.02, P07.07), the adjustment method is same as 6.5.2 “Parameter adjustment in speed mode”.

6.7 Mechanical resonance suppression

Mechanical system has a certain resonance frequency. When the servo gain is increased, resonance may occur near the mechanical resonance frequency, causing the gain to be unable to continue to increase. There are two ways to suppress mechanical resonance:

1）Torque command filter (P07.04，P07.09)

By setting filter time constant, the torque command is attenuated in high-frequency above the cutoff frequency to achieve the purpose of suppressing mechanical resonance.

2）Notch filter:

The notch filter can achieve the purpose of suppressing mechanical resonance by reducing the gain at specific frequencies. After correctly set notch filter, the vibration can get effectively suppress, and try to increase servo gain continuously. The primciple of Notch filter is as shown below.

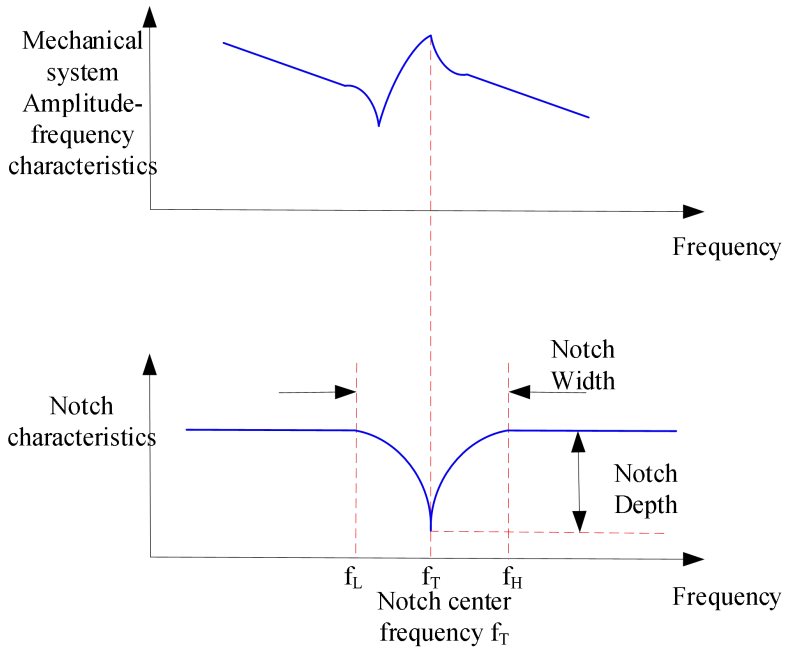


Figure 6.9 Notch filter principle

filter frequency, width level and depth level. This notch is a manual notch, and each parameter is set

manually by the user.

Notch filter width and depth

The notch width is used to express the ratio of the notch width to notch center frequency:

$$\text{The notch width} = \frac{f_H - f_L}{f_T}$$

Among:

f_T：Notch center frequency, that is to say, mechanical resonance frequency

f_H-f_L：f_H-f_L：The notch width represents the frequency bandwidth with an amplitude attenuation rate of -3dB relative to the notch center frequency.

The notch depth level represents the relationship of the ratio of the input to output at center frequency. When the notch depth level is 0, the input is completely be suppressed at center frequency; When the notch depth level is 100, the input is completely pass at center frequency. Therefore, the setting level of notch depth is smaller, the depth of notch is more deeper,the suppression to mechanical resonance is more stronger, but this may cause unstable system, should be attention when using.

Table 6.4 parameters setting of notch

Notch width	Bandwidth/Center frequency		Notch depth	Ratio of input to output	[dB] Gain Represent
0	0.5		0	0	-∞
1	0.59		1	0.01	-40
2	0.71		2	0.02	-34
3	0.84		3	0.03	-30.5
4	1		4	0.04	-28
5	1.19		5	0.05	-26
6	1.41		6	0.06	-24.4
7	1.68		7	0.07	-23.1
8	2		8	0.08	-21.9
			9	0.09	-20.9
			10	0.1	-20
			15	0.15	-16.5
			20	0.2	-14
			25	0.25	-12
			30	0.3	-10.5
			35	0.35	-9.1

			40	0.4	-8
			45	0.45	-6.9
			50	0.5	-6
			60	0.6	-4.4
			70	0.7	-3.1
			80	0.8	-1.9
			90	0.9	-0.9
			100	1	0

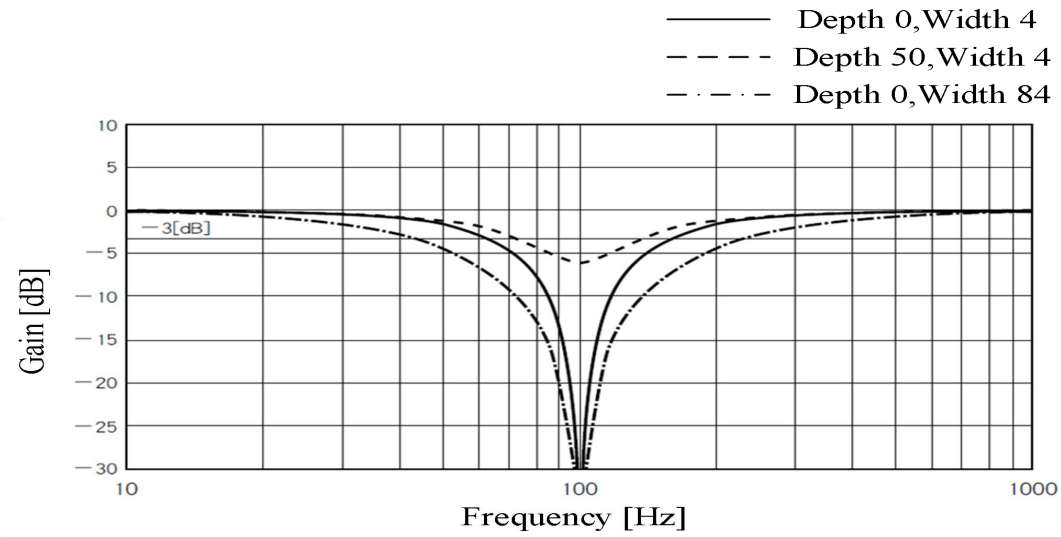


Figure 6.9 Notch frequency characteristic

Notch using steps

① Analyze resonance frequency;

When using a manual notch, you need to set the frenquency of the notch to the actual resonant frequency that occurs. The method to obtain resonance frequency: By setting P08.00=3, the resonant frequency is automatically tested when the servo is running, and the test results are saved in P08.01. After the test is completed, be sure to set P08.00 to 0;

② Enter the of Enter the resonant frequency obtained in step ① into the notch parameters of the selected group, and enter the width level and depth level of the notch group;

③ If the resonance is suppressed, it means the notch is effective, you can continue to adjust the gain. After the gain is increased, if occours new resonance, you can repeat steps ① ~ ②;

④ If the vibration can not be eliminated in a long time, please close the servo function in time;

☆Related function code:

Function code		Name	Setting range	Minimum unit	Factory setting	Effective time	Category	Mode
P08	00	Adaptive filter mode	0~5	1	0	Effective immediately	Running setting	PST
P08	01	Resonance frequency	-	1Hz	0	N/A	Display parampara	PST
P08	02	1st notch filter frequency (manual)	10~4000	1Hz	4000	Effective immediately	Running setting	PST
P08	03	1st notch filter width	0~8	1	2	Effective immediately	Running setting	PST
P08	04	1st notch depth	0~100	1	50	Effective immediately	Running setting	PST
P08	05	2nd notch filter frequency (manual)	10~4000	1Hz	4000	Effective immediately	Running setting	PST
P08	06	2nd notch filter width	0~8	1	2	Effective immediately	Running setting	PST
P08	07	2nd notch depth	0~100	1	50	Effective immediately	Running setting	PST

Chapter 7 Alarm and Processing

Servo drive alarm level is divided into two levels

Table 7-1 Alarm level

Alarm level	Name	Decription
Level 1	Fault	The servo drive has a serious alarm and cannot work normally. It needs to be shut down to cheak. DO terminal outputs ALM singal.
Level 2	Warning	The servo drive warns that the device will not be damaged for the time being, but if it is not handled correctly in time, it may cause a high-level fault output. DO terminal outputs Warn Singal.；

7.1 Fault diagnosis and treatment

Faults can be divided into:

a) Non-resettable NO.1 fault;

b) Resettable NO.1 fault;

c) Resettable NO.2 fault.

Among them, “Resettable” means that after troubleshooting, P11-01 can be set to 1or the DI function can be configured FunIN.2：ALM-RST alarm reset to clear the servo fault state.

“Non-resettable” means that after the fault is handled, drive needs to be re-powered.

The NO.1 and NO.2 faults have different stop modes. The NO.1 fault is free to stop and keeps running freely. The NO.2 fault stop mode is set by P00-12.

The reset method of NO.1, NO2 resettable fault: first turn off the servo enable signal (S-ON is set to OFF), then set P11-01=1or use DI function 2.

7.1.1 Non-resettable NO.1 fault

Table 7-2 Non-resettable NO.1 fault

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
1	System parameter exception	Abnormal data of servo internal parameters	No.1	NO
2	Product model selection failure	Invalid drive model	No.1	NO
3	Motor model selection failure	P01.00 Function code Set an invalid motor code	No.1	NO
4	Fault in parameter storage	1. Parameter storage device failure 2. Parameters read and write too frequently 3. Control power supply instability Drive failure	No.1	NO
5	FPGA fault	1. FPGA initialization exception 2. FPGA logic version is abnormal 3. FPGA detected an exception	No.1	NO
6	Encoder matching fault	In absolute value system mode (P00.06 ≠0), an incremental or single-turn absolute value encoder motor is used	No.1	NO
7	Control power supply undervoltage	Control circuit power supply undervoltage	No.1	NO
9	Overcurrent A	1. The drive or motor parameters are incorrect 2. UVW phase-to-phase short circuit 3. Motor burned out 4. Motor is shorted to ground 5. Drive failure	NO.1	NO
10	Overcurrent B	1. Servo motor wiring is not normal 2. Software detects the overcurrent of power transistor	No.1	NO
11	Encoder disconnection	Incremental encoder A/B/Z three-phase line break	No.1	NO
13	Encoder check exception	1. The bus encoder data check is abnormal 2. the incremental encoder zero check is abnormal	No.1	NO
14	Motor initial angle detection abnormal	The encoder signal interferes when the incremental encoder is powered on	No.1	NO
15	Motor out of control	1. Motor UVW phase sequence setting error 2. UVW wiring error 3. Initial position or encoder parameter setting error	No.1	NO
16	Current sampling fault	The current A/D sampling circuit is faulty	No.1	NO
18	Motor data verification failure	1. No motor parameters are written into the encoder EEPROM 2. Motor parameter verification error	No.1	NO

7.1.2 Resettable NO.1 fault

Table 7-3 Resettable NO.1 fault

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
20	Overvoltage	The main circuit DC voltage abnormally high.	No.1	Yes
21	Undervoltage	The main circuit DC voltage low.	No.1	Yes
22	Overspeed	1. Speed command exceeds the maximum speed value 2. UVW phase sequence error 3. Speed response too high, severely exceed adjustment range 4. Drive failure	No.1	Yes
25	Abnormal motor rotation during bootstrap charging	Abnormal motor rotation during bootstrap charging	No.1	可 Yes
27	DI terminal parameter setting fault	Different DI are assigned with same function repeatedly.	No.1	可 Yes
28	DO terminal parameter setting fault	Different DO are assigned with same output.	No.1	可 Yes
30	Reference position failure	When using the PTP function, P04.00=5 is not set.	No.1	可 Yes

7.1.3 Resettable NO.2 fault

Table 7-4 Resettable NO.2 fault

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
43	Position deviation is too large	In the servo ON state, the position deviation exceeds the position deviation value (P09.09)	No.2	Yes
46	Drive overload	1.The load operation exceeds the inverse time curve of the drive. 2.The UVW output may be lack of phase or phase sequence connected incorrectly.	No.2	Yes
47	Motor overload	1. The load operation exceeds the inverse time curve of the motor. 2.The UVW output may be lack of phase or phase sequence connected incorrectly.	N0.2	Yes
48	Motor stall	1. The mechanical position is stuck, causing the motor current to continuously increase abnormally. 2. Inconsistent response of dual-drive motors	N0.2	Yes

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
		with gantry structure.		
49	Electronic gear setting error	Electronic gear ratio exceeds specification range [0.001, 4000]	No.2	Yes
50	Heat sink (fan) overheating	Servo unit heat sink (fan) exceeds the set fault value	No.2	Yes
51	Encoder battery failure	No battery or battery voltage below 2.6V	No.2	Yes
52	Encoder multi-turn count error	Absolute encoder multi-turn count error	No.2	Yes
53	Encoder multi-turn count overflow	Absolute encoder multi-turn count overflow	No.2	Yes
54	Soft limit setting error	1. The position between positive/negative limit is too short 2. When the motor runs in the counterclockwise direction as the positive direction, the positive limit value is smaller than the negative limit value 3. When the motor runs in the clockwise direction as the positive direction, the negative limit value is smaller than the positive limit value	No.2	Yes
56	node_id setting error	Set the DIP switch or parameter P10.07 (node_id) to 0.	No.2	Yes

7.2 Reasons for warning and treatment

Table 7-5 Reasons for warning and treatment

Fault number Er_	Warning name	Cause of warning
81	Drive overload warning	Warning when 80% of the drive overload limit value is reached.
82	Motor overload warning	The warning before the motor is about to fail, the warning value is determined by P09-05.
83	Changed parameters need to be re-powered to take effect	Changed parameters that need to be re-powered.
84	Reset encoder warning	In enabled state, resets encoder when host computer sending commands.
86	Forward overtravel warning	Positive overtravel switch POT terminal is valid. The motor position exceeds the maximum soft limit.
87	Negative overtravel warning	Negative overtravel switch NOT terminal is valid.

Fault number Er_	Warning name	Cause of warning
		The motor position exceeds the soft limit minimum.
88	Divided pulse output setting fault	The number of encoder divided pulses does not meet set condition or range.
89	AI1\ AI2 zero drift too large	AI1\ AI2 zero drift too large.
90	External regenerative resistor overload	External regenerative resistor power is too small.
91	External regenerative resistor value is too small	The external regenerative resistor is less than the minimum requirement by the drive;Or parameter setting error.
94	DI emergency brake	External emergency brake E_STOP terminal trigger.
95	Absolute encoder battery low	Battery voltage is lower than 3.2V.
96	Homing timeout	1. Home switch failure. 2. Time limit to find home is too short. 3. The high speed for searching home signal is too small.
97	Mechanical home point offset error	1. When the homing mode parameter P16-09=6 or P16-09=8 or P16-09=14, the mechanical home offset parameter P16-14 is set to a value greater than 0. 2.When the homing mode parameter P16-09=7 or P16-09=9 or P16-09=15, the mechanical home offset parameter P16-14 is set to less than 0.
99	Multi-segment position reference position fault	1. In the multi-segment absolute position operation mode, the system does not perform the homing operation or the drive is not set to the absolute value system. 2. In the multi-segment incremental position operation mode, when the function code of P13.07 is set to 1, the system does not perform the zero return operation.
100	The homeing mode is incorrectly set	The homeing mode is set 0, Or the mode is not supported.

Note: The reset method of warning: set P11-01=1 or use DI function 2.

Chapter 8 Parameter List

8.1 Parameter group number

Parameter group number	Parameter group description
P00	Basic control parameters
P01	Servo motor parameters
P02	Digital input and output parameters
P04	Position control parameters
P05	Speed control parameters
P06	Torque control parameters
P07	Gain parameters
P08	Advanced adjustment parameters
P09	Fault and protection parameters
P10	Communication parameters
P11	Auxiliary function parameters
P12	Keyboard display parameters
P13	Multi-segment position function parameters
P14	Multi-speed function parameters
P16	Special function parameters
P17	Drive parameters
P18	Display parameters

8.2 Each group of parameters

P00 group: basic control parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P00	00	Control mode selection	0-position mode 1-speed mode 2-torque mode	1	0	Effective immediately	Setting after motor stop	PST
P00	01	Direction of rotation	0-CCW direction is the forward direction (counterclockwise) 1-CW direction is the forward direction (clockwise)	1	0	Re-power	Setting after motor stop	PST
P00	02	Pulse output positive direction definition	0-CCW direction is forward direction (The corresponding motor rotation direction when the pulse output OA leads the OB) 1- CW direction is forward direction(reverse mode, OA lagOB)）	1	0	Re-power	Setting after motor stop	PST
P00	04	Rigidity level setting	0~31	1	11	Effective immediately	Set when running	PST
P00	05	Inertia ratio	0~3000	0.01倍	100	Effective Immediately	Set when running	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P00	06	Absolute value system selection	0~2	1	0	Re-power	Setting after motor stop	PST
0-incremental position mode 1-Absolute position linear mode （If homing completed when the absolute value encoder is used, power on the encoder immediately. If the value of 6064 is not 0 after the encoder is powered on, set the absolute value mode to this mode.） 2- Absolute value position rotation mode （Mask multi-circle data overflow errors, In the multi-turn data -32768~32767 has been running cycle.）								
P00	07	System maximum speed	0~10000	1rpm	4500	Effective immediately	Setting after motor stop	PST
P00	10	Servo OFF stop mode	0- Freewheel stop, keep free running state 1- Stop at zero speed (deceleration time is controlled by P05.06), keep free running state.	1	1	Effective immediately	Setting after motor stop	PST
P00	11	Fault No.1 stop mode selection	0- Freewheel stop, keep free running state 1- Resaved	1	0	Effective immediately	Setting after motor stop	PST
P00	12	Fault No.2 stop mode selection	0- Freewheel stop, keep free running state 1- Stop at zero speed (deceleration time is controlled by P05.06), keep free running state	1	1	Effective immediately	Setting after motor stop	PST
P00	13	Stop mode when overtravel	0-free running stop. 1- Decelerate the motor with the set torque of the emergency stop torque as the maximum value, and then enter the servo lock state. 2- Decelerate the motor with the set torque of the emergency stop torque as the maximum	1	1	Effective immediately	Setting after motor stop	PST
P00	14	Brake output ON to command receiving delay	0～10000	1ms	200	Effective immediately	Set when running	PST
P00	15	In the rotation state, No.2 fault occurs, or the servo enable is OFF, the brake output OFF delay; In the static state, brake output OFF to the motor not energized delay.	10～10000	1ms	200	Effective immediately	Set when running	PST
P00	16	Rotating state, the speed threshold when the brake output is OFF	0～1000	1rpm	50	Effective immediately	Set when running	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P00	17	NO.1 fault occurs in rotating state delay between servo enable OFF and brake output OFF	0～10000	1ms	500	Effective immediately	Set when running	PST
P00	18	Energy consumption resistance setting	0-Use built-in power consumption resistor 1-Use external power-consuming resistors and cool naturally 2-Use external power-consuming resistors and force air-cooled 3- No energy consumption resistor, relying on capacitance Absorption (The brake tube is always closed at this time)	1	1	Effective immediately	Setting after motor stop	PST
P00	19	External resistor power capacity	1～65535	1W	Model parameters	Effective immediately	Setting after motor stop	PST
P00	20	External resistance value	1～1000	1Ω	Model parameters	Effective immediately	Setting after motor stop	PST
P00	21	External resistance heating time constant	1000~65535	1ms	Model parameters	Effective immediately	Setting after motor stop	PST
P00	22	Energy consumption braking start voltage	0~1000	1V	Model parameters	Effective immediately	Set when running	PST
P00	37	Pulse increment threshold	0~200	1	1	Effective immediately	Set when running	PST
P00	38	Continuous pulseless reception cycle number	1~200	1	3	Effective immediately	Set when running	PST

P01 group: Servo motor parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P01	00	Motor model code	0~65535	1	101	Re-power	Setting after motor stop	PST
P01	01	Motor power line phase sequence direction	0：CCW (The opposite electromotive force U is ahead of V); 0：CCW (The opposite electromotive force V is ahead of U);	1	0	Re-power	Setting after motor stop	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P01	02	Rated voltage	1-1000	1	48	Re-power	Setting after motor stop	PST
P01	03	Rated power	0~65535	0.01Kw		Re-power	Setting after motor stop	PST
P01	04	Rated current	1～10000	0.01A		Re-power	Setting after motor stop	PST
P01	05	Rated torque	0~65535	0.01Nm		Re-power	Setting after motor stop	PST
P01	08	Max speed	0~9000	1rpm		Re-power	Setting after motor stop	PST
P01	09	Rotor inertia	0~10000	0.01 $kgcm^2$		Re-power	Setting after motor stop	PST
P01	10	Number of pole pairs of Permanent magnet synchronous motor	1~50	1 pole pair		Re-power	Setting after motor stop	PST
P01	11	stator resistance Rs	1~65535	0.001 Ω		Re-power	Setting after motor stop	PST
P01	12	Q-axis inductance Lq	1~65535	0.01mH		Re-power	Setting after motor stop	PST
P01	13	D-axis inductance Ld	1~65535	0.01mH		Re-power	Setting after motor stop	PST
P01	14	Back EMF	1~65535	0.01 mV/rpm		Re-power	Setting after motor stop	PST
P01	15	Torque factor	1~65535	0.001 Nm/A		Re-power	Setting after motor stop	PST
P01	18	Encoder selection	0: 2500 lines encoder 1: 17bit incremental encoder 2: 17bit absolute encoder 3: 23bit incremental encoder 4: 23bit absolute encoder			Re-power	Setting after motor stop	PST
P01	20	Encoder resolution	1~1073741824			Re-power	Setting after motor stop	PST
P01	22	Z corresponding Electrical angle	0~3600	0.1°		Re-power	Setting after motor stop	PST
P01	23	U rising edge corresponding electrical angle	0~3600	0.1°		Re-power	Setting after motor stop	PST
P01	24	FPGA upload motor model	0～65535			Effective immediately	Only read parameter	PST

P02 group: digital terminal input and output parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P02	00	FunINL signal unassigned state (HEX)	0～0xFFFF Bit0- corresponds to FunIN.1; Bit1- corresponds to FunIN.2; ..... Bit15-corresponds to FunIN.16	1	0	Re-power	Set when running	PST
P02	01	DI1 terminal function selection	Input function code: 0, 1-45 0: no definition 1~45: FunIN.1 to 45 (refer to the DIDO basic function coding table)	1	1	Effective immediately	Set when running	PST
P02	02	DI2 terminal function selection	Input function code: 0, 1-45 0: no definition 1~45: FunIN.1 to 45 (refer to the DIDO basic function coding table)	1	2	Effective immediately	Set when running	PST
P02	03	DI3 terminal function selection	Input function code: 0, 1-45 0: no definition 1~45: FunIN.1 to 45 (refer to the DIDO basic function coding table)	1	10	Effective immediately	Set when running	PST
P02	10	FunINH signal unassigned state (HEX)	0~0xFFFF Bit0- corresponds to FunIN.17; Bit1- corresponds to FunIN.18; ..... Bit15-corresponds to FunIN.32	1	0	Re-power	Setting after motor stop	PST
P02	11	DI1 terminal logic selection	Input polarity: 0-4 0-low level effective 1-high level effective 2-rising edge is valid 3-falling edge is valid 4- rising and falling edges are valid	1	0	Effective immediately	Set when running	PST
P02	12	DI2 terminal logic selection	Input polarity: 0-4 0-low level effective 1-high level effective 2-rising edge is valid 3-falling edge is valid 4- rising and falling edges are valid	1	0	Effective immediately	Set when running	PST
P02	13	DI3 terminal logic selection	Input polarity: 0-4 0-low level effective 1-high level effective 2-rising edge is valid 3-falling edge is valid 4- rising and falling edges are valid	1	0	Effective immediately	Set when running	PST
P02	21	DO1 terminal function selection	Output code：1～25 0：no definition 1～25：FunOUT.1～25 refer to DIDO function selection code definition	1	1	Effective immediately	Setting after motor stop	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P02	22	DO2 terminal function selection	The DO can only be set to FunOUT11 (Lock release signal output);	1	11	Effective immediately	Setting after motor stop	PST
P02	31	DO1 terminal logic level selection	Output polarity inversion setting: 0-1 0: conduct when active (normally open); 1: not conduct when active (normally closed);	1	0	Effective immediately	Setting after motor stop	PST
P02	32	DO2 terminal logic level selection	Output polarity inversion setting: 0-1 0: conduct when active (normally open) 1: not conduct when active (normally closed)	1	0	Effective immediately	Setting after motor stop	PST

P04 group: position control parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P04	00	Main position command A source	0-low speed pulse command 1- high speed pulse command 2-step amount given 4-multi-segment position command given 5-given through communication	1	0	Effective immediately	Setting after motor stop	P
P04	02	Step amount	-9999～9999	1Unit	50	Effective immediately	Setting after motor stop	P
P04	03	Position command smoothing filter	0 ～65535	0.1ms	0	Effective immediately	Setting after motor stop	P
P04	04	Position command FIR filter	0～1280	0.1ms	0	Effective immediately	Setting after motor stop	P
P04	05	Number of instruction unit required for one round (32-bit) (Only PTP)	16～1073741824	1Unit/Turn	0	Re-power	Setting after motor stop	P
P04	07	Electronic gear 1 numerator (32 bit)	1～1073741824	1	Motor resolution	Effective immediately	Set when running	P
P04	09	Electronic gear 1 denominator (32 bit)	1～1073741824	1	10000	Effective immediately	Set when running	P
P04	11	Electronic gear 2 numerator (32 bit)	1～1073741824	1	Motor resolution	Effective immediately	Set when running	P
P04	13	Electronic gear 2 denominator (32 bit)	1～1073741824	1	10000	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P04	21	Pulse shape	0-pulse + direction, positive logic (Defaults) 1-direction + pulse, negative logic 2-A phase + B phase orthogonal pulse, positive logic 3-A phase + B phase orthogonal pulse, negative logic 4- CCW+CW, positive logic 5- CCW+CW, negative logic	1	0	Re-power	Setting after motor stop	P
P04	22	Position deviation clear function	0- Clear position deviation pulse when fault occurs or servo OFF 1- Clear the position deviation pulse only when a fault occurs 2- Cleared by DI input function (PERR-CLR)	1	0	Effective immediately	Setting after motor stop	P
P04	23	Positioning complete (COIN) output	0-Output when the absolute position deviation value is smaller than positioning completion range. 1- Output when the absolute position deviation value is smaller than positioning completion range and the filtered position command is 0. 2- Output when the absolute position deviation value is smaller than positioning completion range and position command is 0.	1	0	Effective immediately	Setting after motor stop	P
P04	24	Positioning completion range	1～65535	1P	2500 line motor: 7P 17-bit motor: 100P 23-bit motor: 1000P	Effective immediately	Setting after motor stop	P
P04	25	Positioning close range	1～65535	1P	65535	Effective immediately	Setting after motor stop	P

P05 group: speed control parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P05	00	Main speed command A source	0-digit value(P0503) 1- reserved 2- reserved	1	0	Effective immediately	Setting after motor stop	S
P05	01	Auxiliary speed command B source	0-digit value(P0503) 1- reserved 2- reserved 3-multi-speed command	1	3	Effective immediately	Setting after motor stop	S
P05	02	Speed command selection	0-main speed command A source 2-auxiliary speed command B source 3-A/B switching	1	0	Effective immediately	Setting after motor stop	S
P05	03	Speed command keyboard setting	-9000～9000	1rpm	200	Effective immediately	Setting when running	S
P05	04	Jog speed setting	0～9000	1rpm	200	Effective immediately	Setting when running	S
P05	05	Speed command acceleration time	0～10000	1ms	50	Effective immediately	Reserved	S
P05	06	Speed command deceleration time	0～10000	1ms	50	Effective immediately	Reserved	S
P05	07	Speed command limit selection	0-forward P05.08, reverse P05.09, internal limit (default) 1- reserved 2- reserved	1	0	Effective immediately	Setting when running	S

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P05	08	Forward speed limit	0～9000	1rpm	6000	Effective immediately	Setting when running	S
P05	09	Backward speed limit	0～9000	1rpm	6000	Effective immediately	Setting when running	S
P05	14	Speed direction selection	0-direction unchanged 1-direction reversal 2-direction is determined by DI function 25 3-direction is determined by DI function 40/41	1	2	Effective immediately	Setting after motor stop	S
P05	15	Zero fixed speed fixed value	0～6000	1rpm	10	Effective immediately	Setting when running	S
P05	16	Speed threshold of motor running signal output	0～1000	1rpm	20	Effective immediately	Setting when running	PS
P05	17	Speed uniform signal width	0～100	1rpm	10	Effective immediately	Setting when running	PS
P05	18	Speed reaches the specified value	0～6000	1rpm	1000	Effective immediately	Setting when running	PST
P05	20	Zero speed judgment threshold	0～6000	1rpm	10	N/A	Setting when running	PST

P06 group: torque control parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P06	00	Main torque command A source	0-Digit given (P06-05) 1-Reserved	1	0	Effective immediately	Setting after motor stop	T
P06	02	Torque command selection	0-Main torque command A source 1-Auxiliary speed command B source 2- Command A+B source 3-A/B switching	1	0	Effective immediately	Setting after motor stop	T
P06	04	Torque command filter time in torque mode	0～65535	0.01ms	0	Effective immediately	Set when running	T
P06	05	Torque command keyboard setting	--3000～3000(based on rated motor torque)	0.1%	0	Effective immediately	Set when running	T
P06	06	Torque limit source	0-Positive and negative internal torque limit(default) 1-Positive and reverse external torque limit (selected with P_CL, N_CL)	1	0	Effective immediately	Set when running	PST
P06	08	Forward internal torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	09	Backward internal torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	10	Forward side external torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	11	Backward side external torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	13	Speed limit source selection for torque control	0-Internal speed limit (P06.15, P06.16 setting) 1-Reserved	1	0	Effective immediately	Set when running	T

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P06	15	Positive speed limit During torque control	0～9000	1	3000	Effective immediately	Set when running	T
P06	16	Negetive speed limit During torque control	0～9000	1	3000	Effective immediately	Set when running	T
P06	17	Torque arrival command reference value	0～5000 (1000 corresponds to the rated torque of the motor)	0.1%	0	Effective immediately	Set when running	PST
P06	18	Torque arrival effective offset threshold	0～5000 (1000 corresponds to the rated torque of the motor)	0.1%	200	Effective immediately	Set when running	PST
P06	19	Torque arrival ineffective offset threshold	0～5000 (1000 corresponds to the rated torque of the motor)	0.1%	100	Effective immediately	Set when running	PST
P06	20	Speed limit window in torque mode	1~900	1ms	50	Effective immediately	Set when running	PST
P06	21	Multi-segment torque command 1	-3000～3000 (based on motor rated torque）	0.1%	0	Effective immediately	Set when running	T
P06	22	Multi-segment torque command 2	-3000～3000(based on motor rated torque）	0.1%	0	Effective immediately	Set when running	T
P06	23	Multi-segment torque command 3	-3000～3000 (based on motor rated torque）	0.1%	0	Effective immediately	Set when running	T

P07 group: gain parameter

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P07	00	Position loop gain 1	10～20000	0.1HZ	320	Effective immediately	Set when running	P
P07	01	Speed loop gain 1	10～20000	0.1HZ	180	Effective immediately	Set when running	PS
P07	02	Speed loop integral time 1	15～512	0.01ms	3100	Effective immediately	Set when running	PS
P07	03	Speed detection filter 1	0～200	0.01ms	20	Effective immediately	Set when running	PST
P07	04	Torque command filtering 1	0～10000	0.01ms	126	Effective immediately	Set when running	PST
P07	05	Position loop gain 2	10～20000	0.1HZ	380	Effective immediately	Set when running	P
P07	06	Speed loop gain 2	10～20000	0.1HZ	180	Effective immediately	Set when running	PS
P07	07	Speed loop integral time 2	15～51200	0.01ms	51200	Effective immediately	Set when running	PS
P07	08	Speed detection filter 2	0～200	0.01ms	0	Effective immediately	Set when running	PST
P07	09	Torque command filtering 2	0～10000	0.01ms	126	Effective immediately	Set when running	PST
P07	10	DI function GAINSWITCH switching action selection	0-Speed loop regulator P (1) / PI (0) switch, gain is fixed to first group. 1-First gain (0), second gain (1) switching.	1	0	Effective immediately	Set when running	PS

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P07	11	Gain switching mode	0-First gain fixed 1-Second gain fixed 2-Using DI input (GAINSWITCH) 3-Torque command greater 4-Speed command change greater 5-Speed command greater 6-Position deviation greater (P) 7-Receive position command (P) 8-Positioning is not completed (P) 9-Actual speed is greater (P) 10-Receive position command and actual speed (P) 11-Speed loop controller with PDFF control (PS) 12-Reserved 13-Speed loop controller with improved PI control (PS)	1	0	Effective immediately	Set when running	PS
P07	12	Gain switching delay	0~10000	0.1ms	50	Effective immediately	Set when running	PS
P07	13	Gain switching level	0~20000（unit: according to the gain switching mode description）	1	50	Effective immediately	Set when running	PS
P07	14	Hysteresis time of gain switching	0~20000（unit: according to the gain switching mode description）	1	33	Effective immediately	Set when running	PS
P07	15	Position gain switching time	0~10000	0.1ms	33	Effective immediately	Set when running	PS
P07	16	Speed regulator PDFF coefficient	0~1000	0.1%	700	Effective immediately	Set when running	PS
P07	17	Improved speed PI control level	2~9	1	5	Effective immediately	Set when running	PS
P07	18	Anti-integration saturation coefficient	0~1000	0.001	820	Effective immediately	Set when running	PS
P07	19	Speed feedforward control selection	0-No speed feedforward 1-Internal speed feedforward	1	1	Effective immediately	Set when running	P
P07	20	Speed feedforward gain	0～1000	0.1%	0	Effective immediately	Set when running	P
P07	21	Speed feedforward filter time parameter	0～6400	0.01ms	50	Effective immediately	Set when running	P
P07	22	Torque feedforward selection	0-No torque feedforward 1-Internal torque feedforward	1	1	Effective immediately	Set when running	PS
P07	23	Torque feedforward gain	0～1000	0.1%	0	Effective immediately	Set when running	PS
P07	24	Torque feedforward filter time parameter	0～6400	0.01ms	50	Effective immediately	Set when running	PS

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P08	00	Adaptive filter mode	0~5	1	0	Effective immediately	Set when running	PST
P08	01	Resonance frequency	0~65535	1Hz	0	N/A	Display parameters	PST
P08	02	The 1st notch filter frequency	10~4000	1Hz	4000	Effective immediat	Set when running	PST

		(manual)				ely		
P08	03	The 1st notch filter width	0~8	1	8	Effective immediat ely	Set when running	PST
P08	04	The 1st notch filter depth	0~100	1	50	Effective immediat ely	Set when running	PST
P08	05	The 2nd notch filter frequency (manual)	10~4000	1Hz	4000	Effective immediat ely	Set when running	PST
P08	06	The 2nd notch filter width	0~8	1	8	Effective immediat ely	Set when running	PST
P08	07	The 2nd notch filter depth	0~100	1	50	Effective immediat ely	Set when running	PST
P08	15	Damping filter switch	0-OFF 1-ON	1	0	Effective immediat ely	Setting after motor stop	PS
P08	16	Damping filter frequency	10~2000	0.1Hz	2000	Effective immediat ely	Setting after motor stop	PS
P08	17	Damping filter selection	0- Damping filter A 1- Damping filter B	1	1	Effective immediat ely	Setting after motor stop	PS
P08	18	Filter A width	0~200	1	4	Effective immediat ely	Setting after motor stop	PS
P08	19	Filter B gain	0~100	1	100	Effective immediat ely	Setting after motor stop	PS
P08	20	Offline inertia Identification maximum speed	200 ~1000	1rpm	500	Effective immediat ely	Setting after motor stop	PST
P08	21	Offline inertia Identification acceleration and deceleration time	50~800	1ms	100	Effective immediat ely	Setting after motor stop	PST
P08	22	Waiting time after a single offline inertia identification is completed	100~10000	1ms	800	Effective immediat ely	Setting after motor stop	PST
P08	23	Inertia identification mode selection	0-Offline inertia identification: speed command is positive and negative triangle wave 1-Offline inertia identification: Speed command is JOG mode	1	0	Effective immediat ely	Setting after motor stop	PST
P08	24	Motor rotation round for complete single offline inertia identificatio	0~65535	0.01R	83	N/A	Display paramete r	PST
P08	26	High frequency vibration suppression control switch	0-OFF 1-ON	1	0	Effective immediat ely	Setting after motor stop	PS
P08	27	High frequency vibration suppression frequency	10~4000	1 Hz	4000	Effective immediat ely	Set when running	PS
P08	28	High frequency vibration suppression gain compensation	0~1000	0.01	100	Effective immediat ely	Set when running	PS
P08	29	High frequency vibration suppression damping	0~1000	0.01	180	Effective immediat ely	Set when running	PS

		coefficient 1						
P08	30	High frequency vibration suppression frequency compensation 1	-1000~1000	1 Hz	0	Effective immediately	Set when running	PS
P08	31	High frequency vibration suppression frequency compensation 2	-1000~1000	1 Hz	0	Effective immediately	Set when running	PS
P08	32	High frequency vibration suppression damping coefficient 2	0~1000	0.01	0	Effective immediately	Set when running	PS
P08	33	Anti-disturbance compensation switch	0-OFF 1-ON	1	0	Effective immediately	Setting after motor stop	PS
P08	36	Disturbance compensation factor	0~10000	0.1%	0	Effective immediately	Set when running	PS
P08	39	Momentary speed compensation switch	0-OFF 1-ON	1	0	Effective immediately	Setting after motor stop	PS
P08	40	Instantaneous speed compensation gain	0~1000	1 Hz	0	Effective immediately	Set when running	PS
P08	41	Instantaneous speed compensation gain compensation	0 ~1000	0.01	600	Effective immediately	Set when running	PS
P08	45	Model compensation switch	0 - Turn off model compensation 1 - Rigid model 2-Second order vector model	1	0	Effective immediately	Set when running	PS
P08	46	Model compensation gain	10~20000	0.1/s	300	Effective immediately	Setting after motor stop	PS
P08	48	Model Compensation Forward Compensation Coefficient	0~10000	0.1%	1000	Effective immediately	Setting after motor stop	PS
P08	49	Model Compensation Reverse Compensation Coefficient	0~10000	0.1%	1000	Effective immediately	Setting after motor stop	PS
P08	50	Model compensation vibration suppression frequency A	0~2500	0.1Hz	500	Effective immediately	Setting after motor stop	PS
P08	51	Model compensation vibration suppression frequency R	0~2500	0.1Hz	500	Effective immediately	Setting after motor stop	PS
P08	52	Model Compensation Speed Compensation Coefficient	0~10000	0.1%	1000	Effective immediately	Setting after motor stop	PS

P08 group: advanced adjustment parameters

P09 group: failure and protection

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P09	02	Undervoltage detection delay	100 ~20000	0.1ms	700	Effective immediately	Set when running	PST
P09	04	Motor out of control protection	0- Protection open 1- Protection off	1	0	Effective immediately	Set when running	PST
P09	05	Overload warning value	1～100	1%	90	Effective immediately	Set when running	PST
P09	06	Motor overload protection factor	10 ～300	1%	100	Effective immediately	Set when running	PST
P09	07	Undervoltage protection point	50 ～100（100 corresponds to the default undervoltage point）	1%	100	Effective immediately	Set when running	PST
P09	08	Overspeed fault point	50 ～ 120 （ 100 corresponds to maximum motor speed）	1%	120	Effective immediately	Set when running	PST
P09	09	Position deviation excessive threshold (32bit)	1~1073741824	1P	2500 line motor: 32767P 17-bit motor: 393216 23-bit motor: 25165824	Effective immediately	Set when running	PST
P09	12	Low-speed command pulse input pin filter time	0-500	8.33ns	45	Re-power	Setting after motor stop	P
P09	16	Average load rate overload threshold	100~3000	0.1%	1100	Effective immediately	Setting after motor stop	PST
P09	17	Multi-segment position/speed DI input delay detection time	0~1000	1ms	3	Effective immediately	Set when running	PS
P09	23	Motor stall over temperature protection time	10~65535	1ms	200	Effective immediately	Setting after motor stop	PST
P09	24	Motor stall over temperature protection enable	0 - Shield locked rotor over temperature protection monitoring detection 1- Turn on the motor stall and over-temperature protection monitoring and detection	1	0	Effective immediately	Set when running	PST
P09	25	Motor overload protection enable	0- Open motor overload and average load rate overload detection 1- Open motor overload, shield the average load rate overload detection 2- Shield motor overload, open average load rate overload detection 3- Shield motor overload and average load rate overload detection	1	0	Effective immediately	Setting after motor stop	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P09	29	Average load rate protection time	10~65535	160ms	300	Effective immediat ely	Setting after motor stop	PST

P10 group: communication parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P10	00	Communication address	0~247，0 is broadcast address	1	1	Effective immediat ely	Set when running	PST
P10	02	Modbus baud rate setting	0-2400 1-4800 2-9600 3-19200 4-38400 5-57600 6-115200	1	6	Effective immediat ely	Set when running	PST
P10	03	Modbus data format	0 -No parity, 2 stop bits 1-Even parity, 1 stop bit 2-Odd parity, 1 stop bit 3-No parity, 1 stop bit	1	0	Effective immediat ely	Set when running	PST
P10	04	Update writed communication function code to EEPROM?	0-Do not update EEPROM 1-Update to EEPROM except P11 group and P18 group	1	0	Effective immediat ely	Set when running	PST
P10	05	RS232 communication baud rate setting	0-2400 1-4800 2-9600 3-19200 4-38400 5-57600 6-115200	1	6	Effective immediat ely	Set when running	PST
P10	06	RS485 communication address selection	0- Dip switch Settings 1- Host computer communication Settings	1	0	Re-power	Setting after motor stop	PST
P10	07	node_id setting	0-31	1	0	Re-power	Setting after motor stop	PST
P10	08	Now Noid_ID	0-31	1	1	Read-onl y	Read-onl y	PST
P10	09	CAN baud rate	0-500K 1-1M 2-750K 3-500K 4-250K 5-125K 6-100K 7-50K 8-20K 9-10K	1	0	Re-power	Setting after motor stop	PST

P11 group: auxiliary function parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P11	01	Fault reset	0-No operation 1-Fault reset	1	0	Effective immediat	Setting after	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
						ely	motor stop	
P11	02	Soft reset	0-No operation 1-Fault reset	1	0	Effective immediately	Setting after motor stop	PST
P11	03	Inertia recognition function	Effective upon enter	1	0	Effective immediately	Set when running	PST
P11	06	Absolute encoder reset	0-No operation. 1-Absolute encoder related warning and error clearing. 2- Absolute encoder multiturn data reset.	1	0	Re-power	Setting after motor stop	PST
P11	07	Absolute value system soft limit setting	0 - no operation 1-The current position is set to the negative limit 2 - The current position is set as the positive limit	1	0	Effective immediately	Set when running	PST
P11	09	System initialization function	0-No operation 1-Restore factory setting (except P1 and P17 parameters) 2- Clear the fault record	1	0	Effective immediately	Setting after motor stop	PST
P11	10	DIDO forced input and output enable	0- No operation 1- Forced DI enable 2- Forced DO enable 3- Forced DIDO enabled	1	0	Effective immediately	Set when running	PST
P11	11	Set DI forced input	0-0x01FF	1	511	Effective immediately	Set when running	PST
P11	12	Set DO forced output	0-0x001F	1	0	Effective immediately	Set when running	PST
P11	13	Emergency Stop settings	0 – No operation 1- Emergency stop	1	0	Effective immediately	Set when running	PST

P12 group: version parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P12	00	LED warning display selection	0-LED immediately outputs warning message 1-LED does not output warning message	1	0	Effective immediately	Set when running	PST
P12	01	Default display settings	0~100	1	1	Effective immediately	Set when running	PST
P12	03	Speed display filter time	0~10000	0.1ms	50	Effective immediately	Set when running	PST
P12	11	Non-standard version number	VV.B.DD	1	0	N/A	Display parameter	
P12	12	Software version number	VV.B.DD	1	0	N/A	Display parameter	
P12	14	Product series code	PP.XXX	1	3000	N/A	Display parameter	

P13 group: multi-segment position

Function code	Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	00	Multi-segment position operation mode	0-Stop after the end of single operation (P13.01 for segment number selection) 1-Cycle operation (P13.01 for segment number selection) 2-DI switching operation (selected by DI)	1	1	Effective immediately	Setting after motor stop	P
P13	01	Specify starting segment	1～16	1	1	Effective immediately	Setting after motor stop	P
P13	02	Specify end segment	1～16	1	16	Effective immediately	Setting after motor stop	P
P13	03	Interrupted position handle setting	Valid in the other three modes except DI mode 0: Continue to run the segment that has not finished 1: Restart from the first segment	1	0	Effective immediately	Setting after motor stop	P
P13	04	Waiting time unit	0-ms 1-s	1	0	Effective immediately	Setting after motor stop	P
P13	05	Position control method	0-Incremental position control 1-Absolute position control	1	0	Effective immediately	Setting after motor stop	P
P13	08	Segment 1: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	10	Segment 1: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	11	Segment 1: acceleration/deceleration time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	12	Segment 1: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	13	Segment 2: movement displacement (32bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	15	Segment 2: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	16	Segment 2: acceleration/deceleration time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	17	Segment 2: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	18	Segment 3: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	20	Segment 3: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	21	Segment 3: acceleration/deceleration time	0~65535	1ms	1	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	22	Segment 3: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	23	Segment 4: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	25	Segment 4: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	26	Segment 4: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	27	Segment 4: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	28	Segment 5: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	30	Segment 5: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	31	Segment 5: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	32	Segment 5: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	33	Segment 6: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	35	Segment 6: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	36	Segment 6: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	37	Segment 6: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	38	Segment 7: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	40	Segment 7: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	41	Segment 7: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	42	时间 Segment 7: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	43	Segment 8: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	45	Segment 8: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	46	Segment 8: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	47	Segment 8: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	48	Segment 9: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	50	Segment 9: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	51	Segment 9: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	52	Segment 9: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	53	Segment 10: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	55	Segment 10: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	56	Segment 10: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	57	Segment 10: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	58	Segment 11: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	60	Segment 11: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	61	Segment 11: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	62	Segment 11: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	63	Segment 12: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	65	Segment 12: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	66	Segment 12: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	67	Segment 12: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	68	Segment 13: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	70	Segment 13: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	71	Segment 13: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	72	Segment 13: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	73	Segment 14: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	75	Segment 14: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	76	Segment 14: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	77	Segment 14: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	78	Segment 15: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	80	Segment 15: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	81	Segment 15: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	82	Segment 15: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	83	Segment 16: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	85	Segment 16: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	86	Segment 16: acceleration/deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	87	Segment 16: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P

P14 group: multi-speed command

Function code		ame	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P14	00	Multi-speed command operation mode	0- Stop after the end of single operation (P1401 for segment number selection) 1-Cycle operation (P1401 for segment number selection) 2-DI switching operation	1	1	Effective immediately	Setting after motor stop	S
P14	01	Speed command end segment selection	1～16	1	16	Effective immediately	Setting after motor stop	S
P14	02	Running time unit selection	0-s 1-min	1	0	Effective immediately	Setting after motor stop	S
P14	03	Acceleration/decele ration time 1	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	04	Acceleration/decele ration time 2	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	05	Acceleration/decele ration time 3	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	06	Acceleration/decele ration time 4	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	07	Speed segment 1	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	08	Running time of speed segment 1	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	09	Speed up and down time of speed segment 1	0- zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	10	Speed segment 2	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	11	Running time of speed segment 2	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	12	Speed up and down time of speed segment 2	0- zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	13	Speed segment 3	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	14	Running time of speed segment 3	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	15	Speed up and down time of speed segment 3	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	16	Speed segment 4	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	17	Running time of speed segment 4	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	18	Speed up and down time of speed segment 4	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	19	Speed segment 5	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S

Function code		ame	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P14	20	Running time of speed segment 5	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	21	Speed up and down time of speed segment 5	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	22	Speed segment 6	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	23	Running time of speed segment 6	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	24	Speed up and down time of speed segment 6	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	25	Speed segment 7	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	26	Running time of speed segment 7	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	27	Speed up and down time of speed segment 7	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	28	Speed segment 8	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	29	Running time of speed segment 8	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	30	Running time of speed segment 8	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	31	Speed segment 9	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	32	Running time of speed segment 9	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	33	Speed up and down time of speed segment 9	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	34	Speed segment 10	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	35	Running time of speed segment 10	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	36	Speed up and down time of speed segment 10	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	37	Speed segment 11	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	38	Running time of speed segment 11	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	39	Speed up and down time of speed segment 11	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S

Function code		ame	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P14	40	Speed segment 12	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	41	Running time of speed segment 12	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	42	Speed up and down time of speed segment 12	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	43	Speed segment 13	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	44	Running time of speed segment 13	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	45	Speed up and down time of speed segment 13	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	46	Speed segment 14	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	47	Running time of speed segment 14	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	48	Speed up and down time of speed segment 14	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	49	Speed segment 15	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	50	Running time of speed segment 15	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	51	Speed up and down time of speed segment 15	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	52	Speed segment 16	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	53	Running time of speed segment 16	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	54	Speed up and down time of speed segment 16	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S

P16 group: special function parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	00	Interrupt fixed length enable	0-Disable 1-Enable	1	0	Effective immediately	Setting after motor stop	P
P16	01	Interrupt fixed length 1 displacement	0~1073741824	1 Unit	10000	Effective immediately	Set when running	P
P16	03	Interrupt fixed length 1 constant running speed	0~9000	1rpm	200	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	04	Interrupt fixed length acceleration time	0~1000	1ms	200	Effective immediately	Set when running	P
P16	05	Interrupt fixed length deceleration time	0~1000	1ms	200	Effective immediately	Set when running	P
P16	06	Fixed length lock release signal enable	0~1	1	1	Effective immediately	Set when running	P
P16	08	Homing Enable control	0 - Turn off the Homing function. 1- Enable the Homing function by inputting the HomingStart signal through DI. 2-Start return to Home immediately after power-on. 3-Start return to Home immediately. 4- Take the current position as the Home. 5- Set the Home through DI trigger. 6- host computer homing	1	0	Effective immediately	Setting after motor stop	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	09	Homing mode	0-Forward homing, deceleration and home points are home switch; 1- Backward homing, deceleration and home points are home switch; 2- Forward homing, deceleration and home points are motor Z signal; 3- Backward homing, deceleration and home points are motor Z signal; 4- Forward homing, deceleration point is Home switch, home is motor Z signal; 5- Backward homing, deceleration point is Home switch, home is motor Z signal; 6- Forward homing, deceleration and home points are forward overtravel switch; 7- Backward homing, deceleration and home points are backward overtravel switch; 8- Forward homing, deceleration point is forward overtravel switch, home is motor Z signal; 9- Backward homing, deceleration point is backward overtravel switch, home is motor Z signal; 10- Forward homing, no deceleration point, home is home switch; 11-Backward homing, no deceleration point, home is home switch; 12- Forward homing, no deceleration point, home is motor Z signal; 13- Backward homing, no deceleration point, home is motor Z signal; 14- Borward homing, no deceleration point, home is forward overtravel switch; 15- Backward homing, no deceleration point, home is backward overtravel switch; 16-Absolute value homing; 17- Indexing absolute value homing；	1	0	Effective immediately	Setting after motor stop	P
P16	10	High-speed for searching home switch signal	10-3000	1rpm	100	Effective immediately	Setting after motor stop	P
P16	11	searching home switch signal	10 -1000	1rpm	10	Effective immediately	Setting after motor stop	P
P16	12	Limit the accel. / decel. time for searching home	0-65535	1ms	1000	Effective immediately	Setting after motor stop	P
P16	13	Limit time for finding home	0-65535	1ms	10000	Effective immediately	Setting after motor stop	P
P16	14	Mechanical home offset (32 bits)	-1073741824 ~ 1073741824	1Unit	0	Effective immediately	Setting after motor stop	P
P16	28	Encoder origin single-turn absolute position	0~4294967295	1inc	0	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	30	Encoder origin multi-turn count	0~32767	1inc	0	Effective immediately	Set when running	P
P16	31	Homing position arrival wait counter	0~65535	1ms	500	Effective immediately	Set when running	P

P17group: Driver parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P17	01	Drive model code	0~65535	1	20102	N/A	Read-only	PST
P17	02	Voltage class	0~65535	1V	48	N/A	Read-only	PST
P17	03	Rated power	0~65535	0.01KW	40	N/A	Read-only	PST
P17	04	Rated output current	0~65535	0.01A	1000	N/A	Read-only	PST
P17	05	Maximum output current	0~65535	0.01A	2500	N/A	Read-only	PST
P17	12	Current loop proportional gain 1	0~65535	1	2000	Effective immediately	Set when running	PST
P17	13	Current loop integral gain 1	0~65535	1	1200	Effective immediately	Set when running	PST
P17	14	Current loop proportional gain 2	0~65535	1	2000	Effective immediately	Set when running	PST
P17	15	Current loop integral gain 2	0~65535	1	1200	Effective immediately	Set when running	PST

P18 group: display parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P18	00	Servo status	r Rdy 、Run 、Err.00~99 （ error ） ,AL.00~10(warning）	1	-	N/A	Display parameter	PST
P18	01	Motor speed feedback (32 bit)	-9000~9000	1rpm	-	N/A	Display parameter	PST
P18	02	average load rate	0~3000	0.1%	-	N/A	Display parameter	PST
P18	03	Speed command	-9000~9000	1rpm	-	N/A	Display parameter	PST
P18	04	Internal torque command (relative to rated torque)	-5000~5000	0.1%	-	N/A	Display parameter	PST
P18	05	Phase current RMS	0~10000	0.01A	-	N/A	Display parameter	PST
P18	06	Bus voltage value	0~10000	0.1V	-	N/A	Display parameter	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P18	07	Absolute position counter (32 bit)	-1073741824 ~ 1073741824	1Unit	-	N/A	Display parameter	PST
P18	09	Electrical angle	0~3600	0.1°	-	N/A	Display parameter	PST
P18	10	Mechanical angle (relative to encoder zero)	0~3600	0.1°	-	N/A	Display parameter	PST
P18	11	Bus encoder communication check error counter	-	1	-	N/A	Display parameter	PST
P18	12	Corresponding speed information of input position command	-9000~9000	1rpm	-	N/A	Display parameter	PST
P18	13	Position deviation counter (32 bit)	-1073741824 ~ 1073741824	1P	-	N/A	Display parameter	PST
P18	15	Input command Pulse counter (32 bit)	-1073741824 ~ 1073741824	1Unit	-	N/A	Display parameter	PST
P18	17	Feedback pulse counter (32 bit)	-1073741824 ~ 1073741824	1P	-	N/A	Display parameter	PST
P18	19	Position deviation counter command unit (32 bit)	-1073741824 ~ 1073741824	1Unit	-	N/A	Display parameter	PST
P18	21	Digital input signal monitoring	-	-	-	N/A	Display parameter	PST
P18	22	Record the U-phase zero current value during shutdown	-32768~32767	0.01A	-	N/A	Display parameter	PST
P18	23	Digital output signal monitoring	-	-	-	N/A	Display parameter	PST
P18	24	Record the V-phase zero current value during shutdown	-32768~32767	0.01A	-	N/A	Display parameter	PST
P18	25	Total power-on time (32-bit)	0-4294967295	0.1s	-	N/A	Display parameter	PST
P18	27	U-Phase zero sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	28	V-Phase zero sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	29	U-phase current sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	30	V-phase current sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	31	Module temperature value	-	1°C	-	N/A	Display parameter	PST
P18	32	Absolute encoder single-round data	-	pulse	-	N/A	Display parameter	PST
P18	34	Absolute encoder multi-round data	-	turn	-	N/A	Display parameter	PST
P18	35	Maximum load rate	0~3000	0.1%	-	N/A	Display parameter	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P18	40	Display of fault record	0- Current fault 1- The first fault before current fault 2- The second fault before current fault ..... 9- The 9th fault before current fault	1	0	Effective immediatel y	Set when running	PST
P18	41	Error code	-	-	-	N/A	Display parameter	PST
P18	42	Selected fault timestamp (32 bit)	-	0.1s	-	N/A	Display parameter	PST
P18	44	In-time speed at selected fault	-	1rpm	-	N/A	Display parameter	PST
P18	45	In-time current U at Selected fault	-	0.01A	-	N/A	Display parameter	PST
P18	46	In-time current V at selected fault	-	0.01A		N/A	Display parameter	PST
P18	47	Bus voltage at selected fault	-	0.1V	-	N/A	Display parameter	PST
P18	48	Input terminal status at selected fault	-	-	-	N/A	Display parameter	PST
P18	49	Output terminal status at selected fault	-	-	-	N/A	Display parameter	PST

8.3 Object Dictionary Parameters

8.3.1 1000 Group Object Dictionary

Index	Sub-index	Name	Read/Write Properties	Mappable	Unit	Data Type
1000h	-	Device type	RO	NO	VAR	UINT32
1001h	-	Error register	RO	NO	VAR	UINT8
1003h	00h	Number of predefined error fields	RO	NO	ARR	UINT8
	1-5	Error field	RW	NO	-	UINT32
1005h	-	Sync message COB-ID	RW	NO	VAR	UINT32
1006h	-	Sync cycle period	RW	NO	VAR	UINT32
1009h	-	Hardware version	RO	NO	VAR	UINT8
100Ah	-	Software version	RO	NO	VAR	UINT8
100Ch	-	Node guarding time	RW	NO	VAR	UINT16
100Dh	-	Life time factor	RW	NO	VAR	UINT8
1014h	-	Emergency message COB-ID	RW	NO	VAR	UINT32

1016h	00h	Consumer heartbeat time index count	RW	NO	VAR	UINT8
	1	Consumer heartbeat time	RW	NO	VAR	UINT32
1017h	-	Producer heartbeat time	RW	NO	VAR	UINT16
1018h	00h	Device object sub-index count	RO	NO	ARR	UINT8
	01h	Vendor ID	RO	NO	VAR	UINT32
	02h	Product code	RO	NO	VAR	UINT32
	03h	Revision number	RO	NO	VAR	UINT32
	04h	Serial number	RO	NO	VAR	UINT32
1200h	00h	SDO parameter sub-index count	RO	NO	VAR	UINT8
	01h	Client to server COB-ID	RO	NO	VAR	UINT32
	02h	Server to client COB-ID	RO	NO	VAR	UINT32
1400h	00h	RPDO1 communication parameter sub-index count	RO	NO	VAR	UINT8
	01h	RPDO1 COB-ID	RW	NO	VAR	UINT32
	02h	RPDO1 transmission type	RW	NO	VAR	UINT8
	03h	RPDO1 inhibit time	RW	NO	VAR	UINT16
	04h	RPDO1 compatibility entry	RW	NO	VAR	UINT8
	05h	RPDO1 event timer	RW	NO	VAR	UINT16
	06h	RPDO1 sync start value	RW	NO	VAR	UINT8
1401h	00h	RPDO2 communication parameter sub-index count	RO	NO	VAR	UINT8
	01h	RPDO2 COB-ID	RW	NO	VAR	UINT32
	02h	RPDO2 transmission type	RW	NO	VAR	UINT8
	03h	RPDO2 inhibit time	RW	NO	VAR	UINT16
	04h	RPDO2 compatibility entry	RW	NO	VAR	UINT8

	05h	RPDO2 event timer	RW	NO	VAR	UINT16
	06h	RPDO2 sync start value	RW	NO	VAR	UINT8
1402h	00h	RPDO3 communication parameter sub-index count	RO	NO	VAR	UINT8
	01h	RPDO3 COB-ID	RW	NO	VAR	UINT32
	02h	RPDO3 transmission type	RW	NO	VAR	UINT8
	03h	RPDO3 inhibit time	RW	NO	VAR	UINT16
	04h	RPDO3 compatibility entry	RW	NO	VAR	UINT8
	05h	RPDO3 event timer	RW	NO	VAR	UINT16
	06h	RPDO3 sync start value	RW	NO	VAR	UINT8
1403h	00h	RPDO4 communication parameter sub-index count	RO	NO	VAR	UINT8
	01h	RPDO4 COB-ID	RW	NO	VAR	UINT32
	02h	RPDO4 transmission type	RW	NO	VAR	UINT8
	03h	RPDO4 inhibit time	RW	NO	VAR	UINT16
	04h	RPDO4 compatibility entry	RW	NO	VAR	UINT8
	05h	RPDO4 event timer	RW	NO	VAR	UINT16
	06h	RPDO4 sync start value	RW	NO	VAR	UINT8
1600h	00h	RPDO1 mapping count	RW	NO	VAR	UINT8
	01-08h	RPDO1 mapped object	RW	NO	VAR	UINT32
1601h	00h	RPDO2 mapping count	RW	NO	VAR	UINT8
	01-08h	RPDO2 mapped object	RW	NO	VAR	UINT32
1602h	00h	RPDO3 mapping count	RW	NO	VAR	UINT8
	01-08h	RPDO3 mapped object	RW	NO	VAR	UINT32
1603h	00h	RPDO4 mapping count	RW	NO	VAR	UINT8
	01-08h	RPDO4 mapped object	RW	NO	VAR	UINT32
1800h	00h	TPDO1 communication parameter sub-index	RO	NO	VAR	UINT8

		count				
	01h	TPDO1 COB-ID	RW	NO	VAR	UINT32
	02h	TPDO1 transmission type	RW	NO	VAR	UINT8
	03h	TPDO1 inhibit time	RW	NO	VAR	UINT16
	04h	TPDO1 compatibility entry	RW	NO	VAR	UINT8
	05h	TPDO1 event timer	RW	NO	VAR	UINT16
	06h	TPDO1 sync start value	RW	NO	VAR	UINT8
1801h	00h	TPDO2 communication parameter sub-index count	RO	NO	VAR	UINT8
	01h	TPDO2 COB-ID	RW	NO	VAR	UINT32
	02h	TPDO2 transmission type	RW	NO	VAR	UINT8
	03h	TPDO2 inhibit time	RW	NO	VAR	UINT16
	04h	TPDO2 compatibility entry	RW	NO	VAR	UINT8
	05h	TPDO2 event timer	RW	NO	VAR	UINT16
	06h	TPDO2 sync start value	RW	NO	VAR	UINT8
1802h	00h	TPDO3 communication parameter sub-index count	RO	NO	VAR	UINT8
	01h	TPDO3 COB-ID	RW	NO	VAR	UINT32
	02h	TPDO3 transmission type	RW	NO	VAR	UINT8
	03h	TPDO3 inhibit time	RW	NO	VAR	UINT16
	04h	TPDO3 compatibility entry	RW	NO	VAR	UINT8
	05h	TPDO3 event timer	RW	NO	VAR	UINT16
	06h	TPDO3 sync start value	RW	NO	VAR	UINT8
1803h	00h	TPDO4 communication parameter sub-index count	RO	NO	VAR	UINT8
	01h	TPDO4 COB-ID	RW	NO	VAR	UINT32
	02h	TPDO4 transmission	RW	NO	VAR	UINT8

		type				
	03h	TPDO4 inhibit time	RW	NO	VAR	UINT16
	04h	TPDO4 compatibility entry	RW	NO	VAR	UINT8
	05h	TPDO4 event timer	RW	NO	VAR	UINT16
	06h	TPDO4 sync start value	RW	NO	VAR	UINT8
1A00h	00h	TPDO1 mapping count	RW	NO	VAR	UINT8
	01-08h	TPDO1 mapped object	RW	NO	VAR	UINT32
1A01h	00h	TPDO2 mapping count	RW	NO	VAR	UINT8
	01-08h	TPDO2 mapped object	RW	NO	VAR	UINT32
1A02h	00h	TPDO3 mapping count	RW	NO	ARR	UINT8
	01-08h	TPDO3 mapped object	RW	NO	VAR	UINT32
1A03h	00h	TPDO3 mapping count (duplicate)	RW	NO	VAR	UINT8
	01-08h	TPDO3 mapped object (duplicate)	RW	NO	VAR	UINT32

8.3.2 6000 Group Object Dictionary

Index	Sub-index	Name	Read/Write Properties	Mappable	Unit	Data Type
603Fh	-	Error code	RO	YES	VAR	UINT16
6040h	-	Control word	RW	YES	VAR	UINT16
6041h	-	Status word	RO	YES	VAR	UINT16
605Ah	-	Quick stop function selection	RW	NO	VAR	INT16
605Bh	-	Shutdown stop function selection	RW	NO	VAR	INT16
605Ch	-	Disable stop selection	RW	NO	VAR	INT16

605Dh	-	Pause function selection	RW	NO	VAR	INT16
605Eh	-	Fault stop function selection	RW	NO	VAR	INT16
6060h	-	Servo operation mode selection	RW	YES	VAR	INT8
6061h	-	Servo operation mode display	RO	YES	VAR	INT8
6062h	-	Valid target position	RO	YES	Command Unit	INT32
6063h	-	Position feedback value (incremental)	RO	YES	Encoder unit	INT32
6064h	-	Position feedback value	RO	YES	Command Unit	INT32
6065h	-	Following error window	RW	YES	Command Unit	UINT32
6066h	-	Following error window time	RW	YES	ms	UINT16
6067h	-	Position arrival window	RW	NO	Command Unit	UINT32
6068h	-	Position arrival window time	RW	YES	ms	UINT16
606Bh	-	Valid target velocity	RO	YES	Command Unit	INT32
606Ch	-	Velocity feedback	RO	YES	rpm	INT32
606Dh	-	Velocity arrival window	RW	YES	rpm	UINT16
606Eh	-	Velocity arrival window time	RW	YES	ms	UINT16
606Fh	-	Zero speed threshold	RW	YES	rpm	UINT16
6070h	-	Zero speed threshold time	RW	YES	ms	UINT16
6071h	-	Target torque	RW	YES	0.1% (relative to rated torque)	INT16

6072h	-	Maximum torque	RW	YES	0.1% (relative to rated torque)	UINT16
6073h	-	Maximum current	RW	YES	0.1% (relative to rated current)	UINT16
6074h	-	Torque set value	RO	YES	0.1% (relative to rated torque)	INT16
6077h	-	Torque feedback	RW	YES	0.1% (relative to rated torque)	INT16
6078h	-	Current feedback	RO	YES	0.1% (relative to rated current)	INT16
6079h	-	Bus voltage	RO	YES	mV	UINT32
607Ah	-	Target position	RW	YES	Command Unit	INT32
607Bh	00h	Position range sub-index count	RO	NO	VAR	UINT8
	01h	Position range minimum value	RW	YES	Command Unit	INT32
	02h	Position range maximum value	RW	YES	Command Unit	INT32
607Ch	-	Homing offset	RW	YES	Command Unit	INT32
607Dh	00h	Soft position limit sub-index count	RO	NO	VAR	UINT8
	01h	Soft position limit minimum value	RW	YES	Command Unit	INT32
	02h	Soft position limit maximum value	RW	YES	Command Unit	INT32
607Eh	-	Command polarity	RW	YES	VAR	UINT8
607Fh	-	Maximum profile velocity	RW	YES	Command Unit	UINT32
6080h	-	Maximum motor speed	RW	YES	Command Unit	UINT32

6081h	-	Profile velocity	RW	YES	Command Unit	UINT32
6082h	-	End velocity	RW	YES	Command Unit	UINT32
6083h	-	Profile acceleration	RW	YES	Command Unit	UINT32
6084h	-	Profile deceleration	RW	YES	Command Unit	UINT32
6085h	-	Quick stop deceleration	RW	YES	Command Unit	UINT32
6087h	-	Torque slope	RW	YES	0.1% (relative to rated torque)	UINT32
608Fh	00h	Position encoder resolution sub-index count	RO	NO	VAR	UINT8
	01h	Encoder increment	RW	YES	Encoder unit	UINT32
	02h	Motor revolutions	RW	YES	VAR	UINT32
6091h	00h	Electronic gear ratio sub-index count	RO	NO	VAR	UINT8
	01h	Motor revolutions	RW	YES	VAR	UINT32
	02h	Driven shaft revolutions	RW	YES	VAR	UINT32
6093h	00h	Position conversion factor sub-index count	RO	NO	VAR	UINT8
	01h	Position conversion numerator	RW	YES	VAR	UINT32
	02h	Position conversion denominator	RW	YES	VAR	UINT32
6094h	00h	Velocity conversion sub-index count	RO	NO	VAR	UINT8
	01h	Velocity conversion numerator	RW	YES	VAR	UINT32
	02h	Velocity conversion denominator	RW	YES	VAR	UINT32

6095h	00h	Velocity coefficient 1 sub-index count	RO	NO	VAR	UINT8
	01h	Velocity conversion 1 numerator	RW	YES	VAR	UINT32
	02h	Velocity conversion 2 denominator	RW	YES	VAR	UINT32
6097h	00h	Acceleration conversion sub-index count	RO	NO	VAR	UINT8
	01h	Acceleration conversion numerator	RW	YES	VAR	UINT32
	02h	Acceleration conversion denominator	RW	YES	VAR	UINT32
6098h	-	Homing method	RW	YES	VAR	INT8
6099h	00h	Homing speed sub-index count	RO	NO	VAR	UINT8
	01h	Homing high speed	RW	YES	rpm	UINT32
	02h	Homing low speed	RW	YES	rpm	UINT32
609Ah	-	Homing acceleration	RW	YES	ms	UINT32
60C5h	-	Maximum acceleration	RW	YES	Command Unit	UINT32
60C6h	-	Maximum deceleration	RW	YES	Command Unit	UINT32
60F4h	-	Actual following error value	RO	YES	Command Unit	INT32
60FCh	-	Position increment	RO	YES	Command Unit	INT32
60FDh	-	Digital input	RO	YES	VAR	UINT32
60FEh	00h	Digital output sub-index count	RO	NO	VAR	UINT8
	01h	Digital output	RW	YES	VAR	UINT32

	02h	Digital output bitmask	RW	YES	VAR	UINT32
60FFh	-	Target velocity	RW	YES	Command Unit	INT32
6502h	-	Supported operation modes	R0	YES	VAR	UINT32

Appendix A: Alarm light information description

Fault	Decription	Red light state			
		Flicker frequency	Flicker number	Flicker times	Cyclic flicker
Voltage fault	Overvoltage Undervoltage (The main circuit)	1HZ	1	3 seconds	YES
Current fault	Overcurrent (Overcurrent A, Overcurrent B)	1HZ	2	3 seconds	YES
Overload/ Motor stall	Drive overload, Motor overload, Motor stall	1HZ	3	3 seconds	YES
Parameter setting error	DI/DO terminal parameter setting fault，Soft limit setting erro	1HZ	4	3 seconds	YES
Encoder error	Encoder matching fault，Encoder disconnection，Encoder check exception	1HZ	5	3 seconds	YES
Motor verification erro	Motor data verification failure	1HZ	6	3 seconds	YES
Remaining faults	Faults other than the above faults	Stay on			
No faults	-	Stay off			

Appendix C: Brake resistor configuration

When the servo motor is in the state of deceleration braking, the motor will be in the state of generating electricity and feed the electric energy back to the servo driver bus circuit. When the driver bus voltage detection value is higher than the set threshold, the driver will report the overvoltage warning, and the energy generated by the motor needs to be consumed by an external brake resistor. Connect the brake resistance through RB+ and RB- at the power end, and correctly set the brake resistance value and brake resistance power. The resistance value of the optional brake resistance cannot be lower than the recommended resistance value. The recommended parameters of the brake resistance are shown in the following table:

Mode	Brake resistance value (Ω)	Brake resistance power（W）
DSY-200C	10	100
DSY-400C	10	100
DSY-750C	5	100
DSY-1200C	3.5	200

Appendix C: Version change log

Date	Modified version	Content of change
2025.5.15	V1.0	First release